

UNIVERSITY OF PISA  
DEPARTMENT OF PHYSICS  
MASTER'S DEGREE IN PHYSICS OF COMPLEX SYSTEMS

# Anomalous Transport in Soaring Flights

Collective Strategies and Intermittent Search Models

*Candidate*

Matteo Di Sante

*Supervisors*

Prof. Michael Benzaquen

Dr Alexandre Darmon

*Internal tutor (Pisa)*

Prof. Riccardo Mannella

Academic year 2025–2026

Working draft — generated on 2026-08-05.

Sources and code: <https://github.com/matteodisante/soaring-anomalous-transport>

## Abstract

Soaring birds and human pilots stay aloft by extracting energy from atmospheric updrafts, producing long, intermittent trajectories that alternate climbs within thermals with quasi-ballistic glides. Such motion is a candidate for *anomalous transport*, that is, a departure from ordinary Brownian diffusion. This thesis characterizes the transport statistics of real soaring flights from a large empirical dataset and seeks a minimal stochastic transport model able to reproduce those statistics. As the work develops it will also examine how flying in competition changes the motion relative to solo flights.

This document reports the current state of the project. A first dataset has been acquired reproducibly—the archive of the FFVL *Coupe Fédérale de Distance*, in its paraglider and hang-glider disciplines—and Chapter 2 describes its acquisition, its coverage and the pre-processing that turns the raw logs into analysis-ready trajectories.

Chapter 3 measures the transport of the retained ensemble without segmenting a flight. Displacement from take-off turns out to measure the geometry of the launch site rather than the motion, and is retired as a source of exponents in favour of a within-flight estimator that annihilates any polynomial trend. Over 60 s to 2,000 s that estimator gives  $\alpha = 2.02 \pm 0.10$  for paragliders and  $1.94 \pm 0.13$  for hang gliders — strongly super-diffusive, and moved by no stratification the archive allows by more than its own uncertainty. Beyond the second moment, the spectrum of moments is straight, so the motion is monofractal and is *not* a Lévy walk; the propagator is nonetheless not Gaussian; and no leg duration is distinguished, which is a constraint on any segmentation attempted next. Chapter 4 sets out that segmentation and the modelling it enables, and is a plan rather than a result.

# Contents

|          |                                                                     |           |
|----------|---------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                                 | <b>1</b>  |
| 1.1      | Motivation . . . . .                                                | 1         |
| 1.2      | Objectives . . . . .                                                | 2         |
| 1.3      | Scope of this document . . . . .                                    | 2         |
| <b>2</b> | <b>The dataset</b>                                                  | <b>4</b>  |
| 2.1      | Data acquisition . . . . .                                          | 4         |
| 2.1.1    | Source: the FFVL Coupe Fédérale de Distance . . . . .               | 4         |
| 2.1.2    | Collected data . . . . .                                            | 4         |
| 2.1.3    | Pipeline and tooling . . . . .                                      | 4         |
| 2.1.4    | Reproducibility . . . . .                                           | 5         |
| 2.1.5    | Integrity . . . . .                                                 | 5         |
| 2.1.6    | Further data sources . . . . .                                      | 5         |
| 2.2      | The IGC tracklog format . . . . .                                   | 5         |
| 2.3      | Flight metadata and the catalog . . . . .                           | 6         |
| 2.4      | Glider categories . . . . .                                         | 7         |
| 2.5      | Coverage and statistics . . . . .                                   | 8         |
| 2.6      | Data quality and caveats . . . . .                                  | 10        |
| 2.7      | Trajectory pre-processing . . . . .                                 | 11        |
| 2.7.1    | Choice of altitude channel . . . . .                                | 13        |
| 2.7.2    | Fix-level cleaning . . . . .                                        | 16        |
| 2.7.3    | Trimming of ground phases . . . . .                                 | 25        |
| 2.7.4    | Flight-level filtering . . . . .                                    | 27        |
| 2.7.5    | From geographic to local Cartesian coordinates . . . . .            | 30        |
| 2.7.6    | Uniform sampling within a flight . . . . .                          | 32        |
| 2.7.7    | Smoothing and differentiation . . . . .                             | 37        |
| 2.7.8    | Notation . . . . .                                                  | 40        |
| 2.8      | Preliminary characterization . . . . .                              | 40        |
| 2.9      | Pipeline summary . . . . .                                          | 43        |
| <b>3</b> | <b>Global transport</b>                                             | <b>50</b> |
| 3.1      | Displacement from take-off measures the launch geometry . . . . .   | 50        |
| 3.2      | The filtered variation . . . . .                                    | 54        |
| 3.3      | The uncertainty on the exponent . . . . .                           | 58        |
| 3.4      | The exponent from the increment quantiles . . . . .                 | 61        |
| 3.5      | The archive is not a Lévy walk . . . . .                            | 64        |
| 3.6      | The propagator is not Gaussian . . . . .                            | 67        |
| 3.7      | The memory of the heading . . . . .                                 | 68        |
| 3.8      | Speed and vertical velocity . . . . .                               | 69        |
| 3.9      | What this establishes, and what the segmentation receives . . . . . | 70        |

|            |                                                                                    |            |
|------------|------------------------------------------------------------------------------------|------------|
| <b>4</b>   | <b>Flight phases, and what they explain</b>                                        | <b>73</b>  |
| 4.1        | Segmentation into flight phases                                                    | 73         |
| 4.2        | Phase-resolved observables                                                         | 74         |
| 4.3        | Reproducing the reference results                                                  | 75         |
| 4.4        | Modelling and simulation                                                           | 75         |
| 4.5        | Tooling                                                                            | 76         |
|            | <b>Appendices</b>                                                                  | <b>79</b>  |
| <b>2.A</b> | <b>Power spectral density and Welch’s method</b>                                   | <b>80</b>  |
| <b>2.B</b> | <b>Geodetic transforms: from <math>(\varphi, \lambda, h)</math> to ECEF to ENU</b> | <b>82</b>  |
| 2.B.1      | Geodetic coordinates to ECEF                                                       | 82         |
| 2.B.2      | ECEF to local ENU                                                                  | 83         |
| 2.B.3      | Size of the tangent-plane error                                                    | 84         |
| <b>3.A</b> | <b>CTRW with power-law waiting times: EA-MSD vs TA-MSD</b>                         | <b>85</b>  |
|            | <b>Implementation and Computational Details</b>                                    | <b>88</b>  |
| <b>1</b>   | <b>Overview</b>                                                                    | <b>89</b>  |
| <b>2</b>   | <b>The dataset</b>                                                                 | <b>92</b>  |
| 2.1        | Data acquisition                                                                   | 92         |
| 2.1.1      | Pipeline and tooling                                                               | 92         |
| 2.1.2      | Reproducibility                                                                    | 93         |
| 2.1.3      | Integrity                                                                          | 93         |
| 2.2        | Where the data lives                                                               | 93         |
| 2.3        | The IGC tracklog format                                                            | 97         |
| 2.4        | Flight metadata and the catalog                                                    | 97         |
| 2.5        | Glider categories                                                                  | 97         |
| 2.6        | Coverage and statistics                                                            | 97         |
| 2.7        | Trajectory pre-processing                                                          | 97         |
| 2.7.1      | Choice of altitude channel                                                         | 98         |
| 2.7.2      | Fix-level cleaning                                                                 | 101        |
| 2.7.3      | Trimming of ground phases                                                          | 106        |
| 2.7.4      | Flight-level filtering                                                             | 106        |
| 2.7.5      | From geographic to local Cartesian coordinates                                     | 107        |
| 2.7.6      | Uniform sampling within a flight                                                   | 107        |
| 2.7.7      | Smoothing and differentiation                                                      | 108        |
| 2.8        | Preliminary characterization                                                       | 109        |
| <b>3</b>   | <b>Global transport</b>                                                            | <b>111</b> |
| 3.1        | Observables on the whole trajectory                                                | 111        |
| 3.1.1      | The two estimators                                                                 | 111        |
| 3.1.2      | The fit, and what is quoted from it                                                | 112        |
| 3.1.3      | The audit pass                                                                     | 113        |
| 3.1.4      | The shape observables, and what calibrates them                                    | 114        |
| 3.1.5      | Known limitations, with their size                                                 | 116        |

|          |                                             |            |
|----------|---------------------------------------------|------------|
| <b>4</b> | <b>Flight phases</b>                        | <b>117</b> |
| 4.1      | Segmentation into flight phases . . . . .   | 117        |
| 4.2      | Phase-resolved observables . . . . .        | 117        |
| 4.3      | Reproducing the reference results . . . . . | 117        |
| 4.4      | Modelling and simulation . . . . .          | 117        |
| 4.5      | Tooling . . . . .                           | 117        |

# Chapter 1

## Introduction

### 1.1 Motivation

Gliders, paragliders, and soaring birds carry no continuous source of propulsion; to remain airborne they extract energy from air currents in the atmosphere, primarily from *thermals*, the buoyant updrafts that form as sunlight heats the ground. A cross-country flight thus proceeds as a repeating cycle of three phases. In the *transition* phase the flyer glides along a nearly straight path toward the next source of lift. If none is reached at a safe altitude, a *search* phase follows, during which the flyer probes the surrounding air mass to locate lift. Once lift is found, a *climb* phase begins, in which the flyer circles within the rising air to regain altitude. The cycle alternates long, directed displacements with episodes of slow, localized motion.

Intermittent and directionally persistent motion of this kind is one of the settings in which *anomalous transport* arises. Let  $\mathbf{r}(t)$  denote the horizontal position of a flight, measured from take-off. Transport is anomalous when the mean-squared displacement grows as a non-trivial power of time,  $\langle |\mathbf{r}(t)|^2 \rangle \sim t^\alpha$  with  $\alpha \neq 1$ , as opposed to the linear growth ( $\alpha = 1$ ) of ordinary Brownian diffusion. The continuous-time random walk (CTRW) is the standard modelling framework for such regimes [1]: the walker waits a random time  $\tau$ , then makes a random displacement  $\mathbf{x}$ , and the joint density  $\psi(\mathbf{x}, \tau)$  of the two fixes the transport. The Lévy walk is not a separate framework but the particular case in which the two are coupled through a constant speed  $v$ ,

$$\psi(\mathbf{x}, \tau) \propto \delta(|\mathbf{x}| - v\tau) w(\tau), \quad (1.1)$$

with the direction drawn uniformly, so that each waiting time is spent travelling ballistically rather than standing still [2]. Anomalous transport of this kind has been reported in random-search strategies [3], in disordered media [4], and in turbulent flows [5].

The soaring cycle lies somewhere between the uncoupled and the fully coupled case, and it has three parts rather than two: a directed glide, then a search phase, then a localized climb. The glides are legs flown at a nearly constant speed, as in a Lévy walk; the search and the climb that end each leg are episodes of slow, localized motion, closer to the waits of an uncoupled CTRW. Whether those two phases can be collapsed into a single waiting phase is a modelling decision that the data have to support, and it is argued for where the quantitative model is built (Sec. 4.4). One feature is captured by neither limiting case: successive glides need not point in independent directions. The model this thesis works towards is therefore a persistent random walk in continuous time—one whose successive leg directions are correlated, with waiting times between legs.

Which of these ingredients the data actually require is settled by measurement, not assumed in advance, and the order of the measurement is the order of what it needs. Chapter 3 takes the whole trajectory, unsegmented, and asks what class of transport it is: an exponent, whether one exponent governs every moment, how long a heading survives, and whether any leg duration

is distinguished. Chapter 4 then splits the flight into its phases, which is what the leg lengths and waiting times of the model require, and the minimal model is built on what both return. The persistence in particular is a working hypothesis about angular memory, which only the measured heading correlations can confirm or rule out. Whether the trajectories of real soaring flights belong to an anomalous regime, and of which type, is the question addressed in this thesis.

Soaring has previously been studied mainly as a control problem—how to centre and exploit a thermal efficiently, including by reinforcement learning [6]. The present work is statistical instead: it asks which transport properties emerge at the level of the ensemble of observed trajectories. Its direct precedent is the study of Vilpellet et al. [7], who analysed a large set of human soaring flights and found a horizontal transport that is sub-ballistic and approximately independent of aircraft type (Hurst exponent  $H \approx 0.88$ , i.e.  $\langle |\mathbf{r}(t)|^2 \rangle \sim t^{2H}$ , between diffusive  $H = 1/2$  and ballistic  $H = 1$ , so that a value  $1/2 < H < 1$  marks superdiffusive transport). That study also introduced a phase-resolved analysis based on the transition–search–climb cycle. This thesis builds on it: it enlarges the dataset, revisits the cleaning and filtering that precede any analysis, and re-examines both the phase segmentation and the transport analysis.

## 1.2 Objectives

The thesis pursues the following objectives:

1. to assemble a large, reproducible, and documented dataset of real soaring flights;
2. to clean and filter the raw GPS logs into analysis-ready trajectories, and to define, measure and analyze suitable transport observables—the test for anomalous transport is carried out on those observables, as soon as the trajectories exist;
3. to define a segmentation strategy for the transition–search–climb cycle and characterize the motion statistics within each phase;
4. to build a minimal stochastic model able to reproduce the empirical statistics;
5. to extend the dataset to competition flights and examine how flying in competition changes the motion relative to solo flights.

A first dataset—the archive of the *Coupe Fédérale de Distance* (CFD) of the French free-flight federation (FFVL), covering paraglider and hang-glider flights—has been assembled and verified and is described in Chapter 2; data collection continues and will be extended to further sources, in particular sailplanes and competition flights (Sec. 2.1). The dataset (Secs. 2.2–2.5) and its cleaning-and-filtering pipeline (Sec. 2.7) are described in Chapter 2, and the transport of the unsegmented trajectory is measured in Chapter 3; the segmentation, the modelling it enables (Chapter 4), and the extension to competition flights define the work ahead.

## 1.3 Scope of this document

This document reports the current state of the project and is revised as the work proceeds. The statistics it quotes, and the tables and figures that carry them, are generated from the dataset by the analysis code rather than typed, so those numbers cannot drift from the data they describe. Chapter 2 follows the data end to end: its acquisition and the reproducibility of the pipeline (Sec. 2.1); the tracklog format and the flight catalog (Secs. 2.2–2.3); the glider categories and the coverage statistics (Secs. 2.4–2.5); the known data-quality caveats (Sec. 2.6); the pre-processing that turns the raw logs into analysis-ready trajectories (Sec. 2.7); a first

characterization of the cleaned dataset (Sec. 2.8); and a one-map summary of the whole pipeline that closes the chapter (Sec. 2.9). Chapter 3 measures the transport of that ensemble on the whole trajectory, without segmenting a flight: the estimator and the exponent it returns (Secs. 3.2 and 3.3), and what a second moment cannot say (Sec. 3.5). Chapter 4 sets out the phase segmentation and the modelling it enables; none of it is implemented yet.

Throughout, the discussion keeps to the scientific and statistical reasoning behind each choice. Implementation-level detail (code modules, configuration values, sample sizes, computation times, reproduction commands) is collected in [Implementation and Computational Details](#), from p. 89. That part is organised chapter by chapter to mirror the body, and the body links into it at each point where an implementation detail was trimmed out of the text, as “implementation details: Sec. X”.



# Chapter 2

## The dataset

This chapter follows the data end to end: how the raw archive is acquired, verified and kept reproducible (Sec. 2.1); what the records contain and what the collection looks like (Secs. 2.2–2.6); and the pre-processing that turns a raw tracklog into the analysis-ready trajectory the transport analysis consumes (Sec. 2.7), *closed by a first characterization of the cleaned dataset (Sec. 2.8) and a one-map summary of the pipeline (Sec. 2.9).*

### 2.1 Data acquisition

#### 2.1.1 Source: the FFVL Coupe Fédérale de Distance

The data come from the *Coupe Fédérale de Distance* (CFD), the cross-country competition run by the French free-flight federation (FFVL). The federation operates two parallel competitions on separate sites: one for paragliders (`parapente.ffvl.fr`) and one for hang gliders (`delta.ffvl.fr`). Pilots upload their flights; each entry carries metadata—among them pilot, date, take-off and landing sites, scored distance, duration, and glider, with the full set of fields listed in Sec. 2.3—and, in most cases, the raw GPS tracklog in IGC format (the standard flight-recorder format, Sec. 2.2). A “season” is identified by its starting year, e.g. the year 1999 denotes the 1999–2000 season; the paraglider archive starts in 1999–2000 and the hang-glider archive in 2001–2002.

#### 2.1.2 Collected data

For every season the pipeline retrieves two things:

- the **season XML export**, archived verbatim as provenance (`raw_xml/<year>.xml`); it is the authoritative list of flights and links;
- the **IGC tracklog files** referenced by the XML, one file per flight that has a track.

From these, a derived **catalog** is built (`catalog.csv`, one row per flight, its columns detailed in Sec. 2.3) together with a per-season index (`seasons_index.csv`). The raw data (tens of gigabytes) live on an external disk and are never committed to the repository. The catalog is itself rebuildable from the archived XMLs and likewise uncommitted (Sec. 2.3).

#### 2.1.3 Pipeline and tooling

Acquisition is performed by a dedicated, installable Python package, exposed through two command-line tools, one per discipline, sharing the same subcommands for fetching, downloading, cataloguing, and verifying. The design favours robustness for a long-running, large

job: downloads are resumable and safe to interrupt, a response is accepted only once validated as real IGC content, and download requests are rate-limited so as not to load the server (implementation details: Sec. 2.1.1).

#### 2.1.4 Reproducibility

The destination directory is not hard-coded: it is set per machine through an environment variable or a configuration file, of which the repository ships only a placeholder, so no machine-specific path is committed. Archiving the season XMLs verbatim keeps the full provenance (implementation details: Sec. 2.1.2).

#### 2.1.5 Integrity

Because the data sit on an external drive, integrity is re-checked independently after download, rather than trusted at download time (implementation details: Sec. 2.1.3). Both FFVL collections have been verified in full: all 186,052 paraglider and 6,716 hang-glider downloaded tracklogs pass the same content check applied at download — a readable file whose first record is an A header and which carries at least one B fix record (Sec. 2.2). These are the tracklogs actually fetched; they cover a subset of the flights that carry a track at all (Sec. 2.5).

#### 2.1.6 Further data sources

The FFVL CFD is the primary source of the present study, but the dataset is not assumed to be final. The paraglider data are the main analysis target; the hang-glider data are collected in parallel and enable the cross-type comparison of Sec. 2.4 (the counts for both are given in Sec. 2.5). Additional public soaring repositories are being pursued to enlarge and diversify the sample and, above all, to add the third glider category, *sailplanes* (Sec. 2.4). A large sailplane source with a machine-readable interface is *WeGlide*, which exposes flight metadata and IGC tracks through an API. The default access is rate-limited (of order sixty requests per day), which rules out a bulk export of the archive; a request for research access with a higher quota has been sent and is pending. Paraglider and hang-glider coverage may likewise be broadened through further portals such as XContest. A comparable research-access request has been sent to XContest; as of 20 July 2026 neither WeGlide nor XContest has replied. All of these can be acquired with the same approach, each producing a catalog in the same format.

## 2.2 The IGC tracklog format

A flight track is stored as an *IGC file*, the plain-text format defined by the FAI/IGC Technical Specification for IGC-approved GNSS flight recorders [8]. Its layout is fixed by that standard, so a fix can be decoded from fixed character positions; every downloaded track is checked at acquisition to be IGC content — an A record followed by at least one B record (Sec. 2.1).

Each line is a *record* whose type is set by its first character – a code assigned by the standard, not chosen by the logger. The records relevant here are:

- A the flight-recorder/manufacturer identifier (first line of the file);
- H header records (flight date, pilot, glider, geodetic datum, ...);
- I the declaration of the optional extra fields appended to every fix;
- B the GPS fixes – the body of the file.

A B record stores its fields at fixed character positions. Decoded against the standard, the example

B 094329 4933044N 00005193E A 00095 00149

reads as: UTC time 09:43:29; latitude 49°33.044' N and longitude 000°05.193' E on the WGS84 datum; a fix-validity flag; and *two* altitudes in metres – the barometric (pressure) altitude (95 m) followed by the GNSS altitude (149 m).

The validity flag takes two values and concerns the *GNSS* solution only. **A** marks a fix whose GNSS altitude is usable; **V** marks a GNSS drop-out, or a fix for which the receiver could not resolve altitude, in which case the standard requires the GNSS-altitude field to be written as zero. A **V** flag therefore does not make the fix two-dimensional: the latitude and longitude stay usable and, on a logger with a working pressure sensor, so does the barometric altitude, which is the channel this analysis adopts (Sec. 2.7.1).

Decoding a **B** record from fixed character positions can succeed numerically while the result is not a valid encoding at all – a corrupted byte can still parse as a digit. The parser therefore checks, at decode time and independently of any flight-dynamics judgement, that the record is actually a legal encoding – a valid time, valid latitude/longitude fields, a valid hemisphere letter – dropping it otherwise, on the same footing as a record with the wrong length or a non-numeric field (implementation details: Sec. 2.3). This is a check of *format* validity, distinct from the *plausibility* checks of fix-level cleaning (Sec. 2.7.2, e.g. the inter-fix speed bound), which target fixes that decode validly but would be physically implausible given the rest of the trajectory.

The two altitude channels differ in reference and in noise; the analysis adopts the *barometric* one for the vertical dynamics, for reasons quantified in Sec. 2.7.1—a choice that matters because the vertical velocity derived from it feeds the phase segmentation. On older loggers one channel may be absent (recorded as zero).

The sequence of **B** records is thus a time-stamped three-dimensional trajectory. The sampling period is typically a few seconds, is not guaranteed to be uniform, and varies across recording devices – a point the analysis must handle explicitly.

## 2.3 Flight metadata and the catalog

Every flight in the season XML carries a set of metadata as record attributes, which the acquisition preserves. Beyond the identifiers and links, the available fields are collected in Table 2.1.

**Table 2.1.** Flight metadata preserved from the season XML (raw attribute names), beyond the identifiers and links.

| Field(s)                      | Content / caveat                                                                                                                                                       |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| date                          | flight date                                                                                                                                                            |
| pilot, club                   | pilot and club                                                                                                                                                         |
| take-off, landing, department | the two sites and the French department                                                                                                                                |
| aile                          | wing model name (raw XML field; French for “wing”)                                                                                                                     |
| aile_class                    | wing class, one system per discipline (Sec. 2.4)                                                                                                                       |
| flight type                   | the <i>scoring</i> category: free distance, out-and-return, flat or FAI triangle, hike-and-fly, ... ( <i>caveat</i> : a scoring label, not a glider or manoeuvre type) |
| distance, points              | scored distance and points                                                                                                                                             |
| duration, speed               | duration and average speed, where present                                                                                                                              |

From these, the pipeline builds a single derived table, the **catalog** (`catalog.csv`), with *one row per flight* across all seasons: the metadata above plus a `local_path` linking each

flight to its track on disk and a `downloaded` flag. It is the working index of the project – a flight can be traced to its file, and back (through the identifier embedded in the file name) to its metadata and public page, without any external lookup. A smaller **season index** (`seasons_index.csv`), versioned in the repository, summarises each season by the number of flights declared, carrying a track, and downloaded; it is the source of the statistics below. Both tables are regenerated from the archived XMLs on demand (`build-catalog`), so the catalog itself need not be versioned.

## 2.4 Glider categories

Soaring kinematics depend strongly on the aircraft, so the glider must be identifiable for any stratified analysis. Three broad categories exist, in increasing order of performance: **paragliders** (flexible canopy; slowest, glide ratio  $\sim 8\text{--}11$ , tight turns), **hang gliders** (delta wing, spanning flexible framed wings and rigid wings; faster, glide ratio  $\sim 10\text{--}15$ ), and **sailplanes** (rigid aircraft; fastest, glide ratio  $\gtrsim 25$ ). They are, in effect, different aircraft—differing several-fold in cruise speed, glide ratio and turn radius—which is why the reference study [7] analyses each separately rather than pooling them.

In practice the category is set by the *data source*, since the public databases are organised by discipline. Because the FFVL competition is split by discipline (Sec. 2.1), the present dataset spans *both* glider types: the glider-type stratum has two levels, and the cross-type comparison of the reference study [7] is therefore already possible between paragliders and hang gliders. The third category, *sailplanes*, is collected by different portals and is not yet in hand (Sec. 2.1); until it is added the cross-type analysis is limited to the two FFVL disciplines.

Within each discipline the flights can be further stratified by *performance class*, recorded in `aile_class`; the two disciplines use different class systems, collected in Table 2.2. Paragliders follow the EN/LTF certification ladder, which does track performance (higher classes fly faster and flatter) and is therefore the natural within-source stratification variable. The hang-glider scheme is *not* a ladder: the FAI classes are defined by the method of control, so they stratify the fleet by design rather than by performance, and any ordering imposed on them is an assumption to be checked rather than a property of the scheme.

The two ladders are not documented on equal terms, which governs how far each entry of Table 2.2 can be trusted. The FAI classes come from the Sporting Code [9], which the FAI publishes for free: anyone can download the current edition and check a definition against it. The EN classes come from a *European standard*, which is a different kind of document. Standards are drafted by the European Committee for Standardization (CEN) and then published and *sold* by the national standards bodies—AFNOR in France, DIN in Germany, BSI in the United Kingdom—as copyrighted documents, priced per copy at a few hundred euros. There is no public version: unlike a law or a sporting code, the authoritative text of EN 926-2 cannot be consulted without buying it, and it cannot be quoted here.

The EN entries of Table 2.2 are therefore condensed from Table 1 of the *final draft* of the standard, FprEN 926-2:2012, the version submitted to formal vote, which the CEN working group circulated publicly [10]. That draft is the closest thing to the source text that is openly available, and the class descriptions in it are unlikely to have moved in the published EN 926-2:2013, though that has not been checked against the published text. Nothing in this thesis depends on the wording: the class field is used as a stratification label, and what the analysis needs from it is that the labels are distinct and ordered, not what each one certifies.

Two caveats bear on any use of `aile_class`. First, the field must be normalised before it can serve as a stratum: normalisation here means merging the label variants of the same class onto one canonical value per discipline and, where possible, recovering a blank or placeholder entry from the wing model name. Second, the class certifies the wing, not the pilot. The

**Table 2.2.** Class labels recorded in `aile_class`, one system per discipline. The EN ladder is ordered by increasing demand on the pilot, and that ordering is all the stratification needs from it; the FAI classes are *not* a ladder, being defined by the method of control. Sources: the FAI Sporting Code, Common Section 7 [9] (§§1.2, 1.4.1) for the FAI classes and the sub-class ceilings; the EN class descriptions are condensed from Table 1 of the standard, on the caveat explained in the text.

| Class                                                                          | Notes                                                                                                               |
|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Paragliders — EN/LTF ladder</b>                                             |                                                                                                                     |
| EN A                                                                           | maximum passive safety, extremely forgiving; any level of training                                                  |
| EN B                                                                           | good passive safety, forgiving; ceiling of the FAI <i>Standard</i> sub-class                                        |
| EN C                                                                           | moderate passive safety, dynamic reactions; recovery needs precise input; ceiling of the FAI <i>Sport</i> sub-class |
| EN D                                                                           | demanding, potentially violent reactions; for pilots practised in recovery                                          |
| CCC                                                                            | CIVL Competition Class racing wings                                                                                 |
| tandem / non-certified                                                         | two-seater and uncertified wings, pooled in this field                                                              |
| <b>Hang gliders — FAI Class O (paragliders are Class 3 of the same scheme)</b> |                                                                                                                     |
| Delta Class 1                                                                  | weight-shift as the sole method of control                                                                          |
| Delta Class Sport                                                              | sub-class of Class 1: type-certified production wing with a king post                                               |
| Rigide Class 2                                                                 | movable aerodynamic surfaces as the primary control                                                                 |
| Rigide Class 5                                                                 | as Class 2, but in the roll axis only; no fairings                                                                  |

normalisation is done at the pre-processing stage (Sec. 2.7.4) (implementation details: Sec. 2.5); the second point no processing can remove, and it bounds what a class-stratified result can be taken to mean.

## 2.5 Coverage and statistics

The statistics below describe the FFVL CFD dataset currently in hand; further sources may be added later (Sec. 2.1). It comprises 212,602 flights in total, of which 192,971 carry a GPS tracklog and 192,768 have been downloaded and verified, split across two disciplines.

The *paraglider* collection spans 27 seasons, from 1999–2000 to 2025–2026: 203,343 flights, of which 186,225 (91.6 %) carry a tracklog and 186,052 (99.9 %) are downloaded and verified (Table 2.3). The *hang-glider* collection spans 25 seasons, from 2001–2002 to 2025–2026: 9,259 flights, of which 6,746 (72.9 %) carry a tracklog and 6,716 (99.6 %) are downloaded and verified (Table 2.4). The hang-glider archive is roughly twenty times smaller, reflecting the smaller active population of the discipline.

In both disciplines the number of flights carrying a track grows by orders of magnitude over the period, reflecting the adoption of GPS loggers and online scoring: the early seasons contribute a handful of tracks each, while recent seasons contribute thousands of tracks for paragliders and several hundred for hang gliders.

**Table 2.3.** FFVL CFD *paraglider* flights per season (`parapente.ffvl.fr`): total declared, with a GPS track, and downloaded. (implementation details: Sec. 2.6)

| Season       | Flights | With GPS | Downloaded | Coverage (%) |
|--------------|---------|----------|------------|--------------|
| 1999–2000    | 10      | 10       | 10         | 100.0        |
| 2000–2001    | 832     | 1        | 1          | 100.0        |
| 2001–2002    | 1,150   | 3        | 3          | 100.0        |
| 2002–2003    | 1,996   | 123      | 123        | 100.0        |
| 2003–2004    | 1,470   | 337      | 337        | 100.0        |
| 2004–2005    | 1,765   | 574      | 574        | 100.0        |
| 2005–2006    | 2,162   | 894      | 894        | 100.0        |
| 2006–2007    | 2,386   | 1,016    | 1,016      | 100.0        |
| 2007–2008    | 3,081   | 1,667    | 1,667      | 100.0        |
| 2008–2009    | 3,974   | 2,614    | 2,613      | 100.0        |
| 2009–2010    | 4,171   | 3,059    | 3,059      | 100.0        |
| 2010–2011    | 4,872   | 4,185    | 4,169      | 99.6         |
| 2011–2012    | 6,744   | 6,075    | 6,065      | 99.8         |
| 2012–2013    | 7,873   | 7,106    | 7,075      | 99.6         |
| 2013–2014    | 9,804   | 9,157    | 9,127      | 99.7         |
| 2014–2015    | 13,793  | 13,293   | 13,254     | 99.7         |
| 2015–2016    | 13,982  | 13,703   | 13,696     | 99.9         |
| 2016–2017    | 15,760  | 15,534   | 15,528     | 100.0        |
| 2017–2018    | 14,917  | 14,682   | 14,676     | 100.0        |
| 2018–2019    | 16,702  | 16,517   | 16,508     | 99.9         |
| 2019–2020    | 10,966  | 10,868   | 10,863     | 100.0        |
| 2020–2021    | 9,500   | 9,483    | 9,480      | 100.0        |
| 2021–2022    | 14,552  | 14,507   | 14,502     | 100.0        |
| 2022–2023    | 11,590  | 11,561   | 11,560     | 100.0        |
| 2023–2024    | 10,372  | 10,362   | 10,360     | 100.0        |
| 2024–2025    | 10,906  | 10,885   | 10,884     | 100.0        |
| 2025–2026    | 8,013   | 8,009    | 8,008      | 100.0        |
| <b>Total</b> | 203,343 | 186,225  | 186,052    | 99.9         |

**Table 2.4.** FFVL CFD *hang-glider* flights per season (`delta.ffvl.fr`): total declared, with a GPS track, and downloaded. (implementation details: Sec. 2.6)

| Season    | Flights | With GPS | Downloaded | Coverage (%) |
|-----------|---------|----------|------------|--------------|
| 2001–2002 | 248     | 0        | 0          | 0.0          |
| 2002–2003 | 281     | 6        | 6          | 100.0        |
| 2003–2004 | 256     | 28       | 25         | 89.3         |
| 2004–2005 | 244     | 35       | 25         | 71.4         |
| 2005–2006 | 365     | 80       | 76         | 95.0         |
| 2006–2007 | 337     | 100      | 96         | 96.0         |
| 2007–2008 | 388     | 136      | 132        | 97.1         |
| 2008–2009 | 344     | 155      | 155        | 100.0        |
| 2009–2010 | 324     | 136      | 136        | 100.0        |
| 2010–2011 | 425     | 147      | 147        | 100.0        |
| 2011–2012 | 117     | 103      | 103        | 100.0        |

| Season       | Flights | With GPS | Downloaded | Coverage (%) |
|--------------|---------|----------|------------|--------------|
| 2012–2013    | 428     | 406      | 406        | 100.0        |
| 2013–2014    | 423     | 409      | 409        | 100.0        |
| 2014–2015    | 628     | 595      | 591        | 99.3         |
| 2015–2016    | 417     | 413      | 413        | 100.0        |
| 2016–2017    | 468     | 462      | 462        | 100.0        |
| 2017–2018    | 342     | 331      | 331        | 100.0        |
| 2018–2019    | 604     | 594      | 594        | 100.0        |
| 2019–2020    | 378     | 373      | 373        | 100.0        |
| 2020–2021    | 328     | 328      | 328        | 100.0        |
| 2021–2022    | 570     | 568      | 568        | 100.0        |
| 2022–2023    | 376     | 374      | 374        | 100.0        |
| 2023–2024    | 437     | 437      | 436        | 99.8         |
| 2024–2025    | 353     | 352      | 352        | 100.0        |
| 2025–2026    | 178     | 178      | 178        | 100.0        |
| <b>Total</b> | 9,259   | 6,746    | 6,716      | 99.6         |

## 2.6 Data quality and caveats

A few limitations are already known and will shape the analysis:

- **Flights without a track.** About 8 % of paraglider entries and about 27 % of hang-glider entries have no IGC file (older or manually declared flights); they carry metadata only. These are the flights outside the “With GPS” column of Tables 2.3 and 2.4.
- **A small residue of unrecoverable files.** A few flights advertise a track whose download fails or returns something that is not valid IGC content (broken or placeholder links); these are recorded as failures and excluded.
- **Heterogeneous logging.** Sampling rate, availability of the two altitude channels, and per-fix positioning noise depend on the device; trajectories must be resampled and quality-filtered before any ensemble statistic is computed.
- **Altitude channel.** A sizeable minority of loggers record no pressure altitude – almost a third of paraglider and roughly a sixth of hang-glider flights – which fall back to the GNSS altitude, with the channel actually used recorded per flight (Sec. 2.7.1).
- **Glider metadata.** Flights are paragliders or hang gliders, tagged by source (Section 2.4); the class field `aile_class` must be normalised before it is used to stratify.
- **Incomplete dates.** Some historical entries have placeholder dates, written as the sentinel value 0000-00-00 at acquisition. The sentinel makes them recognizable, and such flights are discarded rather than carried along: a flight whose date is unknown cannot be assigned to a season, and no downstream step needs it. The cut is negligible, and measured rather than assumed to be: 8 of the 203,343 paraglider entries carry the sentinel (0.0039 %) and 0 of the 9,259 hang-glider ones. The per-season breakdown is checked too, and not only the total, because a cut that is negligible archive-wide could still empty a single early season, which is where the smallest flight counts sit: the worst-affected season loses 0.17 % of its entries (2001-2002), so no season is materially thinned.

These caveats motivate the pre-processing described in the remainder of this chapter.



## 2.7 Trajectory pre-processing

Each IGC file is parsed into a sequence of fixes  $(t_i, \varphi_i, \lambda_i, h_i)$  – UTC time, geodetic latitude and longitude, and altitude – with both altitude channels retained (implementation details: Sec. 2.7). The pipeline then proceeds in a fixed order:

- (i) choose the altitude channel (Sec. 2.7.1);
- (ii) clean individual fixes (Sec. 2.7.2);
- (iii) trim the ground phases (Sec. 2.7.3);
- (iv) filter out unfit whole flights (Sec. 2.7.4);
- (v) convert to a local Cartesian frame (Sec. 2.7.5);
- (vi) enforce a uniform sampling interval within each flight (Sec. 2.7.6);
- (vii) smooth and differentiate the trajectory (Sec. 2.7.7).

The output is a tidy, analysis-ready representation in a growing `soaring.analysis` sub-package, written to three tables on the data disk: one row per fix, one per segment, one per flight attempted. What each file holds and how the disk is organised is set out in Sec. 2.2.

Steps (i)–(iv) act on the *raw geographic* coordinates, since the only geometric quantities they need (inter-fix speed and spatial extent) are scalar distances, which the great-circle (haversine) formula of Eq. (2.1) gives directly from latitude and longitude, with no coordinate conversion. The conversion (v) to the metric East–North–Up frame (Sec. 2.7.5) is then done on the cleaned, trimmed track, with the origin at the first fix that survives the trimming, so the tangent-plane frame is built from good data.

Steps (vi)–(vii) come last for a different reason: unlike the scalar checks above, they produce *vector* quantities—the components of the 3D position, velocity and acceleration—which require Cartesian axes to be expressed in, and therefore cannot run before the frame of step (v) exists.

Every step below is governed by a handful of numerical thresholds. They are collected once, here, in Table 2.5, so that the value in force at any point of the chapter can be read off a single page instead of being hunted through the text; each subsection then argues for its own entries rather than restating them. All of them are *working values*: fixed a priori, on physical scales rather than on the transport observables, and audited a posteriori on the fraction of data each rule removes (Sec. 2.7.2). The table is typeset from the configuration file itself, so a value changed there changes here.

**Table 2.5.** Working values of the pre-processing pipeline, by stage. Symbols are those used in the text; every value is generated from `configs/preprocessing.yaml`, and the per-parameter rationale is given in the section listed. Config keys and implementation notes: Sec. 2.7.2 onward.

| Symbol                                                                                  | Value                                                    | What it controls                                                                         |
|-----------------------------------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------|
| <b>Fix-level cleaning</b> (Sec. 2.7.2) — removes corrupt fixes, keeps every genuine one |                                                          |                                                                                          |
| $k$                                                                                     | 5                                                        | robust- $\sigma$ multiplier of the Hampel test, Eq. (2.3)                                |
| $w$                                                                                     | 20 s                                                     | half-width of the Hampel window                                                          |
| $\varepsilon_{\min}$                                                                    | 15 m                                                     | absolute floor of the Hampel test: no flag below this residual, whatever the local scale |
| —                                                                                       | 5 fixes                                                  | smallest window population for which the test is evaluated                               |
| $v_{xy}^{\max}$                                                                         | 45 m s <sup>−1</sup> (para), 55 m s <sup>−1</sup> (hang) | absolute inter-fix speed bound; the impossibility gate, on its own (Sec. 2.7.2)          |
| $ v_z ^{\max}$                                                                          | 13 m s <sup>−1</sup>                                     | absolute climb/sink bound                                                                |
| $[h_{\min}, h_{\max}]$                                                                  | [−100, 6000] m                                           | sensor-plausibility window on the altitude                                               |

*continued on the next page*



Table 2.5 (continued)

| Symbol                                                                                                 | Value                                              | What it controls                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\delta_{xy}$                                                                                          | 15 m                                               | diameter within which a frozen-lock run must stay                                                                                                                                                                                                                                 |
| $\delta_z$                                                                                             | 5 m                                                | altitude flatness tolerance of the same rule                                                                                                                                                                                                                                      |
| $\tau_{\text{freeze}}$                                                                                 | 60 s                                               | minimum duration of a run before it counts as frozen                                                                                                                                                                                                                              |
| —                                                                                                      | 95 %                                               | smallest share of a flight’s fixes carrying a non-zero pressure altitude for the barometric channel to count as present (Sec. 2.7.1); below it the flight falls back to GNSS                                                                                                      |
| —                                                                                                      | 30 m                                               | smallest whole-flight barometric range below which the channel is treated as absent (Sec. 2.7.1)                                                                                                                                                                                  |
| —                                                                                                      | 10 %                                               | share of fixes above which the cleaning is declared to have failed and the flight is dropped                                                                                                                                                                                      |
| <b>Ground trimming</b> (Sec. 2.7.3) — keeps only the airborne stretch                                  |                                                    |                                                                                                                                                                                                                                                                                   |
| $v_0$                                                                                                  | $5 \text{ m s}^{-1}$                               | ground-speed threshold separating ground from flight                                                                                                                                                                                                                              |
| $T_0$                                                                                                  | 30 s                                               | how long $v_{xy} > v_0$ must persist for take-off (and, time-reversed, landing)                                                                                                                                                                                                   |
| $T_{\text{ground}}$                                                                                    | 10 min                                             | minimum duration of an interior ground stint before it is excised                                                                                                                                                                                                                 |
| —                                                                                                      | 5 m                                                | altitude flatness tolerance over such a stint                                                                                                                                                                                                                                     |
| <b>Flight-level filtering</b> (Sec. 2.7.4) — discards whole flights that cannot support the statistics |                                                    |                                                                                                                                                                                                                                                                                   |
| —                                                                                                      | 40 min                                             | minimum airborne duration                                                                                                                                                                                                                                                         |
| —                                                                                                      | 20 km                                              | minimum flown path length                                                                                                                                                                                                                                                         |
| —                                                                                                      | 16 h                                               | maximum duration, above which the record is not a single flight                                                                                                                                                                                                                   |
| —                                                                                                      | 600 m                                              | minimum whole-flight altitude range                                                                                                                                                                                                                                               |
| —                                                                                                      | $v_{xy}^{\text{max}}$ (above)                      | plausibility bound on the mean ground speed of the cleaned track; reuses the fix-level bound, read over a whole flight (Sec. 2.7.4)                                                                                                                                               |
| —                                                                                                      | $v_{xy}^{\text{max}}$ (above)                      | reach bound: the extent of the flight may not exceed what that same speed would cover in its duration (Sec. 2.7.4)                                                                                                                                                                |
| <b>Resampling</b> (Sec. 2.7.6) — one uniform grid per flight, holes bridged or split                   |                                                    |                                                                                                                                                                                                                                                                                   |
| $g_{\text{max}}$                                                                                       | $\min(10 \Delta t, \max(20 \text{ s}, 2\Delta t))$ | longest gap that is interpolated instead of splitting the flight                                                                                                                                                                                                                  |
| —                                                                                                      | 10 %                                               | largest share of grid points a retained <i>segment</i> may have had to interpolate                                                                                                                                                                                                |
| —                                                                                                      | 90 s                                               | shortest segment kept: below this a segment cannot carry the smoothing window                                                                                                                                                                                                     |
| <b>Smoothing</b> (Sec. 2.7.7) — Savitzky–Golay filter and its derivatives                              |                                                    |                                                                                                                                                                                                                                                                                   |
| $p$                                                                                                    | 3                                                  | polynomial order of the filter                                                                                                                                                                                                                                                    |
| $\tau_c$                                                                                               | 5 s                                                | correlation time setting the window, on all three channels. Per-channel keys exist— $E/N$ , and the altitude split by <code>alt_source</code> —for the noisy GNSS minority, and all currently hold this same value, because the spectral knees of the three coincide (Sec. 2.7.7) |

continued on the next page

Table 2.5 (continued)

| Symbol | Value     | What it controls                                                                                                                                                                                                      |
|--------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| —      | 5 samples | floor on the window, one sample more than a cubic determines. It is this floor and not $\tau_c$ that fixes the smoothing scale for every logger slower than 1 Hz: in seconds the window is $\max(\tau_c, 5 \Delta t)$ |

### 2.7.1 Choice of altitude channel

The analysis adopts the *barometric* altitude rather than the GNSS altitude for the vertical dynamics. The reason is one of signal quality in the frequency band that matters. The IGC format writes both altitude channels as whole metres, so the resolution of either is fixed at 1 m by the file itself; what distinguishes the barometric channel is that it carries very little high-frequency noise. Its errors are slow: a constant offset from the ICAO reference, tens of metres up to order 100 m (visible in Fig. 2.1a as the gap between the two channels; most of that gap is the barometric offset, though the GNSS altitude contributes a smaller bias of its own), and a drift that already reaches tens of metres within the half-hour window of Fig. 2.1b, and can grow further over a multi-hour flight. Both have the same origin: the barometric altitude is a pressure reading converted through the ICAO standard atmosphere, so it is offset by the difference between the day’s actual sea-level pressure and the ICAO reference of 1,013.25 hPa, and it drifts as that pressure changes over the hours of the flight and as the real temperature profile departs from the standard one. Drift of the sensor itself is a third and much smaller contribution.

Slow errors are annihilated by the differencing that every dynamical quantity involves (a vertical velocity, a displacement increment, the autocorrelation of a velocity). The GNSS altitude, by contrast, is referenced to a fixed geometric surface (the WGS84 ellipsoid), so it carries no atmospheric offset, but it is vertically noisier: the satellite geometry resolves the vertical coordinate less well than the horizontal one<sup>1</sup> (the *vertical dilution of precision*): the GPS performance standard commits to 8 m horizontal against 13 m vertical error at 95 % globally, and 15 m against 33 m at the worst site [11], and *multipath*—the direct signal arriving together with replicas delayed by reflection off terrain or the pilot’s own equipment—biases the range on timescales of minutes rather than adding independent noise from fix to fix [12].

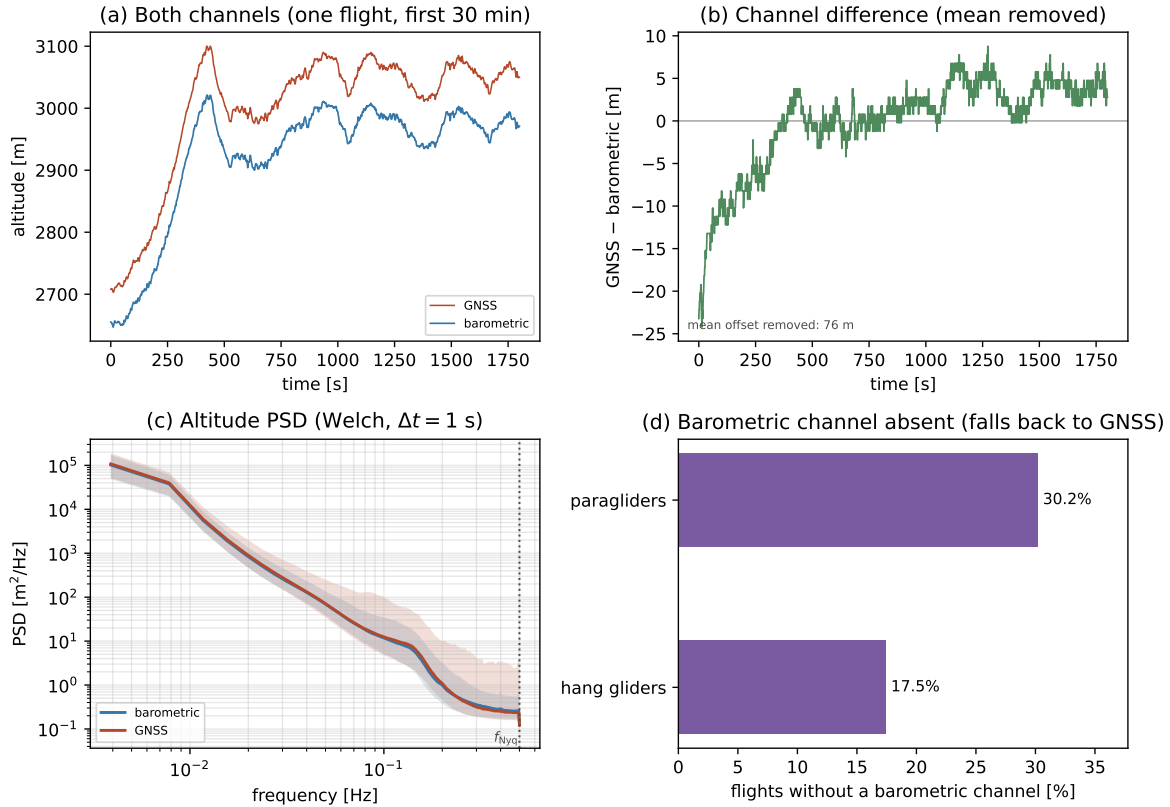
This is why free-flight variometers derive the climb rate from the pressure sensor rather than from GNSS.

**Empirical confirmation.** The data confirm this picture, with one qualification. Figure 2.1 compares the two channels over a subsample sized for a robust spectral estimate (Appendix 2.A): 3,000 flights per discipline, a size chosen on the stability of the median spectrum itself and justified in Sec. 2.7.1 (implementation details: Sec. 2.7.1). At low frequency the two agree, tracking the real climbs and glides. The high-frequency behaviour is where they differ. For the *typical* flight the two are comparable: both approach the same flat high-frequency plateau—the level at which the spectra in panel c flatten towards the Nyquist frequency—set by the metre resolution to which the IGC format rounds *both* channels, so the median spectra (panel c, solid lines) nearly coincide.

What separates the channels is the *spread* across flights, the shaded bands. The barometric band stays tight (that channel is uniformly clean), whereas the GNSS band fans out by one to two orders of magnitude towards the Nyquist frequency: a substantial minority of flights carry

<sup>1</sup>The asymmetry is geometric. A receiver sees satellites only above the horizon, so the ranging directions bracket it in the horizontal plane but all arrive from the same side vertically; the vertical coordinate is correspondingly the worst-determined one. This is what the *vertical dilution of precision* measures.

large GNSS high-frequency noise, from receiver tracking noise amplified by a poor vertical dilution of precision—not from multipath, whose error is correlated over minutes and so raises the low-frequency end of the spectrum rather than the floor—while most do not. Since the vertical velocity is a finite difference, which amplifies exactly that band (differencing scales a spectral component of frequency  $f$  by  $2\sin(\pi f \Delta t)/\Delta t$ , an amplification that grows with  $f$ : it acts as a high-pass filter), and since one cannot know in advance which flights have poor GNSS reception, the uniformly clean barometric channel is the safe choice. This *departs from the reference study* [7], whose pipeline derives the vertical coordinate and velocity from the GNSS altitude; the divergence is deliberate, and one of the points re-examined here.



**Figure 2.1.** Barometric vs GNSS altitude noise. (a) One representative flight over a 30 min window, the two raw channels on a shared axis. The vertical gap between the curves is the offset between their reference surfaces (pressure vs. ellipsoidal). (b) Their difference, mean removed: the offset is not constant but drifts by tens of metres already within this 30 min window. (c) Per-frequency *median* Welch spectrum of each channel (Appendix 2.A), pooled over the two disciplines, with the 10 to 90 percentile band across flights; the medians nearly coincide, while the GNSS band fans out towards the Nyquist frequency  $f_{Nyq} = 1/(2\Delta t)$  (dotted). The median is used because the per-flight spectra are heavy-tailed (Sec. 2.7.1). (d) Fraction of flights whose barometric channel is absent, per discipline. (implementation details: Sec. 2.7.1)

Several practical points follow.

**One channel per flight.** A trajectory uses a single altitude channel end-to-end; the barometric and GNSS values are never spliced within a flight, since a mid-flight switch between two channels offset by tens of metres would inject a spurious jump.

**Source recorded.** Which channel a flight uses is recorded as a per-flight attribute of the processed dataset, so the two source groups remain separable in every downstream analysis;

the raw IGC files are never modified (implementation details: Sec. 2.7.1).

**Missing fixes.** A flight is used with its barometric channel when it has one, and the few individual fixes with a missing barometric value are dropped rather than back-filled from GNSS. Only 4.2 % of barometric paraglider flights (2.1 % of hang gliders) miss any barometric value at all, with a median of 3 (10) missing fixes among them—against  $\sim 10^4$  fixes in a typical track—and a worst case of 1,397 (455), bounded by the presence threshold itself: a flight whose channel covers less than 95 % of its fixes is not treated as barometric at all (Sec. 2.7.1). Whether the dropped fixes are scattered or consecutive, the hole they leave is handled downstream like any other gap (Sec. 2.7.6).

**Absent or stuck sensor: the GNSS fallback.** A logger with *no* pressure sensor writes the whole barometric channel as zero (the presence is essentially all-or-nothing, not a scattering of gaps), so a flight counts as barometric only when the channel covers at least 95 % of its fixes—high on purpose, since a partially filled channel is worse for the vertical analysis than a uniformly sourced GNSS one (Sec. 2.7.1); such a flight falls back to the GNSS altitude, unfiltered – a non-negligible minority of flights (Figure 2.1d). The mere presence of a non-zero barometric channel is not enough: a stuck sensor writes a constant non-zero value and would feed the segmentation a vertical velocity of identically zero, so a barometric channel whose total range over the flight stays below 30 m is treated as absent, and the flight falls back to GNSS. The fallback rate is measured on the full archive: 30.2 % of the 186,025 paraglider and 17.5 % of the 6,716 hang-glider flights scanned carry no barometric channel (implementation details: Sec. 2.7.1).

The liveness test as written catches only a sensor that is dead for the *whole* recording, and it should be read as that and no more. The statistic is a single whole-flight max – min, so a barometer that records normally for the first ten minutes and then freezes keeps a range of hundreds of metres, is judged alive, is adopted, and delivers a vertical velocity of identically zero over most of the flight—which is precisely the outcome the test is here to prevent. Nothing downstream closes the gap: the frozen-lock rule of Sec. 2.7.2 needs the *position* to be frozen, and a cruising wing never trips it; the flight-level altitude-activity cut consumes the same extremum difference and inherits the same blindness. A test that did close it would be a *local* one—a moving-window range, or a longest-constant-run bound—and that is a new threshold with its own failure mode against a wing genuinely holding altitude, so it is not adopted here on the strength of an argument alone. What the present rule guarantees is stated exactly: a channel that is constant throughout is refused, and one that dies partway is not.

**Mixed altitude sources.** The mix is acceptable, for two reasons. The *horizontal*, two-dimensional transport analysis—the main subject of this work, of which the MSD and its scaling are one observable among several—does not use the altitude at all, so every flight is usable there regardless of channel. Altitude enters only the vertical velocity, hence the segmentation and the phase-resolved observables; there the two source groups are kept distinct through the per-flight altitude-source record introduced above. The compatibility check therefore has to be run on the segmentation itself rather than on the altitude statistics: both groups are segmented with the same rules and compared, per discipline, on the fraction of airborne time each phase accounts for. The two estimates of each fraction are compatible when they agree within their uncertainties; a disagreement in any phase is the signal that the noisier channel is biasing the phase assignment. One number per phase per group is enough for it: the fraction of time per phase is what the vertical channel feeds (Sec. 2.8). If the noisier GNSS-only flights were to bias a vertical statistic, that analysis is restricted to the barometric flights – still the large majority – while the horizontal analyses keep the full sample. Forcing GNSS on *all*

flights, to avoid the mix, would be the wrong trade: it would needlessly degrade the vertical signal of the barometric majority to match the worst loggers.

### 2.7.2 Fix-level cleaning

**Geometry on the raw coordinates.** The cleaning runs on the raw geodetic coordinates  $(\varphi_i, \lambda_i)$ , before the Cartesian conversion (Sec. 2.7.5), so distances between fixes cannot be Euclidean; they are great-circle (haversine) distances [13] on a sphere of mean radius  $R_\oplus = 6,371$  km. For two consecutive fixes,

$$\Delta s_i = 2R_\oplus \arcsin \sqrt{\sin^2 \frac{\varphi_{i+1} - \varphi_i}{2} + \cos \varphi_i \cos \varphi_{i+1} \sin^2 \frac{\lambda_{i+1} - \lambda_i}{2}}, \quad (2.1)$$

from which the horizontal speed between the fixes is  $v_{xy,i} = \Delta s_i / (t_{i+1} - t_i)$  and, from the barometric altitude, the vertical speed is  $v_{z,i} = (h_{i+1} - h_i) / (t_{i+1} - t_i)$ . The same distance also yields two whole-flight quantities used below: the *path length*  $L = \sum_i \Delta s_i$ , the total distance flown along the track, and the *extent*  $D = \max_i \Delta s(\text{fix}_0, \text{fix}_i)$ , the farthest any fix lies from the first. The spherical approximation is ample here. Replacing the WGS84 ellipsoid by a sphere perturbs an inter-fix distance by at most a fraction of the order of the ellipsoid’s flattening,  $\lesssim 0.3\%$ <sup>2</sup>—about 3 cm on a 10 m step—whereas the horizontal position error of consumer GNSS receivers is of the order of metres (the GPS performance standard commits to 8 m at 95 % globally [11]); the correction is therefore far below the noise on the very same quantity. Working on the raw coordinates also avoids converting them before the trajectory has been cleaned.

**Design principle.** Corrupt or physically impossible fixes must be removed without discarding the flight around them. At the same time, every removed or altered fix modifies the trajectory whose statistics are about to be measured, and an over-aggressive cleaning biases the result as much as dirty data would. The cleaning therefore assumes as little as possible about the motion: only smoothness at the sampling scale, never a model of the transport itself.

The two possible errors have different costs. A spurious fix that survives the tests below lies at worst tens of metres from the true track—at most the paraglider speed bound of Table 2.6 times the native step,  $\approx 45$  m at 1 Hz—and typically far closer, within the absolute floor  $\epsilon_{\min}$  of the outlier test defined below in Eq. (2.3): a displacement the smoothing absorbs. A genuine fix removed from a slow, horizontally localized phase (a search, a climb) erases a real trapping event, and nothing downstream can restore it. The cleaning therefore defaults to keeping the data: a fix is removed only on positive evidence of a defect, and no detector deletes a fix that still carries a usable horizontal position (at most, a corrupt *altitude* channel is marked missing and restored at resampling, Sec. 2.7.6; a corrupt horizontal position is never reconstructed—the fix carrying it is deleted outright).

**Three classes of defect.** Past the format-validity check made at decode time (Sec. 2.2), which rejects any record that is not a legal encoding and is already implemented and unit-tested, a surviving fix can still be wrong in one of three ways, summarized in Table 2.6.

<sup>2</sup>An arc length on a curved surface is proportional to the radius of curvature used to compute it, so the relative error made by using the mean radius  $R_\oplus$  in place of the ellipsoid’s local radius is of the order by which the two differ. On WGS84 the radii of curvature range from the polar  $b^2/a = 6,335.4$  km to the equatorial  $a^2/b = 6,399.6$  km, a spread of the order of the flattening  $f = (a - b)/a = 1/298.257 \approx 3.4 \times 10^{-3}$ , hence the 0.3 % quoted here [14].

A fix *off the trend* – a GPS position spike, or a jump out and back within one step – is found by a robust<sup>3</sup> local-outlier test, the *Hampel identifier* [15,16]. Let the window of fix  $k$  be the set of fixes within  $\pm w$  of  $t_k$ , and let  $\tilde{\varphi}_k = \text{med}\{\varphi_j : |t_j - t_k| \leq w\}$  and  $\tilde{\lambda}_k = \text{med}\{\lambda_j : |t_j - t_k| \leq w\}$  be the component-wise window medians. The residual of fix  $k$  is its great-circle distance from that median point,

$$r_k = \Delta s((\varphi_k, \lambda_k), (\tilde{\varphi}_k, \tilde{\lambda}_k)), \quad (2.2)$$

with  $\Delta s$  as in Eq. (2.1), and the robust local scale of the residuals is  $\text{MAD}_k = \text{med}\{r_j : |t_j - t_k| \leq w\}$ . The fix is flagged when

$$r_k > \max(k \sigma_k, \varepsilon_{\min}), \quad \sigma_k = 1.4826 \text{MAD}_k, \quad (2.3)$$

Here  $\sigma_k$  is the normalized MAD, the median absolute deviation rescaled to estimate a Gaussian standard deviation [17]: the flag requires a departure of more than  $k$  robust standard deviations *and* more than an absolute floor  $\varepsilon_{\min}$  of a few times the GPS noise (working values in Table 2.5; step-by-step definition and rationale in Sec. 2.7.2). *A flag does not delete. What deletes is the absolute  $v_{xy}$  bound: a fix is removed when it is unreachable from the last accepted fix at that bound and the removal of the block it belongs to lets the track rejoin within it. The identifier's role is to flag and to attribute—to say which fix of a neighbourhood departs from it—and to supply the per-flight anomaly count; the reason it cannot also be required is argued below.*

A genuine but under-sampled tight turn is kept either way. It is what a naive speed-only rule would get wrong and what the identifier was originally there to protect. The Hampel test measures a departure from the local median track, and a tight thermalling circle produces exactly such a departure: at a 10s cadence a circle of a few tens of metres in diameter puts consecutive fixes on opposite sides of their own window median, and the residual clears  $\varepsilon_{\min}$  on geometry alone. Those fixes are the trapping events the transport analysis exists to measure. *What separates them from a spike is not the flag but the speed: a circling wing flies at an ordinary one, whereas a position spike of the same size implies an inter-fix speed no aircraft in this dataset can sustain. The protection is therefore already in the bound that does the deleting, which is where it came from.*

The rule was originally a conjunction—flagged *and* impossible—and running it over the archive exposed a limit in the *flag* that is the reason it no longer is. The Hampel identifier has a breakdown point of 50 %: up to half a window may be arbitrarily wrong without displacing the median, which is what makes it robust—and past that half the median moves onto the corruption instead. The archive contains the case. Some loggers write the null position (0,0) for scattered runs of a few seconds each; where several such runs fall inside one 20s window they exceed half of it, the median follows them, the corrupt fixes score a residual of *zero* and go unflagged, and it is the good fixes around them that are flagged instead. Requiring the flag there does not make the rule conservative, it makes it blind: on one such flight the identifier caught 17 of 99 null fixes, and every survivor left a step of some 5,000 km in the record.

The rule adopted is therefore the *impossibility gate on its own*. A fix is deleted when it is unreachable from the last accepted fix at the  $v_{xy}$  bound *and* the removal of the block it belongs to lets the track rejoin within that bound. Both halves of the argument above survive the change. A genuine tight circle is still never a candidate, because it flies at an ordinary speed—so the protection the flag was there to give is given by the bound itself, which is where it came from in the first place. And the attribution the flag supplied—which endpoint of an impossible step is the wrong one—is supplied instead by the rejoining test, which names the block whose removal restores the trend. What the flag stops being is a *necessary* condition for

<sup>3</sup>*Robust* has its technical sense here: the median and the median absolute deviation have a breakdown point of 50 %, so up to half the fixes in the window may be arbitrarily wrong without displacing the estimate. A mean and a standard deviation have breakdown point zero—a single spike drags both, inflating the scale by just enough to hide itself.



deletion; what it remains is the per-fix record and the per-flight anomaly count. The change bites only where the identifier is outside its own stated range of validity, and that is precisely where the previous rule failed.

A fix *on the trend* agrees with its neighbours yet is still wrong as a record – a duplicated or backward timestamp, or a run of frozen positions from a lost GNSS lock; its residual is near zero, so it escapes the outlier test and needs a structural rule instead.

A value *out of range* concerns the vertical channel: an impossible altitude or climb rate is caught by an absolute bound that needs no context.

**An unreturned vertical step.** The  $|v_z|$  rule recognises two shapes and there is a third. An *out-and-back* spike is censored; a coherent *run* is a sustained manoeuvre and is counted and left (Sec. 2.7.2). A single super-threshold step that is never undone—a barometric re-reference, or one corrupt record—is neither. It was therefore censored by nothing *and counted by nothing*, and since it is a discontinuity rather than an outlier the smoothing stage differentiated it: the written table carried vertical velocities of up to  $5,117 \text{ m s}^{-1}$ . It is not a curiosity. On a sample of the archive, 8.3% of paraglider flights and 6.2% of hang-glider flights carry at least one.

What is done here is the part that is not a judgement: the shape is detected and counted, per flight, as `n_alt_level_shift`, so that it is visible and auditable rather than silently present. What to *do* about it is left open, deliberately. The three candidate treatments—split the flight at the step, on the same footing as an impossible horizontal step; re-reference the altitude after it, restoring continuity; or invalidate a window around it and let the resampling bridge it—each change a rule argued elsewhere in this chapter, and none of them follows from the specification. The vertical channel feeds the phase segmentation of Chapter 4 and not the transport measurement of Sec. 3.1, which is horizontal, so the choice belongs with that analysis rather than ahead of it.

The vertical channel is treated differently from the horizontal one on purpose, and the asymmetry is worth stating, because the obvious symmetric design—a Hampel test on  $z$  too, corroborated by the climb-rate bound—is the wrong one here. Three things differ.

First, an absolute bound is *sufficient* on  $z$  and insufficient on  $xy$ . The vertical envelope is narrow and known: no wing in this dataset sustains  $|v_z| > 13 \text{ m s}^{-1}$ , and no altitude outside  $[-100, 6000] \text{ m}$  is sensor-plausible, so a spike in altitude is almost always out of range on its own. Horizontally there is no such luck: a 50 m spike is an impossible  $50 \text{ m s}^{-1}$  at a 1 s cadence but an unremarkable  $5 \text{ m s}^{-1}$  at 10 s, so the same defect passes or fails the bound depending on the logger. The horizontal case needs a local, scale-free test precisely because its absolute bound is not decisive; the vertical case does not.

Second, a Hampel test on  $z$  would flag the physics. The residual from a local median is largest where the signal turns fastest, and on the altitude channel that means thermal entries and exits—where  $v_z$  swings by several  $\text{m s}^{-1}$  within seconds (Sec. 2.7.6). Those transitions are what the phase segmentation is built on. A horizontal circle, by contrast, can be told from a horizontal spike by its speed, so the local test there has a corroborating witness that the vertical one lacks: the quantity that would have to arbitrate a suspected altitude spike *is* the climb rate, which is the quantity under suspicion.

Third, the costs are not symmetric. A false vertical flag marks one altitude missing and it is interpolated back at resampling; a false horizontal flag would delete a genuine sample of the path. But an interpolation laid across a thermal entry erases the transition rather than restoring it, so a cheap-looking vertical false positive is expensive exactly where it is most likely. Keeping the vertical rule to a bound that only fires on the physically impossible avoids that failure mode altogether.

The thresholds are set in two ways. The absolute bounds are placed on data: the  $v_{xy}$  and  $|v_z|$  bounds sit where the per-fix distributions of Fig. 2.2 stop decaying, and the altitude

window is a sensor-plausibility band audited on the same distributions (lower bound negative for the reference-offset reason in the caption of Table 2.6). The remaining parameters—those of the Hampel test and of the frozen-lock rule—are working values fixed a priori. All of them, with their values, are in Table 2.5; their reasonableness is then checked a posteriori on the fraction of fixes each rule removes (paragraph *Validating the cleaning* below).

**Removal policy.** A speed bound alone cannot attribute a defect: the speed belongs to the segment *between* two fixes and does not identify which endpoint is wrong. The attribution comes from the bounded-removal test—the block whose removal restores the trend is the corrupt one—so an isolated spike is removed without touching its neighbours, and a short burst is removed as one block, the smallest that works. The local-outlier residual, being a property of a single fix, says the same thing where it is valid, and is recorded for that reason; it is not what decides.

What happens to a defective fix depends on *which* of its values is corrupt. Two cases arise—a corrupt altitude, and a corrupt horizontal position or timestamp—and they are handled asymmetrically. The asymmetry follows from a single criterion: keep every genuine horizontal sample, and never hide a hole in the horizontal record. In *either* case the corrupt value itself is simply discarded—nothing is estimated in its place at this stage; whatever needs reconstructing is reconstructed once, at the resampling step (Sec. 2.7.6).

If the corrupt value is the *altitude*, the fix is kept and only its altitude is marked missing. In practice the fix keeps its row in the trajectory table and its altitude entry is set to a missing value, with a per-fix boolean recording that the value was removed by the cleaning rather than never written by the logger; the two are indistinguishable downstream otherwise, and only the first is a statement about data quality. The horizontal position and the timestamp are intact, so the fix still samples the path—exactly the information the transport analysis measures—and discarding it would throw that genuine sample away. The missing altitude is restored by interpolation at resampling, where the filled point is flagged as interpolated in turn; the barometric channel is smooth enough that this costs less than the sensor noise (Sec. 2.7.1).

If the corrupt value is the *horizontal position or the time*, the whole fix is deleted. One could imagine the symmetric treatment—keep the fix and mark the horizontal position missing—but that would be dangerous: the resampling step decides whether to bridge a hole or to split the flight by looking at the *time gap* between surviving fixes. Ten consecutive kept-but-positionless fixes would look like ten samples of a continuously recorded stretch; the gap rule would never fire, and the resampler would lay an invented straight line across an interval that may hold a full thermalling circle. Deleting the fixes leaves the hole visible, and the gap rule then does its job: a short hole is bridged, a long one splits the flight (Sec. 2.7.6). The altitude sample lost with the deleted fix is the cheap part, for the same smoothness reason as above.

The IGC validity flag needs no rule of its own here: a V flag certifies a degraded GNSS altitude (Sec. 2.2), so on a barometric flight the fix is untouched, and on a GNSS-fallback flight it reduces to the corrupt-altitude case above.

**An impossible step is a boundary, never flown path.** The two actions above—delete the block, or split at the step—cover between them what the scan can resolve. What must hold whatever it resolves is a weaker statement, and it is the one the rest of the pipeline leans on: *no step between two surviving fixes may break the  $v_{xy}$  bound*. Where a block is kept because nothing could rejoin it, the step by which the track returns from that block is itself impossible, and left alone it enters a segment as flown motion—in the archive, a jump of tens of kilometres inside one second, carried thereafter into every average taken over that segment. Such a step is a transition whose course is unknown, which is what a long gap is, so it is made



a segment boundary on the same footing (Sec. 2.7.6). It is stated as a property of the output rather than as one more case of the scan, so that it holds whatever the scan enumerated. How often it acts is counted per flight, as `n_boundaried`: a high count would locate the fault in the scan rather than in the data.

**A corrupt block that opens the record.** The blind spot is real and it matters more than its rarity suggests. A scan that carries the last accepted fix as an anchor never questions its own first anchor; and the first fix of the trimmed track becomes the origin of the local frame (Sec. 2.7.5), so a corrupt one displaces every coordinate of the flight rather than one of them, and the error does not average out with length of record. Five paraglider flights left the first full-archive run sitting some 4,500 km from their own origin. Since the ensemble MSD is an average of  $|\mathbf{r}(t)|^2$  over flights, five records out of 156,017 raised it by seven orders of magnitude at the shortest lags.

No fix-level rule attributes it, and that absence is a decision rather than an omission. An impossible step is a statement about a *pair* of fixes and says nothing about which of the two is the wrong one. Away from the ends of a record the bounded-removal test settles the question by evidence—it asks which removal restores the trend—and at the very first fix there is no trend behind the step to ask. Two rules were nevertheless written, and both were wrong, in opposite directions. One fired on the first impossible step, which accuses the earlier end always: applied to an ordinary position spike three seconds into a record it deleted the good fixes and promoted the spike to origin, the same failure with the sign reversed. The other fired on a discontinuity the scan itself declares near the start, which sounds like evidence—a discontinuity survives only where no bounded removal after the step restored the trend—but the split is placed at the first fix of the block that could not be rejoined, and that block lies *after* the step whenever the corruption simply outlasts  $w$ . A 50s corrupt run five seconds in therefore deleted five good fixes, kept fifty bad ones, and left the origin 5,039 km out.

So the discontinuity is left as a discontinuity, and the guarantee is taken one stage later, where it can be stated without attributing blame to either end: a record out of reach of its own first fix is refused whole (Sec. 2.7.4). That bound never depended on a fix-level rule. What such a rule would have bought is the recovery of two flights in 156,017, against a failure mode now demonstrated twice—and the invariant the whole of Chapter 3 rests on,  $|\mathbf{r}(t)| \leq v_{xy}^{\max} t$ , holds either way, checked as such on the written tables (Sec. 2.2).

**Frozen-position runs (GNSS lock loss).** When the receiver loses its GNSS lock, many loggers repeat the last position for tens of seconds while the clock keeps running. Such a run passes every test above: horizontal speed zero, altitude in range, valid timestamps. If kept, it enters the analysis as a long *waiting time* and contaminates the tail of the waiting-time density  $\psi(\tau)$ , the distribution of the durations of the slow, localized phases. If the detector is too aggressive, however, it splits tight thermalling climbs, which are also slow and *horizontally* localized, and destroys real trapping events. The two errors enter the same tail with the same weight: a kept freeze inserts a false long wait, a cut climb removes a real one. They differ in what can be repaired downstream: a kept run with a flat barometric trace is still flagged suspect and swept in the  $\psi(\tau)$  sensitivity checks (Sec. 2.7.3), while an erased trapping event is unrecoverable. A run is therefore cut only when three independent signatures agree, combined as in Eq. (2.4) below:

- (i) *Spatial diameter.* A lost lock collapses the whole run into the logger’s quantisation: the run’s *bounding diameter* – the largest great-circle distance between any two of its fixes – stays below  $\delta_{xy}$ , a tolerance of this rule set a few times above the GPS noise. It is written  $\delta_{xy}$ , and not  $\varepsilon$ , to keep it apart from the Hampel floor  $\varepsilon_{\min}$ : the two carry the same value at present but answer different questions—one is the smallest departure from

a local median worth calling an outlier, the other the largest spread a run may show and still count as motionless (Table 2.5). The run is not a fixed-length window: a candidate run is the maximal stretch of consecutive fixes whose bounding diameter stays below  $\delta_{xy}$ , and the rule then asks how long that stretch lasts. A circling climb instead traces a disc of diameter  $2R \approx 30$  m to 100 m, larger than  $\delta_{xy}$  by at least a factor of two (quantified below), and clears the test on the diameter alone.

- (ii) *An independent witness – the barometric altitude.* A GNSS lock-loss freezes only the GNSS position; the pressure sensor is a *separate* instrument. A wing that is really thermalling keeps gaining height on the barometric channel, whereas a true freeze shows a flat barometric trace. A horizontally motionless run whose barometric altitude is still changing at a plausible rate is a (tight) climb, and is *kept*. One signature, and only one, may overrule this witness: coordinates repeated *byte-identically*, defined in (iii) (the structure of Eq. (2.4) makes this explicit: on a barometric flight the witness is a flat barometer *or* a byte-identical repeat, so nothing else can cut a run whose barometer climbs).
- (iii) *Recorder declarations.* Two signatures declare a defect directly, and they carry different weight. A *byte-identical* repeat of the coordinates – the latitude and longitude fields of consecutive B records written character for character the same – is conclusive on its own: no moving receiver rewrites the very same bits, so the positions are invented whatever the barometer says. The V flag and the zeroed GNSS altitude are weaker: they certify a degraded *GNSS altitude*, not a frozen horizontal fix (Sec. 2.2), so they decide only where no barometric witness exists, i.e. on a GNSS-fallback flight.

A run is classified as a loss of signal only when it is spatially collapsed, *witnessed*, and sufficiently long,

$$\underbrace{\text{diam}(\text{run}) < \delta_{xy}}_{\text{(i) collapsed}} \wedge \underbrace{W}_{\text{witness (ii)/(iii)}} \wedge \underbrace{\Delta t_{\text{run}} \geq \tau_{\text{freeze}}}_{\text{duration}}, \quad (2.4)$$

$$W = \begin{cases} |\Delta z_{\text{baro}}| < \delta_z \vee \text{byte-identical} & \text{barometric flight,} \\ \text{V/zero-alt} \vee \text{byte-identical} & \text{GNSS-fallback flight.} \end{cases}$$

The run is then marked as a gap and the trajectory is *split* there rather than interpolated across, so that no motion is invented over an interval whose true path is unknown; the segments keep their parent’s origin and clock (Sec. 2.7.6).

The witness  $W$  is hierarchical. On a barometric flight *none of the recorder’s own declarations—the V flag, a zeroed GNSS altitude—can* overrule a climbing barometer; the one exception is the byte-identical repeat, which is conclusive on its own. Coordinates repeated bit for bit are invented positions even when the height gain is real, so such a run is cut even while the barometric altitude climbs.

Quantitatively, a real thermalling climb fails the cut on two independent instruments at once. Geometrically, a circling wing turns at  $R = v^2/(g \tan \phi)$ <sup>4</sup>: the tightest paraglider circles ( $v \approx 10 \text{ m s}^{-1}$  at a bank of  $\phi \approx 35^\circ$ , *ordinary values for tight thermalling*) give  $2R \approx 30$  m, and hang gliders, which fly faster, turn wider ( $v \approx 13 \text{ m s}^{-1}$  at the same bank gives  $2R \gtrsim 50$  m). Against the working value of  $\delta_{xy}$  (Table 2.5), test (i) therefore misses by a factor of at least 2 for the tightest paraglider circles and  $\gtrsim 3$  for hang gliders, more for anything wider—ratios to be rescaled if  $\delta_{xy}$  moves. Barometrically, a climb of order  $1 \text{ m s}^{-1}$  sustained over  $\tau_{\text{freeze}} = 60$  s

<sup>4</sup>The coordinated-turn relation: in a steady banked turn the lift  $L$  balances gravity vertically,  $L \cos \phi = mg$ , while its horizontal component provides the centripetal force,  $L \sin \phi = mv^2/R$ ; dividing the two gives  $\tan \phi = v^2/(gR)$ , i.e.  $R = v^2/(g \tan \phi)$ .

moves the barometer by  $\gtrsim 60$  m against the flatness tolerance  $\delta_z$ , which is set well below that scale (Table 2.5).

The duration threshold itself is set on the motion’s timescale:  $\tau_{\text{freeze}}$  is about two thermalling periods and three times the 20 s gap cap of Sec. 2.7.6, so a run long enough to be cut spans at least two full circles of any genuine climb, whose diameter test (i) then sees in full. Cutting a genuine climb on a barometric flight therefore requires two failures at once: the geometric margin of test (i) and the barometric witness of test (ii). The only alternative is a byte-identical repeat, which no *genuinely* moving receiver writes.

The margins make that conjunction unlikely rather than merely improbable in principle: the diameter test is missed by a factor of two or more, and the barometric one by an order of magnitude, so both would have to fail well outside their designed range at the same moment and on the same run. On a barometric flight the rule is, in this sense, conservative by construction—it errs towards keeping.

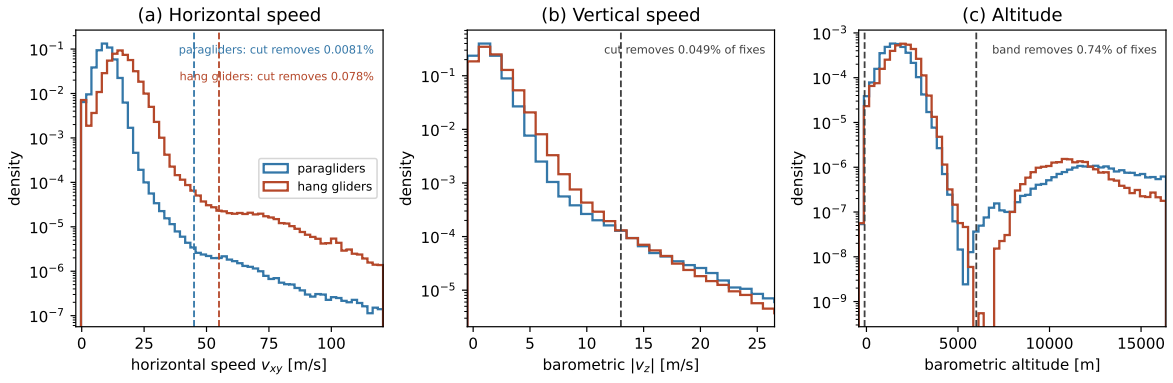
The duration condition is the one place where the rule deliberately abstains. A run that is spatially collapsed and witnessed but lasts less than  $\tau_{\text{freeze}}$  is *kept*, and that is the intended behaviour, not an oversight: at durations of a few tens of seconds a collapsed run and a genuine slow, horizontally localized stretch—a search, the core of a tight climb—are not distinguishable by these signatures, and cutting on them would remove the very events the analysis measures. The cost of the abstention is bounded by the threshold itself. A kept short freeze lengthens one waiting time by at most  $\tau_{\text{freeze}}$ , which is small against the tail of  $\psi(\tau)$  where the anomalous exponent is read, whereas a wrongly cut climb removes a waiting time outright. Short freezes that do get through are still visible downstream: they leave a run of interpolated-looking, near-identical positions, and the  $\psi(\tau)$  sensitivity checks sweep  $\tau_{\text{freeze}}$  precisely to bound how much of the tail depends on this choice.

**Residual corner cases.** The rule assumes a functioning barometric channel, and weakens where that channel is absent. This is not in tension with adopting the GNSS altitude for the *dynamics* of fallback flights (Sec. 2.7.1): there, GNSS is used because it is the only altitude available; here, the question is different—whether an instrument can *witness* a suspected GNSS lock loss—and the GNSS altitude cannot, because it comes from the very receiver whose lock is in doubt and freezes together with the position it would be checking. On GNSS-fallback flights the witness slot of Eq. (2.4) is therefore filled by the recorder’s declarations alone (the V flag or zeroed GNSS altitude, or a byte-identical repeat); the diameter and duration conditions still apply unchanged, so “alone” refers only to the witness, not to the whole rule. The residual error cases are then:

- *Wrongly cut* (GNSS-fallback only): a circling climb tight enough to stay within  $\delta_{xy}$  whose fixes carry a V flag—with no barometer to defend it, the declarations cut it.
- *Wrongly kept* (GNSS-fallback only): an undeclared *jittering* freeze. Some receivers, having lost the lock, do not repeat the last position byte-for-byte but re-estimate it each second, so consecutive positions differ by a metre or two of receiver noise: the run is genuinely frozen, yet neither byte-identical nor declared, and it is kept.
- *Kept, at bounded cost* (barometric flights): a jittered freeze under a *climbing* barometer is kept by construction. The cost is bounded, but it is not nil, and it is worth naming exactly. Every position recorded during the run sits inside the  $\delta_{xy}$ -ball of the last true fix, so the run enters the analysis as a wait. If the wing did move during it—wind drift, or a genuine glide under a climbing barometer—that motion goes unrecorded: the path length is under-counted by the distance actually covered, and a waiting time of up to the run’s duration is inserted where there was none. What is *not* lost is the displacement across the run, because the first position after re-acquisition is correct again; the quantities that

read a position at a given elapsed time (the MSD, the propagator) therefore see the right endpoint, and it is the path-accumulating quantities—path length, phase durations—that pay.

The first two cases—*wrongly cut* and *wrongly kept*—are expected to be rare: each needs a lock loss, a fallback flight, and a run whose bounding diameter happens to sit just on the wrong side of  $\delta_{xy}$ , all at once. Their frequency, however, is not something the pipeline can report, and it would be wrong to claim otherwise: a rule that fails does so silently, and counting how often a detector *fires* says nothing about how often it was right to. What the audit below measures is the firing rate; what measures the error rate is the injected-defect calibration of check (d), where defects of known size are planted in clean tracks and the recovered fraction is counted directly. The default to keep is what makes the two asymmetric in the meantime: an unfired rule leaves real dynamics intact.



**Figure 2.2.** Fix-level cleaning: auditing the absolute bounds. Per-fix distributions of the quantities the absolute bounds test—bounds that serve both the out-of-range rule and, through the  $v_{xy}$  bound, the impossibility gate that deletes an off-the-trend fix (Sec. 2.7.2) – (a) horizontal speed  $v_{xy}$ , (b) barometric  $|v_z|$ , (c) barometric altitude – pooled over a large paraglider subsample and the complete hang-glider population, with the adopted bounds dashed (Table 2.6); the  $y$ -axis is logarithmic so the sparse tail the bounds act on is visible. In panel (a) each discipline shows the flat plateau of sustained speeds with no credible flight explanation that places its own bound:  $45 \text{ m s}^{-1}$  to  $55 \text{ m s}^{-1}$  for paragliders,  $55 \text{ m s}^{-1}$  to  $70 \text{ m s}^{-1}$  for hang gliders. Binning choices and sample sizes: paragraph *Diagnostic sample* of Sec. 2.7.2 (p. 104). This figure audits the *absolute* bounds only; the Hampel and frozen-lock detectors have no per-fix distribution to read a threshold off, and are validated instead by the removal-fraction sweep described below.

**Completeness of the resampled trajectory.** The segmentation (Ch. 4) needs a defined horizontal position *and* altitude at every time, since the vertical velocity is its discriminant. Cleaning does not guarantee this fix by fix – it may delete a fix (a corrupt horizontal position) or mark a channel missing (a barometric spike, a V fix on a GNSS-fallback flight) – because the guarantee is established one step later, at resampling (Sec. 2.7.6). A *segment* is a maximal stretch of the flight between two splits—equivalently, between long gaps. It is the unit of analysis in the sense that every quantity accumulated along the path—a path length, a phase duration, a smoothing window—is computed inside one segment and never across the gap that ends it, since the trajectory between two segments is unknown (Sec. 2.7.6). The guarantee that the segmentation needs is therefore not “complete  $(x, y, z)$  everywhere” but “complete  $(x, y, z)$  within a segment”, not across a split, established by the single audited interpolation of Sec. 2.7.6.

| Defect                                                                                                                                                                                                                                                                                                 | Detected by                                                                                                                                                                                                                                             | Action                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| <b>Off the trend</b> — <i>the horizontal position itself is corrupt</i>                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                         |                                                        |
| Position spike, out-and-back jump                                                                                                                                                                                                                                                                      | impossibility gate: unreachable from the last accepted fix at the absolute $v_{xy}$ bound ( $45/55 \text{ m s}^{-1}$ , para/hang), and its block's removal lets the track rejoin. The Hampel identifier [15, 16] flags and attributes; it does not gate | delete the fix; the short gap is bridged at resampling |
| Position discontinuity (re-acquisition offset)                                                                                                                                                                                                                                                         | impossible step that no bounded removal rejoins                                                                                                                                                                                                         | split at the step, like a long gap                     |
| Impossible step still standing                                                                                                                                                                                                                                                                         | any surviving step past the $v_{xy}$ bound, whatever the scan resolved (postcondition)                                                                                                                                                                  | make it a segment boundary                             |
| Corrupt block closing the record                                                                                                                                                                                                                                                                       | a block still open when the track ends: nothing follows it to rejoin, and by construction it spans at most $w$                                                                                                                                          | delete the block, instead of splitting                 |
| <b>On the trend</b> — <i>record structurally wrong, horizontal position agrees with its neighbours</i>                                                                                                                                                                                                 |                                                                                                                                                                                                                                                         |                                                        |
| Duplicate timestamp                                                                                                                                                                                                                                                                                    | two or more fixes in one UTC second                                                                                                                                                                                                                     | merge to one (centroid)                                |
| Backward timestamp                                                                                                                                                                                                                                                                                     | time decreases (not a midnight roll-over)                                                                                                                                                                                                               | delete the fix                                         |
| Frozen-lock run                                                                                                                                                                                                                                                                                        | $\text{diam} < \delta_{xy}$ for $\geq \tau_{\text{freeze}}$ , witnessed (Eq. 2.4): baro. flat or byte-identical; V/zero-alt on GNSS-fallback                                                                                                            | mark as gap, <i>split</i> there                        |
| <b>Out of range</b> — <i>horizontal position intact, only the altitude value is impossible</i>                                                                                                                                                                                                         |                                                                                                                                                                                                                                                         |                                                        |
| Missing / out-of-band altitude, or $ v_z $ spike                                                                                                                                                                                                                                                       | altitude channel absent, outside $[-100, 6000] \text{ m}$ , or baro. $ v_z  > 13 \text{ m s}^{-1}$                                                                                                                                                      | drop the altitude only; the fix stays                  |
| <b>Invariant:</b> a usable horizontal position is never discarded. A fix is deleted only when its <i>own</i> horizontal position is corrupt or invented (frozen lock); when only the altitude is wrong, the fix is kept and that channel is marked missing, to be restored at resampling (Sec. 2.7.6). |                                                                                                                                                                                                                                                         |                                                        |

**Table 2.6.** Fix-level cleaning, grouped by how each defect presents; in the order of the table, the three groups map onto a robust local test, a structural rule, and an absolute bound. The speed and altitude bounds are audited on the per-fix distributions of Fig. 2.2; the robust-test and frozen-lock parameters ( $k$ ,  $w$ ,  $\varepsilon_{\min}$ ,  $\delta_{xy}$ ,  $\delta_z$ ,  $\tau_{\text{freeze}}$ ) are working values, listed in Table 2.5 and argued in Sec. 2.7.2. The  $v_{xy}$  bound is per-discipline because the speed envelopes differ (a **sailplanes** entry follows once that source is in). The altitude lower bound is  $-100 \text{ m}$ , not  $0$ : the barometric altitude is referenced to the ICAO standard atmosphere rather than to the day's sea-level pressure, so on a high-pressure day a sea-level take-off can read a few tens of metres negative (the offset scale is visible in Fig. 2.1a). (implementation details: Sec. 2.7.2)

**Flight-level integrity gate.** Cleaning acts fix by fix, but it also keeps count of how much it had to do. The counters are taken over the airborne window: defects in a ground phase that trimming removes anyway must not condemn the flight. If cleaning has removed or reconstructed more than a small fraction of a flight’s airborne fixes, the flight is dropped whole.

**Validating the cleaning.** All the detector parameters are validated by one procedure, run once the cleaning routine is built (Ch. 4):

- (a) *Freeze the working values.* The thresholds are fixed a priori at the values of Table 2.5; nothing is tuned on the transport observables themselves.
- (b) *Audit the removal fractions.* The cleaning is run over the full archive with per-rule counters (the same machinery as the integrity gate): for each rule and each discipline, the fraction of fixes flagged and the fraction removed. The expectation is that each rule touches only a small fraction of fixes; this is the criterion to be verified, not a fact assumed in advance.
- (c) *Sweep each threshold.* Each parameter is varied around its working value and the removal fractions are recomputed. “Well placed” is made quantitative rather than left to the eye. Write  $\rho(\theta)$  for the fraction of fixes a rule removes at threshold  $\theta$ , and read the local logarithmic sensitivity

$$S(\theta) = \left| \frac{d \log \rho}{d \log \theta} \right|.$$

A threshold sits on a plateau when  $S \lesssim 1$  over a decade centred on the working value: halving the threshold then at most doubles what it removes, which is the behaviour of a rule biting into a sparse tail of defects.  $S \gg 1$  means  $\rho$  is growing like a power of the threshold with a large exponent—the rule has reached the bulk of the distribution and is removing dynamics, not defects—and the working value is reconsidered. Reporting  $\rho(\theta)$  and  $S$  at the working value, per rule and per discipline, is what the sweep produces. The same sweep covers the split cap of Sec. 2.7.6, monitored on its split fractions rather than on removed fixes.

- (d) *False negatives.* Steps (b)–(c) measure what the cleaning removes, not what it misses—the corrupted fixes that pass undetected. Nothing in an audit of removals can supply that number, so this check is run as a matter of course and not only on suspicion: defects of known size (spikes, frozen runs, time glitches) are injected into clean tracks in known quantity, the cleaning is re-run, and each rule is scored against each defect type on the four counts—defects recovered and missed, clean fixes wrongly removed and correctly kept. The result is a confusion table per rule per defect type, with the recovered fraction as a function of defect size, and it is the only direct measurement of the error rates that the firing counts of step (b) cannot give.

(implementation details: Sec. 2.7.2)

### 2.7.3 Trimming of ground phases

**Outer ground phases (take-off and landing).** The logger starts recording before take-off and stops after landing; these ground phases must be stripped before any kinematic analysis. Detection uses the *horizontal* speed  $v_{xy}$  alone (from (2.1)) – the vertical speed plays no role here. With  $v_0 = 5 \text{ m s}^{-1}$  and  $T_0 = 30 \text{ s}$  (Table 2.5), the retained airborne segment is  $[t_{\text{on}}, t_{\text{off}}]$  with

$$t_{\text{on}} = \min\{t : v_{xy} > v_0 \text{ on } [t, t + T_0]\}, \quad t_{\text{off}} = \max\{t : v_{xy} > v_0 \text{ on } [t - T_0, t]\}. \quad (2.5)$$



Take-off is thus the *first* time the speed stays above  $v_0$  continuously for  $T_0$ , and landing the *last* time it has done so: the same sustained-speed rule read forward and time-reversed. Reading the landing rule backward from the end matters. A rule of the form “cut at the first sustained drop below  $v_0$ ” would fire in the middle of the flight, because a wing soaring into wind can hold a ground speed near zero while genuinely flying. The persistence  $T_0$  prevents a gust or a GPS glitch on the ground from being mistaken for take-off. A flight with no sustained stretch at all has no airborne segment and is discarded, with the reason recorded.

$T_0$  is set between two scales rather than tuned. From below, it must outlast the artefacts it exists to reject: a ground gust that briefly pushes a laid-out wing, or a position glitch that fakes a fast step, lasts a few seconds at most, and  $T_0 = 30$  s is several times that. From above, it must cost little when it is wrong, and it can misfire only by at most  $T_0$  at each end of the flight—negligible against the 40 min minimum duration a retained flight has to reach anyway (Sec. 2.7.4). Between those bounds the result is insensitive to the exact value, which is the reason it can be fixed a priori.

One caveat is structural. The criterion is a *ground* speed, so a slow phase that happens to sit right at the start or end of the flight—a launch straight into ridge lift, say—can be clipped along with the ground phase. The fraction of each flight that trimming removes is recorded per flight and its distribution over the archive reported as a histogram once the pipeline runs in full. That histogram bounds the *exposure* to this failure—how much of a flight the rule can have taken—but it does not detect the failure itself, since a correctly trimmed ground phase and a wrongly clipped ridge-lift launch produce the same trimmed fraction. What separates them is inspection at the tail: the flights in the upper percentiles of the trimmed-fraction distribution are the candidates, and they are few enough to be examined one by one against their altitude trace, which rises during a ridge-lift launch and does not on the ground. Both  $v_0$  and  $T_0$  affect that fraction, so they are made explicit (implementation details: Sec. 2.7.3).

**Interior ground stints (mid-flight landings).** The rule above trims only the *outer* ground phases: a mid-flight landing and relaunch recorded in one log (a top-landing, say) would survive as an interior near-zero-speed stint and read downstream as a false long wait in the tail of  $\psi(\tau)$  (Sec. 2.7.2). An interior-ground detector is therefore specified, built like the frozen-lock rule: it cuts only on corroborated evidence, because its dangerous error is the opposite one—the soaring-into-wind failure mode above. A stint is cut only when *both* of the following hold:

- $v_{xy} < v_0$  continuously for at least  $T_{\text{ground}} = 10$  min—far beyond any search phase, so no genuine slow phase reaches this length;
- the adopted altitude channel flat over the whole stint, the central span of its detrended values staying within  $\delta_z^{\text{gr}} = 5$  m once the slow barometric drift has been removed—a wing at zero ground speed in real air still moves vertically, the same independent-sensor argument as Eq. (2.4).

The detrending is not a detail. The barometric reference itself wanders by tens of metres over an hour (Sec. 2.7.1), which over a stint of  $T_{\text{ground}}$  is already of the order of  $\delta_z^{\text{gr}}$ ; testing the raw range would let a perfectly stationary pilot fail the flatness condition on a pressure change alone. What the condition has to measure is the *residual* vertical motion after that drift is taken out, and  $\delta_z^{\text{gr}}$  is a tolerance on that residual.

That the spread is the central one is not cosmetic, and the alternative was in force until it was measured. A full range is a maximum statistic: over zero-mean noise it grows without bound with the number of samples, as  $2\sigma\sqrt{2\ln n}$ . At the metre-scale barometric noise of this archive it reads 4.6 m over a 60-sample stint and 7.1 m over 3,000, so against a 5 m tolerance the same motionless pilot passed the test on a one-minute stint and failed it on a ten-minute one—on noise alone. Since an interior stint must last  $T_{\text{ground}}$  to be considered at all, it failed



on every stint the rule was ever offered, and the guard could not fire. The central span carries the same units and the same reading and sits at  $3.3\sigma$  whatever the stint's length, so  $\delta_z^{\text{gr}}$  means the same thing on a stint of any duration; a genuine climb of hundreds of metres exceeds it either way.

One consequence is stated rather than hidden. *On a GNSS-fallback flight the guard abstains.* The intended rule is that the vertical witness there uses the GNSS channel with a tolerance widened to match its noise, and in doubt the stint is kept; the widening is not implemented, because the factor it needs is a measured ratio between the two channels' short-timescale noise and a first measurement on the archive did not reproduce the ordering Fig. 2.1 reports. Until that is settled, the single barometric tolerance is applied to both channels, which on a noisier one simply means the flatness condition is not met and the stint is kept—the conservative half of the intended rule, and the safe direction, since the error this guard exists to avoid is excising a genuine climb. What it costs is that a mid-flight landing on a GNSS-fallback flight is neither excised nor flagged.

A stint that is cut is excised, and the flight is split at the excision: the parts before and after are treated from then on as separate segments of the same parent flight, exactly as at a long gap (Sec. 2.7.6). The segment rules apply unchanged—a phase interrupted by the cut is censored rather than counted as complete, and the flight-level cuts of Sec. 2.7.4 are not re-applied to the individual segments.

The order of the two steps is worth stating explicitly, because it is the reverse of the order in which they are described. Trimming runs first (step (iii) of the pipeline) and the flight-level cuts second (step (iv)): a duration or a path length is only meaningful once the ground phases are gone, so the cuts have to see the trimmed track. When trimming has split a flight, those cuts are still evaluated on the *parent*—its airborne duration is the sum of its segments' durations, its path length the sum of their path lengths—and not on each segment separately. The reason is stated in Sec. 2.7.6: a segment is short because a gap happened to fall there, which says nothing about whether the flight is a genuine cross-country one, and re-applying a whole-flight criterion to a fragment would throw away exactly the data recorded around gaps. Segments face only the one requirement that is physically theirs—being long enough to carry the smoothing window (90 s, Table 2.5).

Stints shorter than  $T_{\text{ground}}$  are not cut but *flagged*: a wait with sustained  $v_{xy} < v_0$  and a flat barometric trace is marked suspect whatever its length, and the  $\psi(\tau)$  fits are re-run with and without the suspect waits as a sensitivity check – the same flag-without-deleting pattern as the outlier identifier. Flagged waits are reported, not merely used internally: their count, and the share of the total waiting-time mass they carry, are given per discipline alongside the two  $\psi(\tau)$  fits, so the reader can see how much of the tail rests on them.  $T_{\text{ground}}$  and  $\delta_z^{\text{gr}}$  are parameters of this guard and sit in the `trimming` block of the configuration file, with the values of Table 2.5.

### 2.7.4 Flight-level filtering

After cleaning and trimming, some flights are still unfit for the ensemble analysis and are dropped whole. The step is delicate: too aggressive a cut biases the transport statistics – a high minimum duration, for instance, preferentially keeps successful long flights and *inflates the apparent MSD exponent at long lags, since the flights that survive such a cut are the ones that kept going*. The flight-level filtering therefore uses a *minimal-threshold strategy*: for each criterion, the threshold is set to the smallest value that still clears non-genuine flights (sled runs, tows, aborted or truncated logs), so the ensemble is disturbed as little as possible. In practice “minimal” is read off the data, not guessed: for each criterion the full-archive distribution of its quantity is drawn (Fig. 2.3, top row), together with the fraction of flights that the cut alone would retain as the threshold is moved (bottom row).

The non-genuine population does not announce itself the same way in both criteria. In the path length it forms a distinct cluster at the low end, separated from the genuine flights by a sparse tail, and the threshold can be placed just beyond it. In the duration histogram no such cluster is visible; what locates the cut there is the retained-fraction curve alone, which is flat over a wide stretch before the bulk begins. The threshold is placed in either case on the stretch where the retained-fraction curve is flat—there, moving the cut changes almost nothing, which is precisely the statement that it removes junk and not signal. Five track criteria gate a flight, plus a metadata stratifier: Two of them are sums over the flight, and what they are summed over is part of the rule rather than an implementation detail: the duration and the path accumulate *within* a block and never across the boundary that ends it, since the trajectory there is unknown and a straight line drawn over it was never flown.

A boundary is either a marker an earlier stage left—an excised frozen-lock run, a re-acquisition offset, a mid-flight landing—or an inter-fix gap wider than the  $g_{\max}$  the resampling will split at (Sec. 2.7.6). The second kind is easy to forget, because nothing marks it: a marker records what a *stage* decided, and a hole in the log is nobody’s decision. Forgetting it counted ten minutes of missing record as ten minutes of flight, and the six kilometres of straight line across it as six kilometres flown. Using the same  $g_{\max}$  here as there is what makes a flight measured over the intervals it will actually be analysed over.

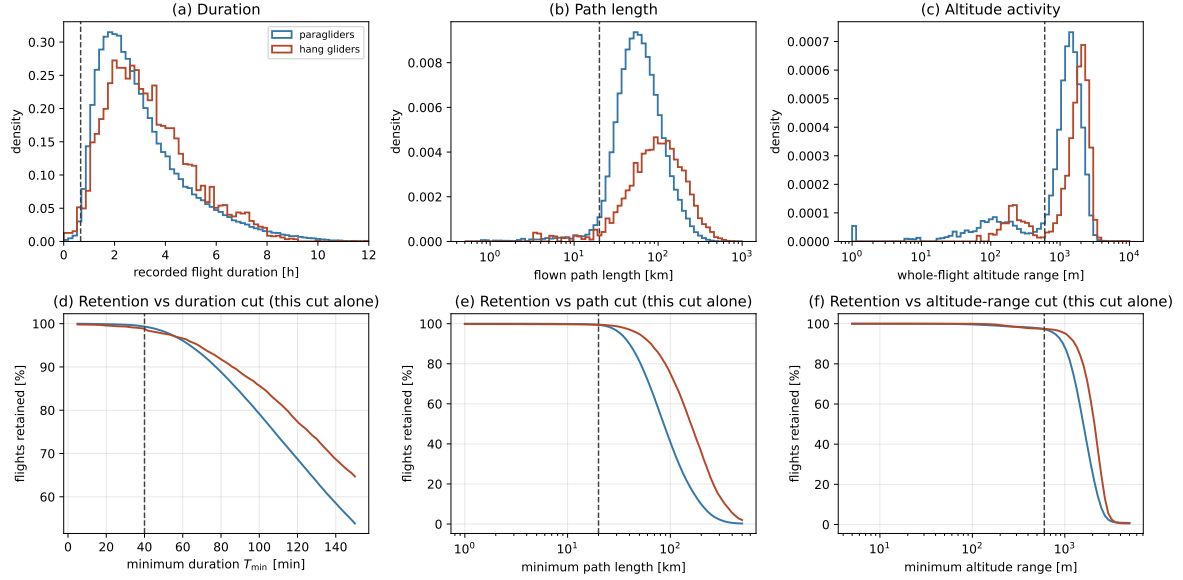
- (1) *Recorded duration*  $T \geq 40$  min. A cross-country flight spans many climb–glide cycles; shorter logs are sled runs or aborted attempts. An upper sanity bound  $T \leq 16$  h removes loggers left running past the flight: the census holds 15 paraglider “flights” of 16 h to 166 h, with paths up to 10,059 km (and none among the hang gliders), which would poison the long-lag analyses generally, the MSD among them.
- (2) *Flown path length*  $L \geq 20$  km, with  $L = \sum_i \Delta s_i$  from the great-circle steps of (2.1), a floor against degenerate logs. On the census, above 40 min the cut catches 236 of 184,583 paraglider flights (median extent 5 km) and 7 of 6,638 hang-glider flights: a fraction of order  $10^{-3}$  of either ensemble. Duration already clears tows and sled runs.
- (3) *Altitude activity*. The total range of the adopted altitude channel over the airborne track—the difference between its maximum and its minimum—must reach 600 m. The threshold is a positive statement about what the flight did, not merely a guard against a dead sensor: a cross-country flight of at least 40 min lives by climbing, and several climbs of several hundred metres each are what keep it airborne, so a record spanning less than 600 m over that time was not soaring—a vehicle log, a tow, a stuck channel. The cut is nonetheless cheap, which is what makes it safe: moving it from 75 m to 600 m changes the retained share of barometric paraglider flights by under two points, so the two values sit on the same plateau and the population between them is thin. Like the other two, it is a *whole-flight* criterion, evaluated on the parent before any splitting; an individual segment may well span less, and that is accepted for the same reason the duration cut is not re-applied per segment—a segment is short because a gap fell there, not because the flight was not a flight. One limitation belongs with it: the cached census records only the *barometric* extremes, so on a GNSS-fallback flight it reads a range of zero whatever the flight actually did, and the diagnostic of Fig. 2.3c,f is drawn over the barometric flights alone. Auditing this cut on the fallback minority needs the GNSS extremes in the scan, which is a full re-read of the archive rather than a re-query of the cache. They are counted and dropped with the reason recorded.
- (4) *Plausibility of the cleaned record*. The mean ground speed of the retained track,  $L/T$ , must not exceed the discipline’s own fix-level bound  $v_{xy}^{\max}$ . This is a sanity bound of the same kind as the upper duration cut above, and it is here for the same reason: a small

number of logs are corrupt beyond what fix-level cleaning can repair—runs of the null position (0,0) scattered through the track—and every jump that survives adds thousands of kilometres of path that were never flown. Asking a *whole-flight average* to stay under the bound that already declares a single step impossible is the weakest statement that catches them, and it is deliberately weak: no genuine flight comes close, since the median mean speed is under  $10 \text{ m s}^{-1}$  and the bound sits at  $45 \text{ m s}^{-1}$ . On the present archive it removes *no* flight at all—0 of either discipline—and that zero is a result rather than a redundancy. The criterion was written when a corrupt jump still entered the flown path; it no longer does, since an impossible step is made a segment boundary at the fix level and a long gap is excluded from the sum here, so by the time the path reaches this test it contains only motion. What such a record now fails is the reach criterion below. The cut is kept because it is the cheapest possible check on that chain: if a future source, or a change upstream, ever lets an impossible length back into the path, this is where it surfaces, and its count stops being zero. No new configuration value is introduced—the bound is the one already set for the fix level, read at the scale of a whole flight.

- (5) *Reach*. No fix may lie further from the first than the bound could have carried it in the time elapsed since:  $\Delta s(\text{fix}_0, \text{fix}_i) \leq v_{xy}^{\max} t_i$  for every  $i$ , because the flight started there. It is stated per fix, and deliberately not as the extent  $D = \max_i \Delta s(\text{fix}_0, \text{fix}_i)$  against the whole duration  $T$ : that weaker form licenses a fix 100 km out ten seconds into the record on the grounds that the flight went on for three hours, and it is not the statement the written table is checked against (Sec. 2.2). One criterion, one form, in both places. Where the previous criterion bounds the path, this one bounds the displacement, and the difference is exactly what makes it necessary. A flight whose *first* fix is the corrupt one has no impossible step at all once that step is made a boundary and its length is no longer summed into  $L$ : it passes on duration, on path, on altitude and on mean speed, and it simply sits thousands of kilometres from where it says it began. Five paraglider flights in the first full-archive run did (Sec. 2.7.2), and since that first fix becomes the origin of the local frame (Sec. 2.7.5), the displacement they carried was not one bad step but every coordinate of the record. On the archive the criterion removes 228 paraglider and 7 hang-glider flights, of order  $10^{-3}$  of either ensemble; on a sample of 4,000 per discipline it touches nothing else. Like the previous one it introduces no configuration value, and it is checked again on the written tables as an invariant of the output (Sec. 2.2), where it is the statement  $|\mathbf{r}(t)| \leq v_{xy}^{\max} t$  that the whole of Chapter 3 assumes.
- (6) *Glider class* (metadata). For class-stratified analyses, flights whose `aile_class` cannot be normalised onto the discipline's system (Sec. 2.4) are dropped from the class-resolved sub-analyses but kept in the pooled one. Stratification by glider *type* (paraglider vs hang glider) is already possible.

Every count quoted above is a generated macro, recomputed against the thresholds in force from a cached full-archive scan; how, and by which code, is in Sec. 2.7.4. The thin tail of very slow loggers is handled by the  $\Delta t$  stratification (Sec. 2.7.6); a separate minimum-fix-count cut would be redundant, since above the duration cut even a 10 s logger records hundreds of fixes. A further criterion, sampling regularity, is applied later, where a flight's time base is examined (Sec. 2.7.6) (implementation details: Sec. 2.7.4).

Figure 2.3 shows these diagnostics for the three criteria that carry a tunable threshold, (1)–(3), and both disciplines, computed on the full census (no subsampling). The duration and path cuts sit far out in the tail, so the retained fraction is insensitive to their exact value.



**Figure 2.3.** Flight-level filtering diagnostics, per discipline. *Top:* distributions of (a) recorded flight duration, (b) flown path length and (c) whole-flight altitude range. *Bottom:* the fraction of flights retained versus (d) the duration cut, (e) the path-length cut and (f) the altitude-range cut, each computed for *that cut alone* (marginal, not cascaded). Dashed lines mark the adopted cuts ( $T \geq 40$  min,  $L \geq 20$  km, range  $\geq 600$  m). The duration and path cuts sit far out in the tail. Panels (c) and (f) are drawn over the *barometric* flights only: the cached scan stores no GNSS altitude extremes, so a fallback flight would enter them at a range of zero for a reason unrelated to how it flew (Sec. 2.7.4). (implementation details: Sec. 2.7.4)

### 2.7.5 From geographic to local Cartesian coordinates

The cleaned, trimmed track is next mapped from geographic to local Cartesian coordinates. The two remaining steps are different in kind (Sec. 2.7): resampling and smoothing operate on the coordinates *componentwise*—and the smoothing returns their derivatives in the same *componentwise* way (Sec. 2.7.7)—and therefore need actual Cartesian components to act on. Latitudes and longitudes cannot play that role—they are angles on a curved surface, and differencing them componentwise does not yield displacements. The mapping proceeds in two steps.

**Step 1: geodetic to ECEF.** Each fix is converted to Earth-Centred, Earth-Fixed (ECEF) coordinates on the WGS84 ellipsoid (semi-major axis  $a = 6,378,137$  m, flattening  $f = 1/298.257223563$ ,  $e^2 = f(2 - f)$ ) [14, 18],

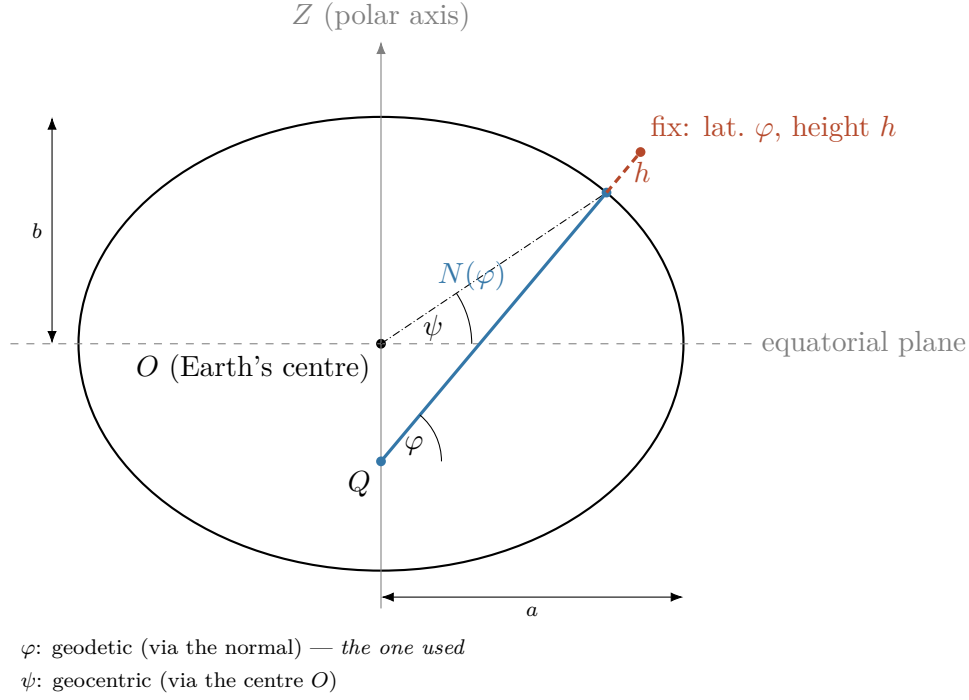
$$\begin{aligned} N(\varphi) &= \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}}, \\ X &= (N + h) \cos \varphi \cos \lambda, \\ Y &= (N + h) \cos \varphi \sin \lambda, \\ Z &= (N(1 - e^2) + h) \sin \varphi, \end{aligned} \quad (2.6)$$

where  $N(\varphi)$  is the prime-vertical radius of curvature and  $h$  the ellipsoidal height, measured along the ellipsoid normal; the transform is derived step by step in Appendix 2.B.

**Geodetic versus geocentric latitude.** The latitude  $\varphi$  in (2.6) is the *geodetic* latitude: the angle between the equatorial plane and the ellipsoid *normal* at the point (Figure 2.4). It must not be confused with the *geocentric* latitude  $\psi$ , the angle to the line joining the point to the Earth’s centre  $O$ . On an ellipsoid the two differ, because the normal does not pass through  $O$ : the gap  $\varphi - \psi \approx (e^2/2) \sin 2\varphi$  (derived in Appendix 2.B) peaks at  $\varphi = 45^\circ$ , where  $\sin 2\varphi = 1$ ,

and there reaches  $\approx 0.19^\circ$  (about  $11.5'$ ), which if the two were confused would misplace a fix by up to  $\sim 20$  km of ground distance.

We use the geodetic latitude throughout, and no conversion is needed: WGS84 – hence the GPS receiver and the IGC B records (Sec. 2.2) – reports geodetic latitude, and (2.6) with  $N(\varphi)$  is precisely the geodetic-to-ECEF transform that consumes it. These are also the  $(\varphi_0, \lambda_0)$  that orient the local frame of Step 2 (Figure 2.5).



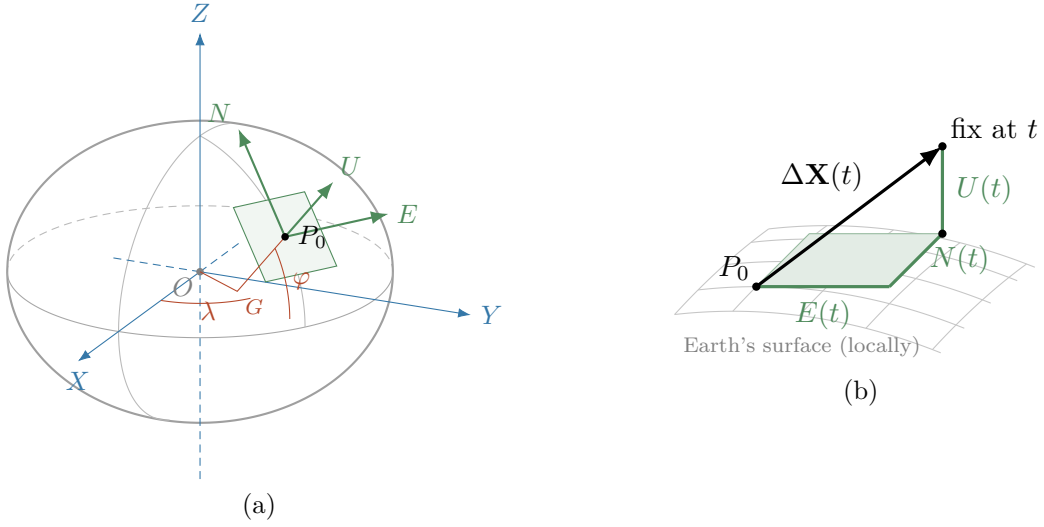
**Figure 2.4.** Geodetic latitude and height on the reference ellipsoid, in meridian cross-section (flattening exaggerated; the true WGS84  $f \approx 1/298$  would be indistinguishable from a circle): the semi-axes  $a$  and  $b$ ; the geodetic latitude  $\varphi$ , the angle between the ellipsoid normal and the equatorial plane; the height  $h$  along that normal; and  $N(\varphi)$ , the distance along the normal from the surface to the point  $Q$  on the polar axis. The normal does not pass through the centre  $O$ , so  $\varphi$  differs from the geocentric latitude  $\psi$  (dash-dotted). Longitude  $\lambda$  is out of the page here; it is shown in Figure 2.5.

**Step 2: ECEF to local ENU.** Taking the first fix  $(\varphi_0, \lambda_0, h_0)$  of the *trimmed track* as the origin—the first fix of free flight, which is close to but not the take-off point on the ground, since the ground phase has been removed (Sec. 2.7.3), the ECEF displacement  $\Delta \mathbf{X} = \mathbf{X} - \mathbf{X}_0$  is *rotated* into the local East–North–Up (ENU) frame tangent to the ellipsoid at that point—the word is exact, not loose: the origin shift has already been taken out by forming  $\Delta \mathbf{X}$ , and what is left is an orthogonal matrix of unit determinant, so lengths and angles are preserved and only the axes change,

$$\begin{pmatrix} E \\ N \\ U \end{pmatrix} = \begin{pmatrix} -\sin \lambda_0 & \cos \lambda_0 & 0 \\ -\sin \varphi_0 \cos \lambda_0 & -\sin \varphi_0 \sin \lambda_0 & \cos \varphi_0 \\ \cos \varphi_0 \cos \lambda_0 & \cos \varphi_0 \sin \lambda_0 & \sin \varphi_0 \end{pmatrix} \Delta \mathbf{X}. \quad (2.7)$$

with the rotation matrix derived in Appendix 2.B. The ENU frame is the local Cartesian frame in which the whole analysis is carried out:  $E$  points east,  $N$  north, and  $U$  along the local vertical (the ellipsoid normal at the origin). The ECEF coordinates of Step 1 are only an intermediate representation of the computation—no observable is ever expressed in them. Over the spatial extent of a single flight the tangent-plane approximation is accurate to well

below the GPS noise for the horizontal coordinates: projecting onto the tangent plane shortens a distance  $d$  along the surface by  $\approx d^3/(24R_\oplus^2)$ , about 1 cm at  $d = 20$  km and 13 cm at 50 km, against metre-scale GPS noise (Appendix 2.B).



**Figure 2.5.** The ENU rotation: it turns a vector’s global ECEF components into local East, North, Up ones. (a) The global ECEF frame  $(X, Y, Z)$  in blue, centred at the Earth’s centre  $O$ ; the two red angles that locate the origin fix  $P_0$  (the first fix of the trimmed track, Sec. 2.7.5): longitude  $\lambda$ , measured in the equatorial plane from the prime meridian, and geodetic latitude  $\varphi$ , the tilt of the ellipsoid normal, measured at the point  $G$  where the normal meets the equatorial plane; and the local ENU frame in green, tangent at  $P_0$ . Flattening exaggerated as in Figure 2.4:  $U$  follows the normal  $G-P_0$ , not the radius from  $O$ . The green arrows are the rows of the rotation matrix, as unit vectors. (b) Close-up of the tangent plane: a later fix’s ECEF displacement  $\Delta \mathbf{X}(t)$  is the space diagonal of the box whose edges are its ENU components  $E(t)$ ,  $N(t)$ ,  $U(t)$ .

The altitude  $h$  that enters this transform is the adopted channel of Sec. 2.7.1. The horizontal coordinates are insensitive to that choice, and the mechanism is visible in Eq. (2.6):  $h$  enters the horizontal ECEF components only through the radius factor  $(N + h)$ , so switching channel—which changes  $h$  by the inter-reference offset, at most  $\sim 100$  m (Fig. 2.1a)—rescales them by a relative  $\delta h/(N + h) \lesssim 2 \times 10^{-5}$ . Across the full extent of a flight ( $\sim 50$  km) that is under a metre, below the GPS noise on the same coordinates.

For the vertical, the analysis does not use the rotation’s  $U$  component as its working coordinate: it keeps the adopted altitude channel itself (Sec. 2.7.1; barometric for most flights, GNSS on the fallback minority),  $z(t) = h(t)$ , at its measured value—the altitude of the first fix is *not* subtracted. Nothing is gained by re-zeroing: every vertical quantity that matters downstream (the vertical velocity, the climb/sink discrimination of the segmentation) depends on increments, which are invariant to the reference, while keeping the measured value preserves the absolute height, an observable in its own right (potential energy, convective-boundary-layer structure) (Sec. 2.7.8).

### 2.7.6 Uniform sampling within a flight

The smoothing and differentiation below assume a *uniform* sampling interval, so each surviving flight’s time base is examined next. Loggers record at a fixed nominal rate – paragliders mostly at 1 s (69% of flights), hang gliders spread across several rates from 1 s to 10 s with a thin tail beyond (Fig. 2.7) – but the realised series can carry jitter and drop-outs. We keep each flight at its *native* rate; a common cadence is unnecessary, because the derivatives are not taken by finite differences across flights but read off the local polynomial that the smoothing fits to each



flight separately, and the flight's own  $\Delta t$  enters only as the factor that turns that polynomial's coefficients into physical units (Sec. 2.7.7). What matters is uniformity *within* a flight, not across flights.

Per flight, let  $\Delta t$  be the median inter-fix interval. Three cases arise:

**Uniform.** All intervals equal  $\Delta t$  to tolerance: the series is used as is.

**Mildly irregular.** Isolated jitter or single drop-outs – no gap beyond the split bound  $g_{\max} = \min(10 \Delta t, \max(20 \text{ s}, 2 \Delta t))$  (its two scales are motivated in the paragraph *The split bound has two scales* below) and less than 10 % of the grid missing (Table 2.5): the flight is resampled onto the uniform grid at its native  $\Delta t$ , interpolating across the small gaps only, and the interpolated fraction is recorded.

**Broken by a long gap.** A gap longer than  $g_{\max}$  – native, or opened where cleaning removed a frozen-lock run (Sec. 2.7.2), or a position discontinuity such as a re-acquisition offset, where the track jumps and stays: both sides are genuine and only the transition between them is unknown, so it is handled exactly like a long gap – is not bridged: interpolating across it would fabricate dynamics the smoothing could not tell from real motion. The flight is *split* at the gap into independently analysed segments, each carried forward on its own; what a split means for the analysis is stated below.

The missing-fraction check runs here, and not among the flight-level cuts of Sec. 2.7.4, because it needs the time base and the splits to exist first. It needs the time base because “missing” is only defined once there is a grid to be missing from: the quantity counted is the share of points of the uniform grid at the flight's own  $\Delta t$  that carry no recorded fix, and neither the grid nor its spacing exists before this step. It needs the splits because without them a single long gap—which is not going to be interpolated at all—would dominate the count and condemn a flight whose segments are each perfectly well sampled. It is evaluated *per segment*, after splitting: a segment whose residual missing fraction (the grid points that would have to be interpolated) exceeds 10 % is dropped alone, and the flight is lost only if no segment survives. A flight with no long gap is a single segment, so a gap-free flight missing more than 10 % of its grid is dropped whole by the same rule.

**The split bound has two scales.** The relative part,  $10 \Delta t$ , tolerates *isolated dropped fixes* in proportion to the cadence and caps the number of interpolated grid points at ten. Alone, however, it would let a slow logger bridge very long holes: 30 s at  $\Delta t = 3 \text{ s}$ , 100 s at  $\Delta t = 10 \text{ s}$ . The harm of a bridged gap is set by the motion's own timescales, not the logger's: an interpolated stretch is a straight line, so a gap approaching the thermalling period ( $\sim 30 \text{ s}$ ) could replace a full circle, and the segmentation downstream would read the invented straight stretch as a glide. The absolute cap of 20 s is pinned by three constraints:

- at least  $2\times$  the slowest common cadence (10 s, Fig. 2.7), so a single missed fix never splits a flight;
- at most about two thirds of a thermalling period, so a bridged gap can never hide a full circle;
- well below  $\tau_{\text{freeze}} = 60 \text{ s}$ , for consistency: an excised frozen-lock run of that length is split, so a native gap of comparable length cannot be bridged.

One refinement keeps the first constraint true at *every* cadence: the cap is floored at  $2 \Delta t$ , i.e.  $g_{\max} = \min(10 \Delta t, \max(20 \text{ s}, 2 \Delta t))$ , so the thin tail of very slow loggers (Fig. 2.7) also tolerates a single missed fix. Which of the three terms sets  $g_{\max}$  depends on the cadence: at the dominant 1 s rate it is the relative bound,  $g_{\max} = 10 \text{ s}$ ; from  $\Delta t = 2 \text{ s}$  to 10 s it is the absolute cap,  $g_{\max} = 20 \text{ s}$ ; above that it is the  $2 \Delta t$  floor.



On the census, the two-scale bound  $g_{\max}$  just defined leaves 15.3 % of paraglider and 40.5 % of hang-glider flights with at least one split; had the relative part  $10 \Delta t$  acted alone, without the absolute cap, the shares would be 11.8 % and 14.5 %. The increase sits at the slow cadences, where a 20 s hole is only a few missed fixes (generated macros, Sec. 2.7.6). A split is cheap for the ensemble observables (below) but costly for the *phase* statistics: every split censors the phase in progress when the gap opens, so splitting too eagerly preferentially removes long phases from the duration distributions—the same tail the cleaning protects. For this reason the absolute cap is treated as a working value like the detector thresholds: it is swept in the validation procedure of Sec. 2.7.2 and frozen where the split fractions plateau. The sweep is reported as a figure, not only as a number: the fraction of flights with at least one split, and the fraction of phases censored, plotted against the cap for each discipline, with the adopted value marked. That plot is what shows the plateau, and therefore what justifies the choice.

The bound is the same for every channel, altitude included. One might expect the altitude to tolerate wider gaps, since the vertical speed is about an order of magnitude smaller than the horizontal ( $v_z \sim 1 \text{ m s}^{-1}$  against  $v_{xy} \sim 10 \text{ m s}^{-1}$ ); it does not, for two reasons. First, the interpolation error over a gap is set by the *curvature* of the signal, not by its speed: a fast but straight motion is interpolated exactly, a slow but turning one is not, so a slower channel is not automatically a smoother one. Second, and more decisively, the altitude feeds  $v_z$ , the segmentation’s discriminant. A long altitude gap interpolated linearly returns one average slope, which erases any climb–glide transition inside it—and transitions are exactly where  $v_z$  swings fastest: a paraglider’s typical glide sink is  $1\text{--}1.5 \text{ m s}^{-1}$  and its climb in a working thermal  $2\text{--}4 \text{ m s}^{-1}$  (hang gliders sink faster, at  $1.5\text{--}2 \text{ m s}^{-1}$ , and climb over the same range); both sit well inside the bulk of the  $|v_z|$  distribution of Fig. 2.2b, far from the tail where the absolute bound is placed, so a thermal entry swings  $v_z$  by several  $\text{m s}^{-1}$  within a couple of seconds, the worst case for interpolation. A real asymmetry between the channels does exist, but it concerns *isolated* missing samples, not gap width: a single missing altitude is reconstructed far more reliably than a single missing position, because the barometric signal is much smoother—that is what justified “drop the altitude only” at cleaning (Sec. 2.7.2); it does not justify a wider gap bound for  $z$ .

**Consequences of a split.** The observables of Ch. 3 are affected differently by a split. In practice a split is bookkeeping rather than a new trajectory: the conversion to ENU (step v) runs before this step, so every fix already carries coordinates and elapsed time referred to the parent origin, and the split only assigns a segment identifier. **The key fact is that a segment keeps its parent’s origin and clock** (Sec. 2.7.8): a GPS position is absolute, so the positions after the gap remain exactly as valid as those before it. Ensemble quantities (the MSD, the propagator) need the horizontal position at a given elapsed time, not the path travelled in between, so a split costs them nothing—the flight simply contributes no data at the lags that fall inside the gap. Quantities that instead *accumulate along the path*—the path length, a phase duration, a step length, any time-averaged window—cannot be evaluated across a stretch whose true course is unknown: sums and windows for these quantities are always taken inside one segment, never straddling a gap. Three cautions follow.

- *Censoring at the boundary.* A phase in progress when the gap opens is truncated: its duration is a lower bound, not a measurement. Boundary-truncated durations are flagged as censored and simply *excluded*—from the  $\psi(\tau)$  fits and from the phase-duration histograms alike—rather than counted as complete, which would let every split masquerade as a spuriously short wait. The censored phases are identified, then, exactly as a survival analysis would identify them; what is not done is to feed them back in through a censoring-aware estimator—a Kaplan–Meier survival curve, or a likelihood with a right-censored term, which would use the information that the true duration

exceeds the observed one instead of discarding the observation. As long as the censored fraction stays small (it is recorded per discipline), plain exclusion keeps the estimators transparent at negligible cost, and the recorded fraction tells us if that assumption ever fails—if it does, the machinery to switch is the standard one just named.

- *A segment is not a fresh flight.* For observables referenced to the time since the flight began—the time-averaged MSD stratified by duration, for instance (Ch. 3)—a segment starting at parent time  $t_0 > 0$  is not re-zeroed into a new flight that starts at  $t = 0$ . It is used for what it is: an observation window  $[t_0, t_1]$  onto a process already  $t_0$  old. Re-zeroing would mix statistics of young and aged trajectories, which an aging analysis must keep apart.
- *No second pass of the flight-level cuts.* The duration and path cuts of Sec. 2.7.4 were passed by the parent flight and are not re-applied to its segments. Re-applying them would be biased in a specific way: gaps concentrate in particular conditions (weak reception late in long flights, for example), so discarding every sub-40 min segment would throw away precisely the data recorded around gaps. A segment has to satisfy only what the downstream machinery physically requires—enough duration to support the smoothing window and the statistics computed within it. That minimum has a value and a place in the pipeline, and it belongs here rather than only in the configuration: a segment shorter than  $T_{\text{seg}} = 90$  s (Table 2.5) cannot carry a smoothing window of  $\tau_c = 5$  s with enough points on either side of it, and is dropped alone; the rest of the flight is retained.

**Interpolation across short gaps.** The scope of this step is set by the split bound: a gap longer than  $g_{\text{max}}$  splits the flight and is never interpolated, so what is filled here are only the *short* holes, up to  $g_{\text{max}}$ , that remain inside a segment. The fill is done per channel, where a channel is one of the three scalar series on the uniform grid: the horizontal coordinates  $E(t)$  and  $N(t)$  and the adopted altitude  $z(t)$ .

The sentence above is true of the *time base* and not of a channel, and the two come apart in exactly one place.  $g_{\text{max}}$  bounds the interval between two consecutive *fixes*; it says nothing about an interval over which a fix exists but carries no value. That case is not hypothetical and is not rare: the vertical channel can be absent while the horizontal record is perfect—a logger that writes no barometric value, or a value the fix-level cleaning removed (Sec. 2.7.2, where the whole point of the asymmetry is that such a fix is *kept*). No time gap opens there, so no split is declared, and the linear fill spans the hole whatever its length.

Two things follow, and both are decided here rather than left implicit. First, such a hole does *not* split the flight. The horizontal trajectory, which is what Chapter 3 measures, is intact across it; cutting it because the vertical is missing would discard good data for a defect that does not touch it. Second—and this is the correction the check demanded—the reconstruction must be visible. It was not: a grid point sitting on a real fix whose altitude had to be invented was reported as *measured*, because the *interpolated* flag is a statement about the time base alone. On a test flight missing 500 s of altitude the filled  $z$  departed from the truth by up to 540 m, the smoothing differentiated the straight line into a vertical velocity that was pure invention, and the record read `frac_interpolated = 0` with `was_resampled` false—the flight declaring itself uniform as recorded. The vertical therefore carries its own per-fix flag, `z_reconstructed`, with the per-segment share `frac_z_reconstructed` and, per flight, the longest unbroken run `z_gap_max_s`. The run length is reported next to the fraction because a fraction cannot separate the two cases that matter: a hundred holes of two seconds, which a straight line bridges well, from one hole of twelve minutes, which it does not. Any analysis of the vertical—the climb/sink discriminant of the segmentation above all—excludes on that flag; the horizontal analyses are untouched by it, and that is the reason the flag is per channel

rather than one for all three. The two horizontal coordinates are interpolated independently, which is the componentwise interpolation of the horizontal position; the barometric and GNSS altitudes are never mixed here—only the flight’s adopted channel is filled (Sec. 2.7.1).

The horizontal coordinates, which the transport analysis measures directly, are interpolated with a *monotone piecewise cubic* (a shape-preserving Hermite interpolant): one cubic per interval, whose slopes are chosen so that the interpolant reproduces the monotonicity of the samples themselves—where the data increase, the curve increases, and it does not turn back on itself between two points. The construction needs no derivative data: the slope at each sample is estimated from the neighbouring samples and then limited to whatever value keeps the monotonicity condition satisfied, which is what distinguishes it from an unconstrained cubic spline and is the whole reason for using it here. A spline free to choose its slopes can swing outside the range of the points it passes through, and such an excursion would be indistinguishable, to the smoothing step, from a real manoeuvre. The altitude is interpolated linearly<sup>5</sup>. The linear error over a hole of width  $g$  is bounded by  $\max |\ddot{z}| g^2/8$ ; holes are bridged inside a phase far more often than across a transition between phases, and within a phase the barometric signal has a nearly constant slope—a steady climb or a steady glide—so  $|\ddot{z}|$  is small there (the low high-frequency content of the barometric channel, Fig. 2.1c, is the same statement in the frequency domain). So the error is at the metre scale even at the 20 s cap and far smaller for typical holes—transitions, where  $\ddot{z}$  is large, are handled by the flag below.

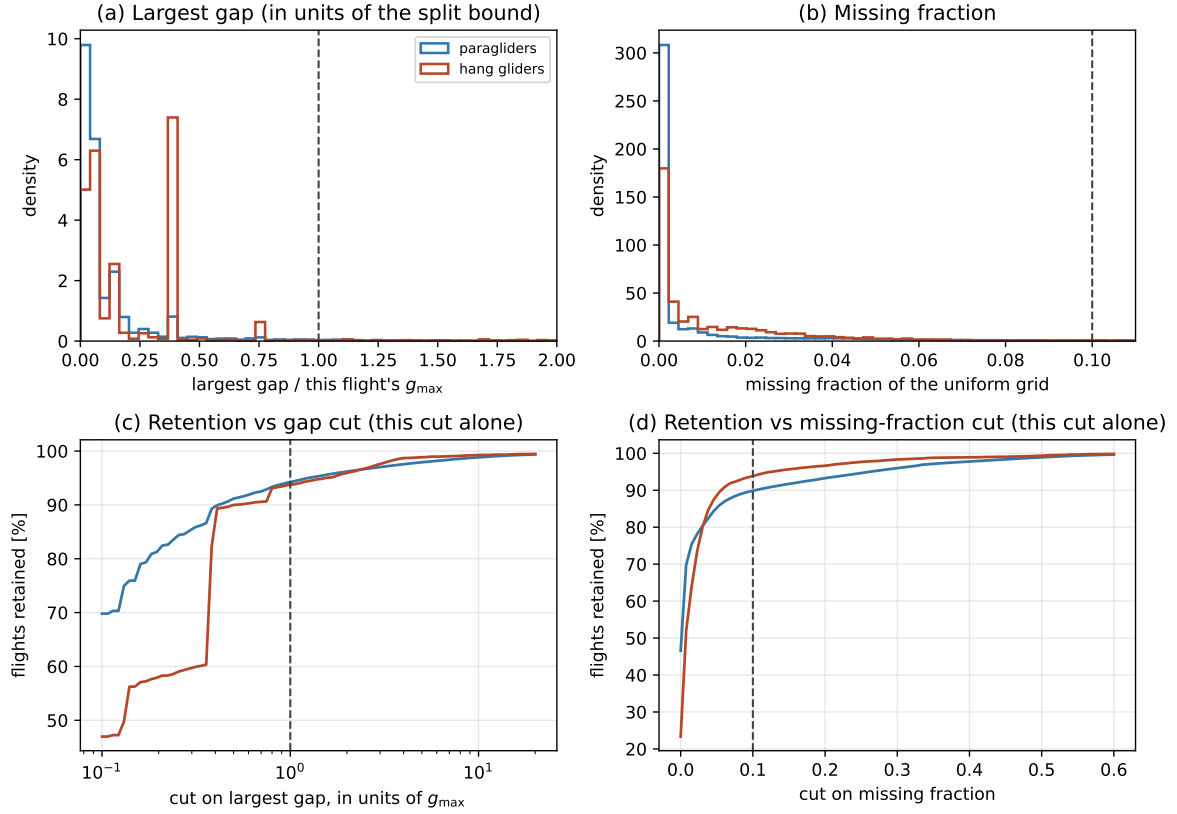
The altitude is interpolated linearly and the horizontal coordinates by a monotone cubic, and the asymmetry follows from what each channel is used for. The altitude is differentiated once more than the horizontal coordinates before it is read— $v_z$  is the segmentation’s discriminant—so an interpolation artefact in  $z$  propagates into a phase assignment, whereas the same artefact in  $E$  or  $N$  is absorbed by the smoothing. Linear interpolation cannot overshoot, and its error over a hole of width  $g$  is bounded by  $\max |\ddot{z}| g^2/8$ ; a cubic is more accurate where the second derivative is smooth and worse exactly where it is not, which is at the transitions a phase assignment turns on.

The result is precisely the completeness invariant promised at cleaning (Sec. 2.7.2): *within a retained segment, every grid time carries a defined  $(E, N, z)$* , so the segmentation, and the analysis more generally, is never handed a missing value. The altitude values dropped at cleaning (a barometric spike, a V fix on a GNSS-fallback flight) are the ones restored here. Every filled point carries a flag—*interpolated* where the time base had no fix, *z\_reconstructed where a fix was there but its altitude was not*—with a concrete consequence: downstream statistics can distinguish measured from reconstructed samples, and a phase boundary that the segmentation places inside a bridged stretch is treated like one at a segment boundary—the true transition could lie anywhere inside the hole, so the adjoining phase durations count as censored and are excluded from the duration fits, exactly as at a split.

The gap bound  $g_{\max}$  is the criterion called “sampling regularity” in the flight-level filtering (Sec. 2.7.4); Figure 2.6 makes the fraction of flights it leaves whole auditable. One caveat carries downstream: a few observables are intrinsically  $\Delta t$ -dependent, the turning angle above all. The dependence is in the observable’s own definition: a turning angle is the heading change over one step, so two loggers recording the *same* motion at 1 s and 10 s produce systematically different distributions of it. The velocity is a partial exception, because it is not formed as a finite difference here but read off the smoothing polynomial (Sec. 2.7.7); what remains  $\Delta t$ -dependent in it is the effective width of that polynomial’s window, which is set in seconds and therefore spans a different number of samples at each cadence. Pooling all flights would blend this instrument artefact into the physics; where such observables are compared across

<sup>5</sup>The bound follows from the standard error formula for linear interpolation. On an interval of width  $g$  the difference between a  $C^2$  function  $z$  and the chord through its endpoints is  $z(t) - p(t) = \frac{1}{2} \ddot{z}(\xi)(t - t_0)(t - t_0 - g)$  for some  $\xi$  in the interval; the quadratic factor is largest at the midpoint, where it equals  $g^2/4$ , giving  $|z - p| \leq \max |\ddot{z}| g^2/8$ .

flights, the ensemble is therefore stratified by rate or restricted to the dominant one.

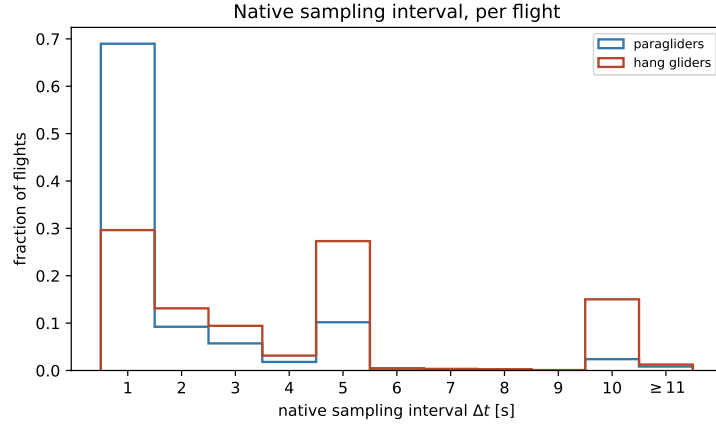


**Figure 2.6.** The two sampling-regularity criteria and their retention curves, per discipline. Computed on the same full track scan as Figure 2.3—every parsed flight, not only those passing the flight-level cuts, because a cut must be audited on the population it acts on, before its own effect hides part of the picture—what would hide it is the flight-level cleaning applied first, not this cut. (a)/(b) Distributions of the two tested quantities: the largest single gap of the flight and the missing fraction of a uniform grid at its native interval. The gap is plotted in units of *that flight's own*  $g_{\max}$ , not of  $\Delta t$ , so the criterion actually in force sits at 1 (dashed) for every flight; because  $g_{\max}$  is the two-scale bound and not a fixed multiple of  $\Delta t$ , a plot against  $\Delta t$  alone would draw a cut that binds at no cadence but 1 s. The distribution is discrete below the cut, since a gap is usually a run of wholly missed fixes and hence an exact multiple of the native interval. (c)/(d) The fraction of flights each cut alone would retain as the threshold is moved: *marginal* curves, i.e. each cut applied by itself to the full population, not after the other cuts. Both panels are recomputed once the fix-level and flight-level stages run in full; the census behind them predates that pipeline. (implementation details: Sec. 2.7.6)

### 2.7.7 Smoothing and differentiation

Velocity and acceleration drive the kinematics and the segmentation, but differencing raw GPS [coordinate series](#) amplifies the high-frequency noise characterised in Sec. 2.7.1. A **Savitzky–Golay** filter [19] solves both problems at once: it fits a low-order polynomial to a sliding window by least squares and reads off that polynomial, or its derivatives, at the window centre, returning smoothed position, velocity and acceleration in one pass per component.<sup>6</sup> The fit is done in units of samples, so its coefficients come out per sample; the velocity and acceleration

<sup>6</sup>Concretely: in the window of  $w$  samples centred on  $t_i$ , the coefficients of the polynomial  $P(t) = \sum_{k=0}^p a_k (t - t_i)^k$  minimize  $\sum_j (x_j - P(t_j))^2$ ; the smoothed value is  $a_0$ , the velocity  $a_1$ , the acceleration  $2a_2$ . Because the grid is uniform, the  $a_k$  are fixed linear combinations of the window samples, so the whole filter is a convolution with precomputed coefficients [19].



**Figure 2.7.** Native sampling interval  $\Delta t$  (median inter-fix time) per flight, on the full dataset per discipline.  $\Delta t$  is in practice discrete – it lands on an exact whole second for nearly every flight (Sec. 2.7.6) – so bins are one second wide, centred on each integer; each bar is the fraction of flights at exactly that rate. Paragliders concentrate at 1 s; hang gliders spread across several rates, with a thin tail beyond 11 s (last bin, aggregated). A thin tail is present in both disciplines; what differs is the bulk, with the 10 s rate far commoner among hang gliders than among paragliders. Each flight keeps its own  $\Delta t$  (Sec. 2.7.7). (implementation details: Sec. 2.7.6)

in physical units are  $a_1/\Delta t$  and  $2a_2/\Delta t^2$ , with the flight’s own  $\Delta t$  (Sec. 2.7.6). This is the only place the cadence enters, and it is why flights of different cadences need no common grid.

The window never crosses a segment boundary. Near a boundary, where a centred window no longer fits, the polynomial is fitted on the nearest full window inside the segment and evaluated off-centre (the standard boundary treatment of the filter). An off-centre evaluation is an extrapolation of the fitted polynomial rather than an interpolation, and the variance of a least-squares polynomial grows towards the ends of its fitting interval, so the first and last few samples of a segment carry a larger uncertainty than the interior ones. They are marked as such—the same per-sample flag mechanism as the interpolated points—so that an observable sensitive to it (an acceleration extreme, say) can be computed on interior samples only. Whether the exclusion is worth making is an empirical question: with windows of five to a few tens of samples and segments of at least 90 s, the affected samples are a small fraction of a segment, and the check is simply to recompute the observable with and without them.

The filter runs on the resampled grid of Sec. 2.7.6, so it sees a uniform time base with the short holes already filled. The filled points carry no independent information—each is a smooth function of its neighbours—so within a window they mildly correlate the samples; over holes this short that introduces no bias, and their fraction is recorded per flight. Flights that needed a great deal of reconstruction never reach this step: the interpolated share of a segment is capped at 10 % at resampling (Sec. 2.7.6) and the airborne share a flight may lose to cleaning at 10 % at the integrity gate (Sec. 2.7.2); what is *recorded* here, and not merely thresholded, is the per-flight value of that share, so a downstream statistic can be stratified on it.

The filter has two hyperparameters, the window length  $w$  (odd, in samples) and the polynomial order  $p$ , fixed from the measured noise spectrum rather than by eye:

- (i) From the ensemble Welch power spectrum of the raw ENU components on a sample of flights (the diagnostic of Sec. 2.7.1 applied to  $E$ ,  $N$ ,  $U$ ), read the frequency  $f_c$  of the knee where the dynamical signal meets the flat GPS noise floor. Its reciprocal is a smoothing timescale  $\tau_c = 1/f_c$ : dynamics slower than  $\tau_c$  are signal, faster fluctuations are noise. Then set the window per flight from that timescale. Making  $w$  span  $\tau_c$  is a *choice*—the natural one, since it puts the filter’s cut-off at the measured knee—not



a result:  $w$  is  $\tau_c/\Delta t$  rounded up to the nearest odd integer (odd, so that the window has a centre sample to evaluate at), with a floor at  $w = 5$ , the smallest window on which a cubic fit is meaningfully determined. At the dominant 1 s cadence the two rules coincide:  $w = 5$  samples spans exactly  $\tau_c$ , and the smoothing acts at the measured noise scale. At every slower cadence the floor is the binding rule (already at  $\Delta t = 2$  s, where  $\tau_c/\Delta t < 5$ ), and the five-sample window then spans *more* than  $\tau_c$ —about 10 s for a 2 s logger, 25 s for a 5 s one—so slow loggers are smoothed somewhat more than the noise scale requires. The cost is accepted: at those cadences the noise band sits at or beyond the Nyquist frequency  $1/(2\Delta t)$ , so there is little to remove at the noise scale in the first place, and these flights are in any case excluded from the  $\Delta t$ -sensitive observables by the rate stratification (Sec. 2.7.6).

- (ii) Fix  $p$  as the lowest order that works. An acceleration requires curvature in the fit, so  $p \geq 2$ . One order more is needed for the sake of the *velocity*: on a symmetric window, adding an even-order term leaves the odd-order coefficients untouched (a parity property of least squares on a symmetric grid), so the velocity of a  $p = 2$  fit is identical to that of a straight-line fit. What  $p = 3$  adds is the cubic coefficient  $a_3$ , which is proportional to the third derivative—the rate of change of the acceleration—and which shifts  $a_1$ : the velocity of a cubic fit is the first one that responds to the acceleration *varying* across the window, rather than assuming it constant. Going higher only lets the fit follow more noise. Hence  $p = 3$ .
- (iii) Treat the vertical separately from the horizontal, and condition it on the altitude *source*. The a priori expectation was asymmetric: the smooth barometric channel would need a *shorter* window than the horizontal coordinates, and the noisy GNSS vertical a *longer* one than them. The measurement said otherwise: the three knees coincide at  $f_c \approx 0.2$  Hz ( $\tau_c = 5$  s). This is a measured value and it carries the status of one: it comes from the per-channel Welch spectra of Sec. 2.7.7, read on a sample of the *present* archive, and the plot that shows the three knees falling together is the evidence for it. Two qualifications belong with the number. It is read on a sample, not on one flight, because a single spectrum is too noisy to locate a knee; and it is read on data that have not yet passed the fix-level and flight-level stages, so it is provisional—the spectra are recomputed, and  $\tau_c$  re-read, once the cleaned ensemble exists. The reason is trivial: the flat high-frequency floor of every channel comes from the rounding the IGC format applies when the file is written—altitudes to whole metres, coordinates to thousandths of an arc-minute—rather than from receiver noise, and rounding noise flattens the spectrum at the same frequency whatever channel it sits on. What distinguishes the noisy GNSS minority is how *high* its floor sits, not *where* the knee is (Sec. 2.7.7). So the answer to whether  $E$ ,  $N$  and  $z$  get different smoothing timescales is currently no: the per-channel machinery exists, for the noisy GNSS minority, but the adopted windows are equal.

The choice is then checked on a held-out sample—flights not used to place  $f_c$ —with three acceptance criteria: the residual spectrum (raw minus smoothed) must be flat above  $f_c$ , showing that what was removed is the noise band and nothing below it; the smoothed kinematics must stay within physical bounds—the acceleration first, since it is the highest derivative and therefore the most sensitive to the window, but the speed and the climb rate on the same footing, against the same envelopes the fix-level bounds use (Table 2.5); and the results must be stable when  $w$  is changed by one step (implementation details: Sec. 2.7.7).

**A fourth criterion, on the observable itself.** The three above are properties of the filter. A fourth is a property of what the filter is for: the smoothing must not remove the displacement the transport analysis is about to measure. The concern is not visible in  $\tau_c$  alone,

because  $\tau_c$  is not what sets the window for most of the archive. The window is floored at 5 samples—one more than a cubic needs, so that it stays a least-squares fit rather than an interpolation—and that floor binds for every logger slower than 1 Hz. In *seconds* the window is therefore  $\max(\tau_c, 5 \Delta t)$ : 5 s at a 1 s cadence, 50 s at 10 s, 150 s at 30 s—which reaches the lower end of the range over which Chapter 3 fits its exponent.

Measured rather than argued, on synthetic trajectories of known exponent passed through the adopted filter and read the way the pipeline reads them—displacement from the frame origin, which is the *raw* first fix (Sec. 2.7.5), not the smoothed one—the answer is that the filter is transparent to the motion. For a superdiffusive trajectory ( $\alpha = 1.8$ , the regime Chapter 3 finds) and no measurement noise, the filtered MSD sits within 2.3 % of the true one at a lag of two samples and within 1 % at every lag from three samples upwards. The reason is structural rather than fortunate: a cubic Savitzky–Golay filter reproduces any locally cubic trajectory *exactly*, and a strongly correlated trajectory is locally close to cubic, so it passes through untouched.

What does move the short-lag MSD is the opposite of smoothing, and in the opposite direction. Independent measurement noise of standard deviation  $\sigma$  per component adds  $\approx 2\sigma^2$  to the MSD at every lag; where the true displacement is comparable to that, the measured curve is a noise floor rather than motion. With  $\sigma = 3$  m the same experiment gives a measured-to-true ratio of 8.1 at one sample, 1.26 at five, 1.05 at twelve and 1.003 at sixty-four—and the smoothing is what pulls it down, the one-sample ratio falling from 8.1 to 3.6 when the window goes from 5 samples to 11. So the filter does not erase the observable, it de-biases it; the floor it leaves is set by  $2\sigma^2 \approx 18 \text{ m}^2$ , which the measured MSD exceeds by orders of magnitude everywhere in the fitted range. The consequence for reading Fig. 3.1 is that its first decade is instrument, not transport, and the fit does not begin there. A second effect points the same way and belongs beside it: a flight answers a lag with the position at *its own* nearest fix, so the curve’s tread at short lags is that flight’s  $\Delta t$  rather than the 1 s the lag grid is snapped to. For every logger slower than 1 Hz the short-lag curve is a staircase, and the local slope of a staircase alternates between zero and a spike. Both effects die out well below the lower end of the fit range, which is why they are a caveat on reading panels (a) and (c) at short lag rather than on the exponent.

### 2.7.8 Notation

Throughout,  $\mathbf{r}(t) = (x(t), y(t)) = (E(t), N(t))$  denotes the *horizontal* position in this local frame and  $z(t)$  the adopted altitude channel (Sec. 2.7.1), kept at its measured value (Sec. 2.7.5). By construction each trajectory starts at the horizontal origin,  $\mathbf{r}(0) = \mathbf{0}$ , with the clock reset to  $t = 0$  at the start of free flight (the first fix of the trimmed track, Sec. 2.7.3), so  $t$  is the elapsed flight time (a segment created by a split at a long gap, Sec. 2.7.6, keeps its parent flight’s origin and clock). The vertical is *not* re-zeroed: the measured height at the start of free flight is retained (Sec. 2.7.5). The horizontal velocity is  $\mathbf{v}(t) = \dot{\mathbf{r}}(t)$ , with magnitude the horizontal speed  $v_{xy} = |\mathbf{v}|$ ; the vertical velocity is  $v_z = \dot{z}$ . The horizontal displacement over a lag  $t$  is  $\Delta \mathbf{r}(t) = \mathbf{r}(t_0 + t) - \mathbf{r}(t_0)$ . A terminological rule holds throughout the document: when a statement concerns  $\mathbf{r}$  it says *horizontal position* explicitly, and when it concerns the full three-dimensional point it says *3D position*; a bare “position” is avoided.

## 2.8 Preliminary characterization

This section concerns the dataset that the analysis actually consumes: what the whole of Sec. 2.7 leaves behind. Three of its stages discard: the fix-level cleaning, through the integrity gate; the flight-level filtering, where a flight is lost outright against a stated criterion; and the resampling, which is easy to overlook because it is a *re-expression* of a trajectory rather



than a cut, and which is nonetheless responsible for the single largest loss in the cascade—a flight whose every segment fails the sparsity or the minimum-duration rule leaves nothing behind (Table 2.7). The conversion and the smoothing drop nothing. Its point is to establish the basic statistics on that retained ensemble—which is why, for instance, the native-rate distribution reappears here: Figs. 2.6 and 2.7 were computed on *every* parsed flight, as a filtering diagnostic, whereas the analysis needs the same quantities on the flights actually used. None of the checks below requires the phase segmentation, so all of them run as soon as the pipeline does; two of them constrain what the transport analysis of Chapter 3 is allowed to do, and both come out against the simpler option.

**Trajectory statistics.** For each retained flight: airborne duration  $T$ , total horizontal path length, number of valid fixes and native sampling interval  $\Delta t$ . Their distributions (Fig. 2.8) describe the dataset the results rest on, expose any outlier that survived the filtering, and, through  $\Delta t$ , fix the per-flight smoothing window (Sec. 2.7.7). Over the 155,788 retained paraglider flights the median airborne duration is 2.63 h (interquartile range 1.78 h to 3.93 h), the median flown path 87 km (57 km to 134 km) and the median record 7,129 fixes; 69.8 % of them log at 1 Hz. The 6,132 hang-glider flights are longer and faster—3.07 h and 151 km in the median—and slower-logging, with a median  $\Delta t$  of 3 s against 1 s and only 27.6 % at 1 Hz.

These are not the raw-archive figures of Sec. 2.5: the retained median duration exceeds the recorded median 2.6 h because the duration floor and the altitude-activity floor remove short and inactive records, and the retained median path is close to the recorded 86 km because the 20 km path floor bites on few flights (Table 2.7). Nothing in the retained ensemble is an outlier in the sense the filtering was meant to catch: the table carries 0 non-finite coordinates and 0 non-monotone clocks, and the fastest step between two *measured* samples is  $49 \text{ m s}^{-1}$  against the fix-level bound of  $45 \text{ m s}^{-1}$ —an excess of under a tenth, on 32 steps out of  $1.4 \times 10^9$ , which Sec. 3.1 accounts for.

**What the cascade costs, and whether it costs it evenly.** Table 2.7 reads the discards as a cascade, and one criterion dominates: 21,089 paraglider flights, 11.9 % of those still standing, leave nothing behind after resampling. That is four times the next largest cut, it is a rule about record quality rather than about flight, and it is not a threshold Sec. 2.7 argues for at all — so it deserves a check the others do not.

It fails that check in one respect and passes it in another. Across native cadences the sparsity rule is markedly uneven: it removes 0.1 % of the flights at its most forgiving cadence and 22.6 % at 2 s, so the retained ensemble is enriched in the loggers whose grids are easiest to fill. Across time it is even: over the 20 seasons carrying at least a thousand flights the retention rate stays between 70.3 % and 91.6 % (Fig. 2.10b), so two decades of changing logger technology have not left the pipeline eating one era of the archive. The seasons below that size are the ladder’s first years, when it carried tens of flights, and their rates are noise rather than trend.

**Where the flights are.** The take-off sites need three frames rather than one, because the CFD is a French ladder flown partly outside France. Figure 2.9 gives them, on coastlines and borders rather than on bare axes: metropolitan France, La Réunion, and the rest of the world. Of the retained paraglider flights 94.7 % launch in metropolitan France, 3.2 % (4,939 flights) on La Réunion, and 2.2 % (3,359) elsewhere—the Canaries, Andalusia, the Italian Alps, and further out Colombia, Nepal, India, South Africa and Brazil. The hang gliders are more domestic still, at 96.7 %. The overseas and foreign flights are kept: they are the same discipline flown under the same rules, and the geographic stratification below is where a difference between terrains would show up.

Within metropolitan France the ensemble does not sample the country evenly. It occupies 5,332 distinct take-off sites at a hundredth of a degree, and 58.8% of it launches inside the Alpine box alone, against 7.8% in the Pyrenees, 8.2% in the Massif Central and 19.8% elsewhere in France. The density is drawn on a  $0.15^\circ$  cell with the count on a logarithmic colour scale, since at 155,788 flights individual markers saturate over the Alps and stop answering the question the map is drawn for; La Réunion, seventy kilometres across, gets a cell fifteen times finer. The orographic groups are longitude and latitude boxes, drawn on Fig. 2.9a and listed with their edges in the mirror of this section (implementation details: Sec. 2.8); they are used below to define strata, and the map itself is not the test.

**Isotropy, and what it costs.** The transport analysis would like to study the one-dimensional marginal of the propagator in place of the full two-dimensional density (Sec. 3.1), a substantial simplification. The reduction is legitimate only if the process is isotropic, and isotropy cannot be assumed here. The CFD flights are not races flown together—each is a solo cross-country flight, later entered into a scoring ladder—but a single flight is strongly directed: a chosen route, the orography, the valley winds. Isotropy can therefore hold at best at the level of the *ensemble*, and the check is run there.

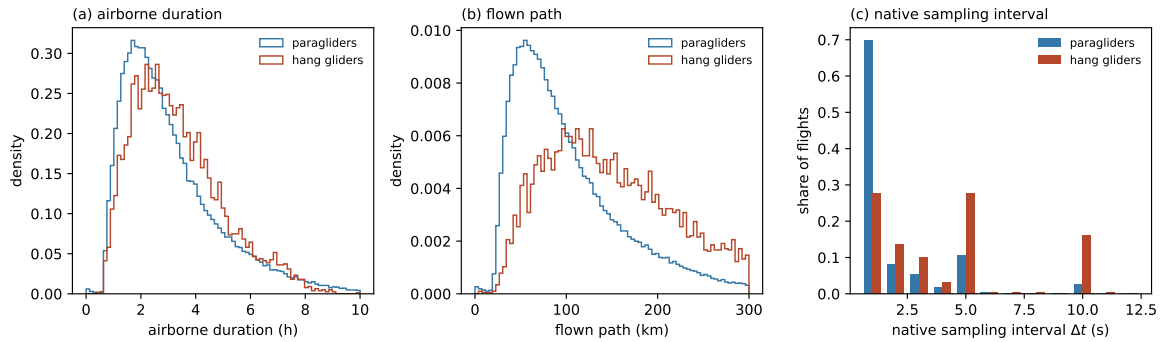
The primary test is the per-component variance ratio: for an isotropic process  $\langle E(t)^2 \rangle = \langle N(t)^2 \rangle = \text{MSD}(t)/2$  at every  $t$ , so the ratio  $\langle E(t)^2 \rangle / \langle N(t)^2 \rangle$  should be identically 1. Figure 2.11a shows it is not. Over the lags the transport analysis reads it sits at 0.80 for paragliders, between 0.70 and 0.86 and never touching unity: the north–south mean square exceeds the east–west by about a quarter, steadily and across three decades. The hang gliders are anisotropic the other way, at 1.19. A Kolmogorov–Smirnov comparison of the empirical distributions of  $E(t)$  and  $N(t)$  at 4 lags agrees, with distances from 0.038 to 0.125; at this sample size the associated  $p$ -values are numerically zero and carry no information, which is why the distance is quoted and the test is not.

The consequence is operative. The two marginals  $P(E(t) = x)$  and  $P(N(t) = x)$  are analysed *separately* rather than pooled, and the pooled 1-D marginal is not used as a stand-in for the 2-D propagator. That is the natural fallback and is informative in itself: unequal scaling exponents  $H_E \neq H_N$ , or equal exponents with different scaling functions, signal geometrically anisotropic transport, which for flights channelled along Alpine valleys is the expected outcome rather than a nuisance. The anisotropy is also a reason to read the ensemble MSD with the care Sec. 3.1 gives it, since  $\langle |\mathbf{r}|^2 \rangle$  sums two components that do not grow alike.

**Compatibility across strata.** The dataset spans more than two decades, two glider types and a range of performance classes (Sec. 2.4), from recreational EN A/B wings to CCC racing wings, whose kinematics differ substantially. Before treating the flights of one discipline as draws from a single statistical process, the subpopulations are checked. The test is the horizontal MSD computed within each stratum and read against the pooled curve over the same lags; the two disciplines are never pooled, and a stratum below 2,000 flights is not drawn as if it were a measurement. The class normalisation of Sec. 2.4 runs first, since as long as one wing class appears under several spellings a stratum is not a well-defined set of flights.

Figure 2.11b–d gives three axes and two answers. Across the 6 wing classes the curves separate: the typical stratum departs from the pooled curve by 40% and the extreme by 90%, with the CIVL competition wings above the pooled curve throughout and the EN A wings well below it. Across the 5 orographic groups they separate further, by 62% typically and 130% at the extreme, with the Massif Central at half the pooled curve over the intermediate lags and the Alps above it. Across the 17 seasons they do not: the typical departure is 12% and the largest 21%, and the latter belongs to the smallest season in the archive.

So the ensemble may be pooled over time and may not be pooled over wing class or over terrain. Season is not a relevant stratum and the archive’s two-decade span is not a confound. Class and geography are relevant strata, and the difference is itself a result rather than a nuisance: a competition wing covers ground faster than a recreational one, and thermals over a mountain range are anchored to slopes and organised along ridges whereas over flat terrain they drift with the wind, so the persistence of the glides and the spacing of the climbs need not be the same. The exponent of Chapter 3 is therefore reported by wing class as well as pooled, alongside cuts by native cadence, season and declared task (Table 3.2); the orographic cut is made here, on the ensemble MSD, which is the statistic it separates. A pooled exponent is read as an average over strata that are known to differ, not as a property of a homogeneous population.



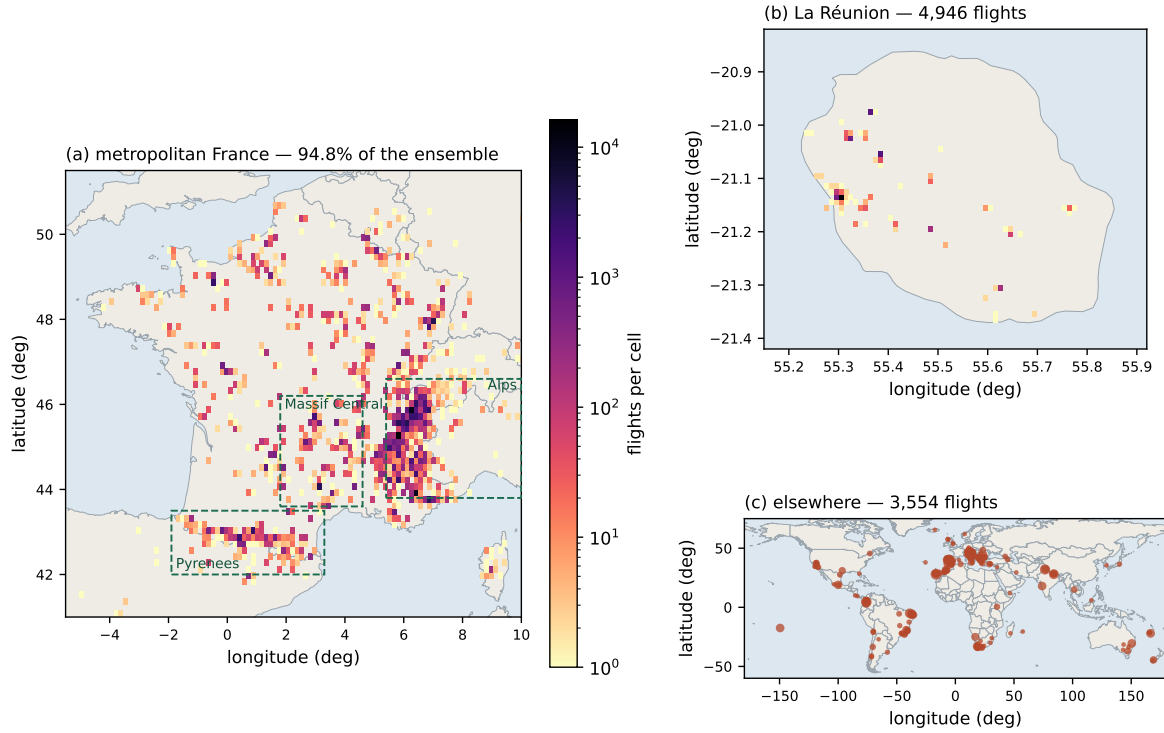
**Figure 2.8.** The records the pipeline retains. (a) Airborne duration, (b) flown path and (c) native sampling interval  $\Delta t$ , per discipline, over the retained flights rather than over every parsed flight—which is what distinguishes these from Figs. 2.6 and 2.7. Computed by `scripts/reporting/generate_prelim_figure.py`.

## 2.9 Pipeline summary

Figure 2.12 closes the chapter with the whole processing chain in one map: every step the data undergo between the raw IGC tracks of Sec. 2.2 and the ensemble handed to the transport analysis, in the order the pipeline applies them. Table 2.8 pairs each stage with the level it acts at, its effect, and the adopted parameters; its last column leads back to the section where each choice is argued.

Two decisions organise the chain. First the order: every check that runs on the scalar record as logged — times, geodetic coordinates, altitude — comes first (stages 1–4), and the vector geometry is built only from what survives (stages 5–7); a coordinate conversion neither repairs a corrupt record nor has to be undone when one is dropped (Sec. 2.7). Second the level: each stage acts per fix, per flight or per segment, and never on more than one flight at a time. The point is not that pooling flights would be tempting—it would not—but that each stage’s parameters are set from the flight it is acting on. The smoothing window comes from that flight’s own  $\Delta t$ , the outlier scale from its own local MAD, the resampling grid from its own cadence; none is an archive-wide constant. What is shared across flights is only the thresholds of Table 2.5, and those are fixed a priori, not estimated from the ensemble—so no flight’s cleaning can depend on which other flights happen to be in the dataset.

The parameters in the table are the adopted operating point of `configs/preprocessing.yaml`, typeset through macros that `generate_census_stats.py` regenerates from that same file (implementation details: Table 1.1), so the summary follows the configuration in force.

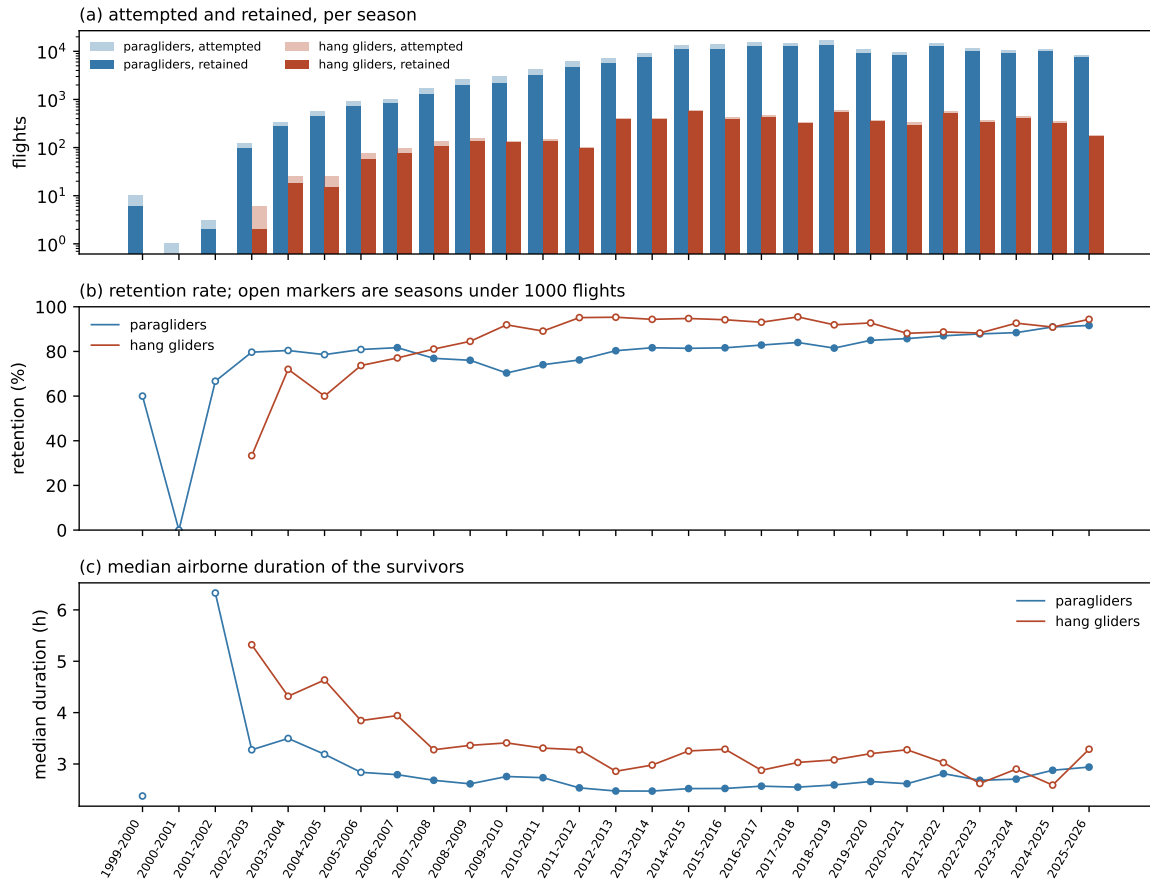


**Figure 2.9.** Take-off sites of the retained ensemble, on three frames. (a) Metropolitan France, density on a  $0.15^\circ$  cell and a logarithmic colour scale; the dashed boxes are the orographic groups used as strata in Fig. 2.11. (b) La Réunion, on a  $0.01^\circ$  cell, the one overseas department with a substantial share. (c) Everything else, one marker per  $0.5^\circ$  cell with the area set by its count. Coastlines and borders are Natural Earth admin-0, cropped and committed by `scripts/reporting/build_basemap.py` so that the figure regenerates offline (implementation details: Sec. 2.8).

**What the chain retains.** Run over the whole archive, the cascade retains 155,788 of 186,052 paraglider flights (83.7%) and 6,132 of 6,716 hang-glider flights (91.3%). Table 2.7 says where the rest went. The numbers are a *cascade*, not a set of marginal effects: a flight counted against one criterion was still standing when that criterion tested it, so the table reads in pipeline order and the counts do not add up to what each cut would remove on its own (those are the `\StatScan*` numbers quoted in Secs. 2.7.4 and 2.7.6, measured on the raw population).

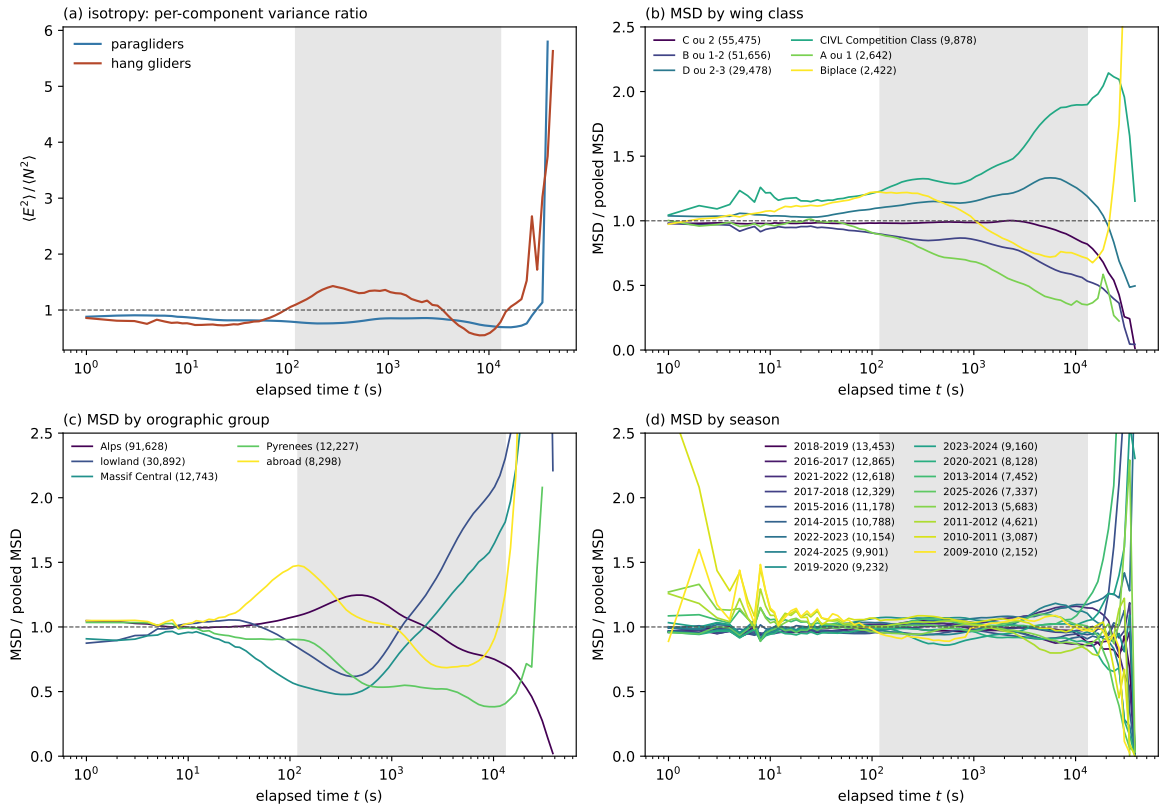
Inside the retained flights the cleaning is light, which is what the design asked for: it deletes 1.84% of paraglider fixes and 0.75% of hang-glider ones, and invalidates an altitude on a further 0.016% and 0.009%. The local outlier test flags far more than is deleted, and by design: it never gates, so 0.68% and 0.97% of fixes are flagged and *kept*. The flag is a per-fix record and a per-flight anomaly count, and what condemns a fix is the impossibility gate (Sec. 2.7.2). Resampling reconstructs 0.58% and 0.32% of grid points. A split is common but not the rule: 6.9% of retained paraglider flights and 16.3% of hang-glider ones carry at least one, the difference following the cadences of Fig. 2.7 as Sec. 2.7.6 anticipated.

**The analysis ensemble, defined.** Everything downstream reads one object, and this is it. The *analysis ensemble* is the set of trajectories written to `fixes.parquet`: one row per grid point of every *retained* segment of every *retained* flight, carrying the elapsed time  $t$  in the parent clock and the local Cartesian coordinates  $(E, N, z)$  with their first two derivatives. A flight is retained when it passes stages 1–4 of Table 2.8 and at least one of its segments survives stage 6; a segment is retained when it is long enough and dense enough on its own uniform grid. That leaves 155,788 paraglider flights of 186,052 (83.7%) in 179,395 segments, and 6,132



**Figure 2.10.** The archive season by season. (a) Flights attempted and retained, on a logarithmic scale: the ladder grows by three orders of magnitude over its life. (b) Retention rate and (c) median airborne duration of the survivors; open markers are seasons of fewer than 1,000 flights, whose rates are noise. Retention is flat across the substantial seasons, which is what says the cascade does not discard one era of the archive preferentially. Computed by `scripts/reporting/generate_dataset_stats.py`.

hang-glider flights of 6,716 (91.3%) in 6,951. No consumer re-applies a retention rule: a flight or segment that should not be analysed has no rows to read. Every segment keeps its parent's origin and clock, so a position is always measured from that flight's take-off and a time is always elapsed since it, whether or not the record is continuous up to that point. Section 2.8 characterises this ensemble, Chapter 3 analyses it, and `docs/guide/data-on-disk.md` states which columns each estimator reads and what a segment boundary means to it.

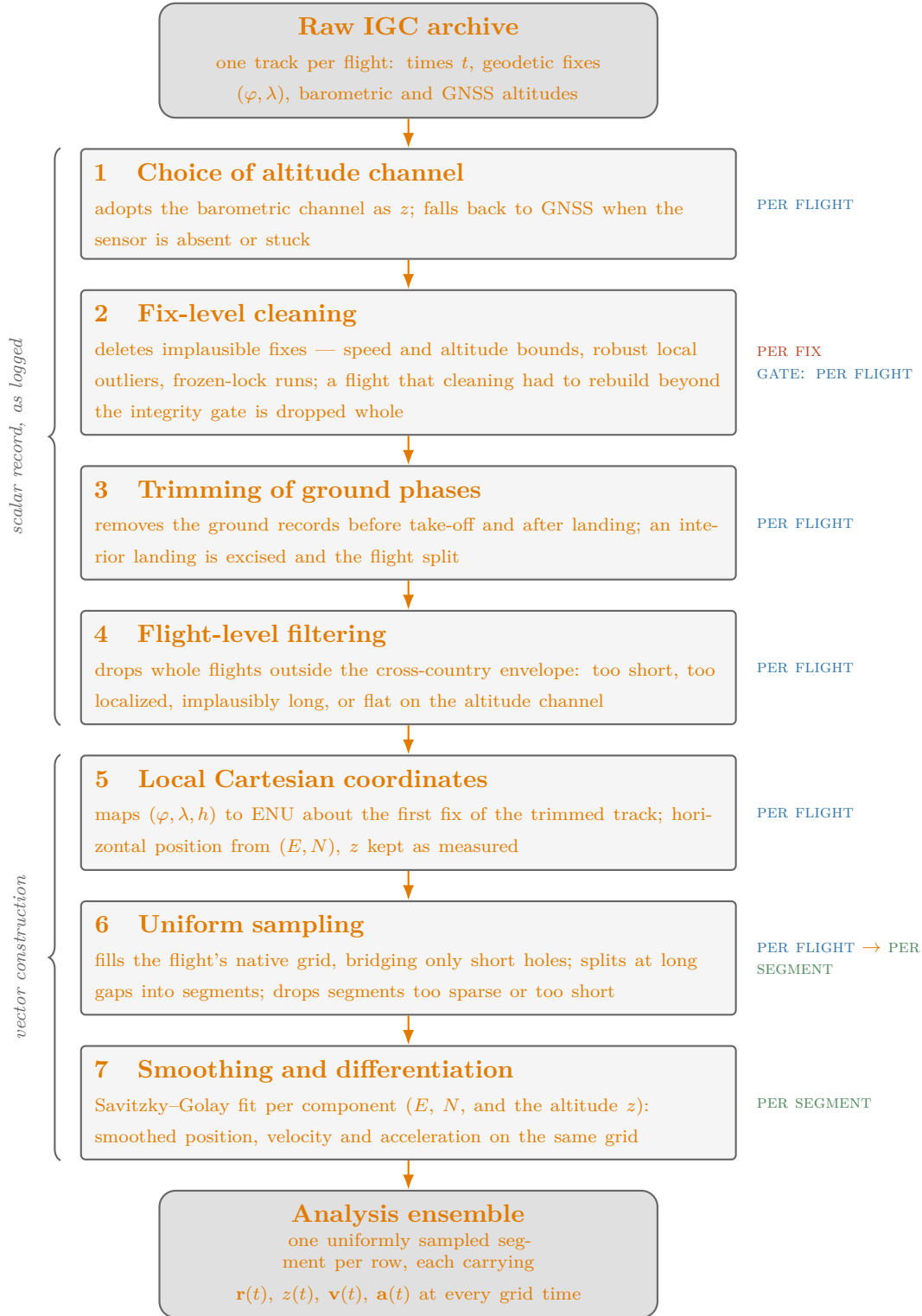


**Figure 2.11.** Whether the retained ensemble may be pooled. (a) The per-component variance ratio  $\langle E^2 \rangle / \langle N^2 \rangle$  against elapsed time: unity is isotropy, and neither discipline reaches it over the shaded band, which is the lag range the transport analysis reads. (b–d) The ensemble MSD computed within each stratum and divided by the pooled curve, for paragliders, by wing class, by orographic group and by season; strata below 2,000 flights are not drawn. Class and terrain separate the curves, season does not. Computed by `scripts/reporting/generate_prelim_figure.py`.



**Table 2.7.** The pre-processing cascade over the full archive, in the order the pipeline applies it. *Standing* is how many flights had survived every earlier criterion when this one ran and *share* is the removal as a fraction of those, not of the archive: a criterion late in the cascade tests a population the earlier ones have already thinned, so a percentage of the archive would understate it. Read this way one cut dominates, and it is not one the text argues a threshold for—21,089 paraglider flights, 11.9% of those still standing, leave nothing behind after resampling. The marginal effect of each cut on the raw population is a different quantity and is what the `\StatScan*` numbers of Secs. 2.7.4 and 2.7.6 report. Generated by `generate_dataset_stats.py` from the `flights_meta` table the pipeline itself writes.

| Why the flight was dropped              | paragliders  |         |           |                |
|-----------------------------------------|--------------|---------|-----------|----------------|
|                                         | standing     | removed | share (%) | remaining      |
| unreadable track                        | 186,052      | 27      | 0.01      | 186,025        |
| never airborne                          | 186,025      | 225     | 0.12      | 185,800        |
| integrity gate                          | 185,800      | 1,442   | 0.78      | 184,358        |
| shorter than the duration cut           | 184,358      | 2,107   | 1.14      | 182,251        |
| longer than the duration cut            | 182,251      | 3       | 0.00      | 182,248        |
| path below the floor                    | 182,248      | 182     | 0.10      | 182,066        |
| altitude activity below the floor       | 182,066      | 4,960   | 2.72      | 177,106        |
| out of reach of its own first fix       | 177,106      | 228     | 0.13      | 176,878        |
| no segment survived resampling          | 176,878      | 21,089  | 11.92     | 155,789        |
| no segment carries the smoothing window | 155,789      | 1       | 0.00      | 155,788        |
| <b>retained</b>                         |              |         |           | <b>155,788</b> |
| Why the flight was dropped              | hang gliders |         |           |                |
|                                         | standing     | removed | share (%) | remaining      |
| never airborne                          | 6,716        | 15      | 0.22      | 6,701          |
| integrity gate                          | 6,701        | 73      | 1.09      | 6,628          |
| shorter than the duration cut           | 6,628        | 86      | 1.30      | 6,542          |
| path below the floor                    | 6,542        | 1       | 0.02      | 6,541          |
| altitude activity below the floor       | 6,541        | 153     | 2.34      | 6,388          |
| out of reach of its own first fix       | 6,388        | 7       | 0.11      | 6,381          |
| no segment survived resampling          | 6,381        | 249     | 3.90      | 6,132          |
| <b>retained</b>                         |              |         |           | <b>6,132</b>   |



**Figure 2.12.** The processing chain, from the raw archive to the analysis ensemble. Stage numbers match Table 2.8; the badge to the right of each stage is the level it acts at (per fix, per flight, per segment). Every check on the scalar record (stages 1–4) precedes the construction of vector quantities (stages 5–7), so no coordinate conversion is computed for a record that is about to be repaired or dropped.

|   | Stage                         | Level                        | Effect                                                                                                                                                                           | Adopted parameters                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Sec.  |
|---|-------------------------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1 | Choice of altitude channel    | flight                       | adopt the barometric channel as $z$ ; fall back to GNSS when the sensor is absent or stuck                                                                                       | stuck sensor: whole-flight barometric range $< 30$ m                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2.7.1 |
| 2 | Fix-level cleaning            | fix (gate: flight)           | delete implausible fixes; drop the flight when the share of airborne fixes this stage removes, or leaves for the resampler to rebuild, exceeds the integrity gate                | <ul style="list-style-type: none"> <li>• <math>v_{xy} \leq 45/55 \text{ m s}^{-1}</math> (para/hang)</li> <li>• <math> v_z  \leq 13 \text{ m s}^{-1}</math></li> <li>• <math>z \in [-100, 6000] \text{ m}</math></li> <li>• Hampel: <math>\pm 20 \text{ s}</math>, <math>k = 5</math>, <math>\varepsilon_{\min} = 15 \text{ m}</math></li> <li>• frozen lock: <math>\delta_{xy} = 15 \text{ m}</math>, <math>\delta_z = 5 \text{ m}</math>, <math>\tau_{\text{freeze}} = 60 \text{ s}</math></li> <li>• integrity gate <math>\leq 10 \%</math></li> </ul> | 2.7.2 |
| 3 | Trimming of ground phases     | flight                       | cut ground records before take-off and after landing; excise interior landings, splitting the flight                                                                             | airborne: $v_{xy} > 5 \text{ m s}^{-1}$ sustained for 30 s; interior stint $\geq 10$ min with the barometric altitude flat to within 5 m                                                                                                                                                                                                                                                                                                                                                                                                                  | 2.7.3 |
| 4 | Flight-level filtering        | flight                       | drop flights outside the cross-country envelope                                                                                                                                  | duration $\geq 40$ min and $\leq 16$ h; path $\geq 20$ km; altitude range $\geq 600$ m                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2.7.4 |
| 5 | Local Cartesian coordinates   | flight                       | map $(\varphi, \lambda, h)$ to ENU about the first fix of the trimmed track; horizontal position from $(E, N)$ ; $z$ kept as measured                                            | exact WGS84 transform; no thresholds                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2.7.5 |
| 6 | Uniform sampling              | flight $\rightarrow$ segment | fill the native grid $\Delta t$ , bridging short holes; split at long gaps; drop a segment that is too short, or too sparse—too many of its grid times having to be interpolated | <ul style="list-style-type: none"> <li>• split at gap <math>&gt; \min\{10 \Delta t, \max(20 \text{ s}, 2\Delta t)\}</math></li> <li>• interpolated fraction of a segment <math>\leq 10 \%</math></li> <li>• segment duration <math>\geq 90 \text{ s}</math></li> </ul>                                                                                                                                                                                                                                                                                    | 2.7.6 |
| 7 | Smoothing and differentiation | segment                      | Savitzky–Golay fit per component $(E, N, z)$ : position, velocity, acceleration                                                                                                  | <ul style="list-style-type: none"> <li>• polynomial order <math>p = 3</math></li> <li>• window <math>w = \max\{\text{odd}(\tau_c/\Delta t), 5\}</math> samples</li> <li>• <math>\tau_c = 5 \text{ s}</math>, measured (provisional until the spectra are re-read on the cleaned ensemble)</li> </ul>                                                                                                                                                                                                                                                      | 2.7.7 |

**Table 2.8.** The pipeline stage by stage: the level each stage acts at, its effect, and the adopted parameters. Every value is typeset through a macro regenerated from `configs/preprocessing.yaml` by `generate_census_stats.py` (implementation details: Table 1.1), so the summary follows the configuration in force. Stage numbers match Figure 2.12.

## Chapter 3

# Global transport

This chapter measures the transport of the retained ensemble without segmenting a single flight. What can be established that way is a *class* of transport: how far a wing gets in a given time, how that growth is distributed, how long the motion remembers its direction, and how many distinct regimes the answer needs. What cannot be established is which flight behaviour produces it, and that is Chapter 4.

The measurements are made on the analysis ensemble assembled in Chapter 2 and defined at the close of Sec. 2.9: 155,788 paraglider and 6,132 hang-glider flights, holding 179,395 and 6,951 retained segments. Every statistic below is computed within a retained segment and never across the gap that ends one (Sec. 2.7.6): a boundary removes the pairs of fixes that straddle it rather than joining them, and a flight answers a given lag only when it holds a fix within half its own native interval of that lag (implementation details: Sec. 3).

Section 2.8 constrains how that ensemble may be averaged, in two ways that hold throughout. The horizontal components are not equivalent: the ratio of their mean squares sits at 0.80 for paragliders over the lags read here, between 0.70 and 0.86 and never touching unity, and at 1.19 for hang gliders. The two marginals are therefore measured separately, and a pooled one-dimensional marginal is never substituted for the two-dimensional propagator.

The second constraint is heterogeneity within a discipline. Read against the pooled ensemble MSD over the same lags, the typical wing class departs by 40 % and the typical orographic group by 62 %, against 12 % across seasons. Wing class and terrain are relevant strata and season is not, so the two disciplines are never pooled, every exponent is reported by stratum as well as pooled, and a pooled figure is read as an average over populations known to differ rather than as a property of a homogeneous one.

Over 60 s to 2,000 s, on an estimator that annihilates any polynomial trend in the trajectory, cross-country soaring gives  $\alpha_2 = 2.02 \pm 0.10$  for paragliders and  $1.94 \pm 0.13$  for hang gliders. It is not the Lévy walk this thesis was framed around: the spectrum of moments is straight,  $\nu(q)$  moving by 0.019 and 0.009 across the range of  $q$  it is read over, and the run lengths such a description would rest on have no scale, their tail index falling from 4.92 to 2.24 as the threshold that declares a turn is scanned. It is not Gaussian either: the non-Gaussian parameter is +0.042 for paragliders and +0.181 for hang gliders in the median, rising to an interior maximum near 266 s, so one exponent governing every moment states how the increment distribution scales and nothing about its shape.

### 3.1 Displacement from take-off measures the launch geometry

Every flight starts at the origin,  $\mathbf{r}(0) = \mathbf{0}$ , so the position after an elapsed time  $t$  and the displacement from take-off are the same vector. Its mean square admits two averages, and

they are not two attempts at one quantity. The ensemble average

$$\text{MSD}(t) = \langle |\mathbf{r}(t)|^2 \rangle, \quad (3.1)$$

taken over flights at a fixed elapsed time, answers how far from its launch site the population has got after flying for  $t$ . Its growth law  $\text{MSD}(t) \sim t^\alpha$  fixes the transport regime —  $\alpha = 1$  Brownian,  $\alpha > 1$  super-diffusive,  $\alpha < 1$  sub-diffusive, with  $H = \alpha/2$  — and it is the observable the anomalous-transport question is normally posed with. The time average

$$\overline{\delta^2(\tau)} = \langle |\mathbf{r}(t_0 + \tau) - \mathbf{r}(t_0)|^2 \rangle_{t_0}, \quad (3.2)$$

taken over starting times  $t_0$  inside one record, answers how far a wing travels in a window of duration  $\tau$  placed anywhere in its flight.

The ensemble average is synchronised at take-off, so a feature it carries at a fixed elapsed time is a property of that common origin. The time average slides its window along each record and meets a lag wherever it occurs, with no privileged origin. A feature the first shows and the second does not belongs to the origin; a feature both show is a property of the motion. Neither is a proxy for the other, and their difference is a measurement in its own right.

Both are computed on the analysis ensemble defined at the close of Sec. 2.9 and characterised in Sec. 2.8: 155,788 paraglider and 6,132 hang-glider flights, 179,395 and 6,951 retained segments. Every flight enters at the elapsed times its retained segments cover, with no re-zeroing at a split (Sec. 2.7.6); the time average is taken inside a segment and never across the gap that ends one. A flight contributes to a lag only when it holds a fix within half its own native interval of it, so a slow logger says nothing about a lag below its own cadence rather than answering with its fix at  $t = 0$ . Figure 3.1 shows both estimators.

**The residual each fit leaves.** Least squares returns a slope whatever the curve underneath it does, so each fit is reported with the residual it leaves. Over the range each estimator is fitted on, the ensemble curve departs from its own fitted power law by 15.2 % rms for paragliders and 31.2 % for hang gliders; the time-averaged curve departs from its own by 3.8 % and 3.6 %.

The size of a residual is half of what decides whether an exponent describes a curve. The other half is whether it exceeds the noise on the points, and here it does by a wide margin. Each lag's value is a mean over the flights covering it, so it carries a standard error that the stored inter-decile band and contributing count give directly. Over the fitted range the residual is 51 times that error on the paraglider ensemble curve and 22 times on the hang-glider one, against 28 and 5 on the two time averages. A departure from a fitted line that exceeds the error on each point many times over is not scatter about a power law; it is a shape. Those ratios are the evidence that these curves are not power laws.

The arrangement of the residuals is an arch. The ensemble curve leaves 40 of them in 3 sign runs where independent signs would give 21: they go negative, positive and negative once, which is a bend with the fitted line drawn as a chord across it. That count is a description of the arrangement and not a test. A Wald–Wolfowitz  $z$  presumes independent signs, and every lag of these curves averages the same flights, which inflates the rejection rate of that rule on an exact power law to most of its draws (implementation details: Sec. 3).

**The shape of the ensemble curve.** The shape is definite and reproducible. The local slope falls to a minimum of 1.31 near 137 s, rises to 2.21 near 1,504 s, and settles back towards 2 beyond it; the hang gliders do the same between 1.10 and 2.52. The maximum is the informative part, because it lies *above* the ballistic value. No sum of power laws with positive coefficients and exponents at or below 2 can produce a local slope above 2: for  $f = \sum_i a_i t^{\alpha_i}$  with  $a_i > 0$ ,

$$\frac{d \log f}{d \log t} = \frac{\sum_i a_i \alpha_i t^{\alpha_i}}{\sum_i a_i t^{\alpha_i}} \quad (3.3)$$

is a weighted mean of the  $\alpha_i$  and cannot exceed the largest of them. A slope above 2 requires the ballistic component to switch on at a finite time rather than to be present from  $t = 0$ .

The arch is larger than the estimator's own uncertainty. The local slope is not a difference between neighbouring lags but a Theil–Sen regression over a window, reported with the scatter of that window about its own line; over the fitted range that scatter sits well below the lag-to-lag variation of the slope itself (implementation details: Sec. 3). The curve moves more between neighbouring lags than the noise inside a window accounts for.

**Three controls on that shape.** Three mechanisms would produce the same shape without any of it belonging to the motion, and each is measured rather than argued away.

The first is the coverage rule meeting a changing population: the ensemble behind  $\text{MSD}(t)$  grows as slow loggers reach their own cadence and shrinks as flights land, and an average taken over a drifting population can bend. Splitting the archive by native cadence removes the drift, since within one cadence class every flight answers every lag above that cadence. Across the 3 paraglider classes the local slope at a given lag spreads by 0.06, against the 0.91 the shape itself covers. The hang-glider classes are far smaller and their spread, 0.78 against a swing of 1.43, settles nothing; that archive's version of this control is the next one, whose cohorts are larger.

The second is survivorship: flights that last longer are not a random sample, and if they also travel faster the curve steepens for a reason that is not motion. Fixed-duration cohorts control it, since within a cohort of flights lasting at least  $T$  every member is present at every lag up to  $T$ . Inside a cohort the same minimum and the same overshoot are there, and the local slope spreads across cohorts by 0.11 and 0.21. A synthetic ensemble in which every flight is exactly Brownian and only the *speed* is correlated with the duration reproduces neither feature: its pooled exponent is biased upwards while every cohort of it recovers the Brownian value (implementation details: Sec. 3).

The third is a heading shared across the ensemble, which would make  $\langle |\mathbf{r}(t)|^2 \rangle$  grow ballistically for a trivial reason. It is bounded by the part of the mean square the mean vector carries,  $|\langle \mathbf{r}(t) \rangle|^2 / \langle |\mathbf{r}(t)|^2 \rangle$ , which over the fitted range never exceeds 1.8 % for paragliders and 10.0 % for hang gliders. The flights do not share a heading. They share a starting point.

**The origin of the crossover.** A shared starting point is enough to produce it. A wing does not leave its launch site immediately: it climbs out, often circling close to it, and only then commits to a course. Until it does, its distance from take-off is bounded by the size of the local soaring area however fast it is flying, which is what depresses the local slope at intermediate lags; once the population has committed, the distance grows with the flown speed, which is what returns the slope towards 2. Because the commitment happens at a spread of times rather than at  $t = 0$ , the transition overshoots.

The time at which it happens is measured rather than fitted. Taking the first crossing of 5 km from take-off as the departure onto the cross-country leg — well beyond a thermalling excursion, well inside the median flown path of 89 km — 95.1 % of paraglider flights depart, with a median departure time of 1,725 s and an interquartile range of 1,222 s to 2,565 s, against a slope peak at 1,504 s. For hang gliders 96.4 % depart, at a median of 1,530 s against a peak at 1,334 s. In both disciplines the overshoot sits on the scale of the departure and slightly before it, which is where a transition driven by a spread of onset times should sit, the slope being steepest while most of the population is still crossing. The radius is chosen rather than fitted, so the agreement corroborates the account and does not measure it.

Two consequences follow. The ensemble MSD about take-off is a statement about the geometry of the flight relative to its launch site as much as about the transport, and over the decade where the overshoot sits it is mostly the former. And an exponent fitted across



that crossover is a straight line through a bend: over 121 s to 13,021 s least squares returns  $\alpha_{\text{EA}} = 1.918$  and 1.896, and that is what a chord across a bend gives rather than a growth law the motion obeys.

**The same displacement, averaged over starting times.** Panel (b) is Eq. (3.2) on the same ensemble. Over 121 s to 6,341 s and 121 s to 7,149 s it gives  $\alpha_{\text{TA}} = 1.732$  ( $H = 0.866$ ) for paragliders and 1.688 ( $H = 0.844$ ) for hang gliders, both well above the Brownian  $\alpha = 1$  and below the ballistic  $\alpha = 2$ . The residual it leaves is the smaller one quoted above and its local slope covers 0.40 and 0.55, against 0.91 and 1.43 on the ensemble curves. That is enough to make an exponent a useful summary and not enough to make it a description, so both figures are reported as *effective* exponents over a stated range.

The arch is absent. The ensemble slope peaks at an interior lag and falls back; the time-averaged slope reaches its minimum at 1,504 s and 1,334 s and rises from there to the end of its range, with no interior maximum at all. Since the two are the same statistic on the same ensemble, differing only in what they average over, a feature that survives one average and not the other belongs to what distinguishes them. The crossover is a property of the common origin, and its absence from panel (b) is what identifies it rather than a defect of either estimator.

**The gap between the two averages.** Fitted over their respective ranges the ensemble slope exceeds the time-averaged exponent by 0.187 for paragliders and 0.208 for hang gliders, and a systematic gap of that kind is one of the signatures of a non-ergodic, aging process [20]. It is not evidence of one here, and this observable as computed cannot be made into evidence of one: reading a gap between two exponents presupposes that both are exponents, and the ensemble figure is a chord across a curve whose local slope covers 0.91, so changing the fitted range moves it by more than the gap. What the measurement supports is weaker — displacement measured *from take-off* grows faster, over the upper part of the fitted range, than displacement over a window of the same duration placed anywhere in the flight — and separating aging from launch geometry needs the phase decomposition of Sec. 4.1 and the ergodicity-breaking parameter of Appendix 3.A, which compares the *distribution* of  $\overline{\delta^2}$  across flights rather than two fitted slopes and so needs neither curve to be a power law.

**What the two estimators return.** The two estimators return two results of different kinds. From the time average, an effective exponent, on a curve whose departure from a single power law is small in size and systematic in arrangement. From the ensemble average, a timescale and not an exponent: displacement from take-off crosses over near 1,504 s for paragliders and 1,334 s for hang gliders, where the population leaves its launch area, and no single power law spans that crossover. The mean-squared displacement about the origin, which is the quantity the anomalous-transport question is normally put in, does not deliver an exponent on this archive.

Neither figure carries an interval. A bootstrap over flights would treat the archive's records as independent, which they are not, and it would report the sampling term alone, where on the estimator for which both have been measured the movement of the exponent with the fitted range is far the larger. A curve that is a systematic arch has, in any case, no single exponent for a sampling error to be an error *of*. The resampling unit and the range dependence are both measured below, on the estimator that replaces this one, and the exponents that carry an interval in this chapter are that estimator's.

**The two ends of the range.** Below the foot of every fit range the curves carry small bumps, from the coverage rule meeting a lag grid of consecutive integers and from the smoothed

samples at a segment's ends (implementation details: Sec. 3). Nothing below 121 s in Figure 3.1 is read as motion. At the other end the ceiling is set by coverage: the ensemble collapses onto a handful of long flights at the longest lags, which is why each fit stops well inside the plotted range. One limitation belongs to the ensemble estimator alone. A segment keeps its parent's origin (Sec. 2.7.6), so a boundary at a re-acquisition offset leaves an unknown instrument constant in every subsequent absolute position; it enters  $\langle |\mathbf{r}(t)|^2 \rangle$  and cancels out of  $\overline{\delta^2(\tau)}$ , and it is carried by 0.029 % of paraglider and 0.457 % of hang-glider flights (implementation details: Sec. 3).

**Three contaminants of a displacement statistic.** Three things contaminate a displacement statistic on this archive, and one estimator disposes of all three.

The first is the common origin, measured above: over the decade where the crossover sits,  $\langle |\mathbf{r}(t)|^2 \rangle$  is governed by where flights start rather than by how they move.

The second is each flight's own course. A wing that crosses country has a mean velocity, and a mean velocity contributes  $|\mathbf{v}_d|^2 \Delta^2$  to any mean square displacement whatever the underlying process. Here it does not simply inflate: the CFD scores closed courses and 56 % of retained paraglider flights fly a triangle, which returns towards its start, so the displacement saturates at lags set by the scoring rule rather than by the air.

The third is the population. Section 2.8 measured the strata of this archive as incompatible on the ensemble MSD, by 40 % across wing classes and 62 % across orographic groups. A superposition of ordinary processes with different persistence times imitates a single anomalous one, which is the most economical way a measurement of this kind can be wrong.

None of the three is repaired by subtracting an estimate of it. That is the specification the next section meets, by filtering.

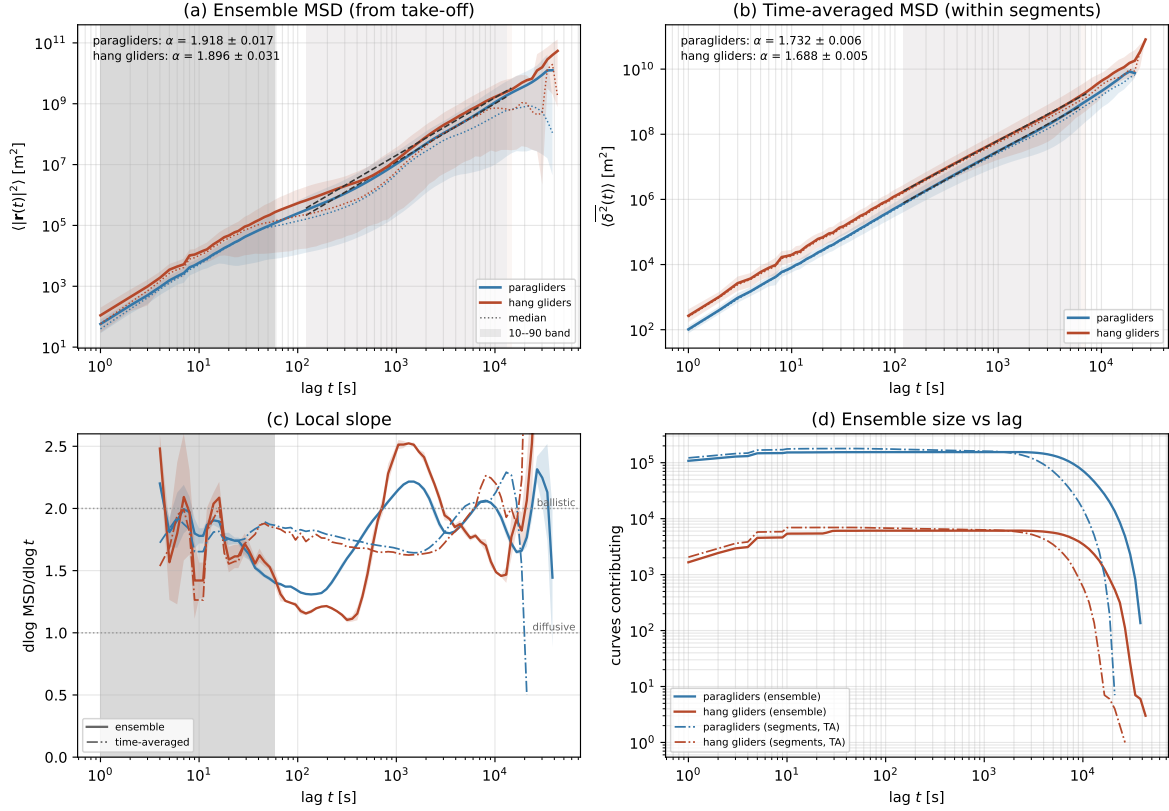
## 3.2 The filtered variation

The displacement statistic leaves a specification: a quantity measured from no common origin, requiring no estimate of the flown course, and not turned into a single anomalous law by a mixture of populations. The first two are met by construction, the third by measurement in the section that follows.

**Filtering rather than subtracting.** The course can be subtracted rather than filtered, and subtraction fails on this archive in a way that is measured rather than argued. A wing that crosses country has a mean velocity  $\mathbf{v}_d$ , and a mean velocity contributes  $|\mathbf{v}_d|^2 \Delta^2$  to any mean square displacement whatever the process underneath it, so removing it looks like a precondition for reading an exponent at all. Removing it means estimating it. The natural estimate  $[\mathbf{r}(T) - \mathbf{r}(0)]/T$  is built from the record's own endpoints, and subtracting it turns the trajectory into a bridge whose increments must sum to zero by construction. On synthetic fractional Brownian motion of known exponent the operation returns a value well below the truth, and there is no lag at which the distortion is absent: the centred variation already sits below the raw one at the shortest lags of a record and falls far below it towards the record's own length (implementation details: Sec. 3).

**The estimator.** A finite difference removes a trend without estimating it, because it annihilates the trend identically. Writing  $A_p$  for the difference of order  $p$  taken at spacing  $\Delta$ ,

$$A_p \mathbf{r}(t; \Delta) = \sum_{k=0}^p (-1)^k \binom{p}{k} \mathbf{r}(t + (p-k)\Delta), \quad (3.4)$$



**Figure 3.1.** Mean-squared displacement of the retained ensemble, by two averages. (a) The *ensemble* MSD against elapsed flight time, log–log, with the fitted line dashed, its range shaded, and the 10% to 90% band across flights with the median dotted. (b) The *time-averaged* MSD, over starting times within each segment and cut at half a segment’s length. (c) The local logarithmic slope of both over a window of  $\pm 0.15$  decades, with the ballistic and diffusive values marked. (d) The flights, or segments, contributing at each lag, on which the upper end of each fit range is set.

the operator vanishes identically on every polynomial trajectory of degree  $p - 1$ , and the *filtered variation* is its mean square,

$$V_p(\Delta) = \left\langle |A_p \mathbf{r}(t; \Delta)|^2 \right\rangle \sim \Delta^{\alpha_p}, \quad (3.5)$$

the average running over every admissible  $t$  inside a segment and over every flight, and the modulus taken on the two horizontal components of one vector rather than by pooling the two marginals, which Sec. 2.8 forbids. Summing the components leaves  $V_p$  invariant under a rotation of the frame, so an anisotropic ensemble does not return a different exponent for having been measured with north somewhere else. For a self-similar process with stationary increments  $\alpha_p = 2H$  at every order, with only the prefactor depending on  $p$ . What differs between orders is what each removes before measuring:  $p = 1$  is the plain increment and is blind to the origin,  $p = 2$  to a constant velocity,  $p = 3$  to a constant acceleration. No drift is estimated at any order, so none can be estimated badly. The order is not free, in that a difference of order  $p$  spans  $p\Delta$  of record: a segment answers at a lag only when it is longer than that, so each order costs coverage at the long end of the grid, and the scan stops at  $p = 3$ .

The scan over orders is a measurement and not a check on one. The difference  $\alpha_1 - \alpha_2$  is the part of the apparent exponent that was the flown course, read off directly rather than modelled, and the agreement between  $\alpha_2$  and  $\alpha_3$  is the statement that nothing polynomial is left for a higher difference to find. On synthetic data with a constant course imposed on a process of known exponent, order 1 returns a value displaced from the truth and orders 2

and 3 both recover it (implementation details: Sec. 3).

Everything below is computed inside a retained segment, never across the gap that ends one (Sec. 2.7.6), and never on displacement from take-off. The pass keeps one curve per flight, so every stratification of this section and of the next is a row selection rather than a further traversal of the fix table (implementation details: Sec. 3).

**The fitted range.** 60 s to 2,000 s, which is 1.5 decades, declared in advance and used for both disciplines and all three orders. Both ends are set by the system rather than chosen to suit the curve. Below, the Savitzky–Golay window has a floor of 5 samples (Sec. 2.7.7), so a trajectory has been made smooth over a window of 5 of its own logger’s sampling intervals, and 60 s sits above that window for the slowest logger retained. Above, the declared task takes over: the scoring rule rewards a closed course, a closed course comes home, and it comes home at a time set by the task rather than by the air. Every exponent below is quoted with the range it was fitted over.

The fit is ordinary least squares on  $\log V_p$  against  $\log \Delta$  across the lags of that range, applied to the mean of the per-flight curves rather than to per-flight exponents averaged afterwards. A flight therefore contributes to every lag its segments cover, and no flight contributes an exponent of its own.

**Table 3.1.** The order scan over 60 s to 2,000 s. Order 1 is the plain increment and still carries each flight’s course; orders 2 and 3 annihilate a constant velocity and a constant acceleration, and agree with each other, which is what says that nothing polynomial is left. The quoted interval combines a day-and-site bootstrap with the movement of the exponent when the fitted range is halved; the two are given separately because they are not of the same size, and the section that follows takes them apart.

|              | order $p$ | $\alpha_p$      | sampling | range |
|--------------|-----------|-----------------|----------|-------|
| paragliders  | 1         | $1.75 \pm 0.06$ | 0.001    | 0.062 |
|              | 2         | $2.02 \pm 0.10$ | 0.001    | 0.104 |
|              | 3         | $2.04 \pm 0.14$ | 0.001    | 0.140 |
| hang gliders | 1         | $1.70 \pm 0.06$ | 0.002    | 0.063 |
|              | 2         | $1.94 \pm 0.13$ | 0.003    | 0.128 |
|              | 3         | $1.97 \pm 0.20$ | 0.003    | 0.198 |

**The order scan.** Table 3.1 gives the scan and Figure 3.2 the curves behind it. At second order the archive returns  $\alpha_2 = 2.02 \pm 0.10$  for paragliders and  $1.94 \pm 0.13$  for hang gliders, over 60 s to 2,000 s in both. The two disciplines agree inside their intervals, which was not guaranteed, since they differ in speed, in glide ratio and in where they fly.

Figure 3.2 carries the local slope of  $V_2$  and that of the ensemble MSD on one pair of axes, over the lags both cover. The arch that dominates the ensemble curve near 1,504 s, and that carries its local slope through a swing of 0.91 for paragliders and 1.43 for hang gliders, is three times smaller in the filtered variation and of the opposite sign. The local slope of  $V_2$  swings by 0.28 and 0.48 inside the fitted range, and it turns twice: a maximum of 2.17 near 117 s, a minimum of 1.89 at 613 s, and a rise after it. Where the ensemble curve is an arch the filtered variation is a dip, so the two are not the same structure attenuated but different structures — which is what the removal of the common origin and the flown course was supposed to achieve. How large that swing is, in exponent units, is the range dependence reported in the next section.

**Removing the course raises the exponent.**  $\alpha_1$  and  $\alpha_2$  differ by  $-0.27$  for paragliders and  $-0.24$  for hang gliders, and the sign is the opposite of the one a residual drift would leave. A drift inflates a displacement statistic; on this archive the declared course *suppresses* it. The CFD scores closed tasks, and 56 % of retained paraglider flights and 63 % of hang-glider flights fly a triangle. A triangle returns towards its start, so displacement saturates by construction, and it saturates at lags set by the task rather than by the air. The plain increment sees that and reads an exponent that is too low; the second-order variation does not see it at all.

The reading is confirmed where it can be tested directly rather than inferred from the scan. Fitting closed and open tasks separately, the gap in the exponent, taken as open minus closed, is  $+0.11$  at order 1 for paragliders and  $+0.03$  for hang gliders — closed tasks reading lower, which is what a course that comes home requires — and it reverses to  $-0.10$  and  $-0.07$  at order 2. Inside the declared range the two orders separate the two populations by comparable amounts, and what discriminates between them is the behaviour of each gap when the range is extended past its ceiling: the order-1 gap grows substantially and the order-2 gap does not move. A difference of curvature does not care where the course ends.

The scan therefore fixes the sign of a comparison made later in this chapter and by anyone comparing an exponent measured elsewhere with the one measured here. An exponent read off displacement with no trend removed is, on this archive, an exponent read partly off the scoring rule, and it is expected to sit below one read with the course filtered out, by the difference between the two orders:  $-0.27$  for paragliders and  $-0.24$  for hang gliders.

**Orders two and three agree.** By 0.02 for paragliders and 0.02 for hang gliders, well inside the interval either exponent carries. Nothing polynomial survives the second difference for the third to remove, so the exponent is not the residue of a trend of any degree.

**Order one reproduces the time-averaged MSD.** It should: the first-order filtered variation is  $\overline{\delta^2(\Delta)}$  under another name. The comparison is a check on two implementations and not on the physics, and it is worth making because the two were written separately, traverse the fix table separately, and are fitted over different ranges — 60 s to 2,000 s here against 121 s to 6,341 s for the time average. They return  $\alpha_1 = 1.75$  against  $\alpha_{\text{TA}} = 1.732$  for paragliders and 1.70 against 1.688 for hang gliders: closer, in both disciplines, than either moves when its own range is halved.

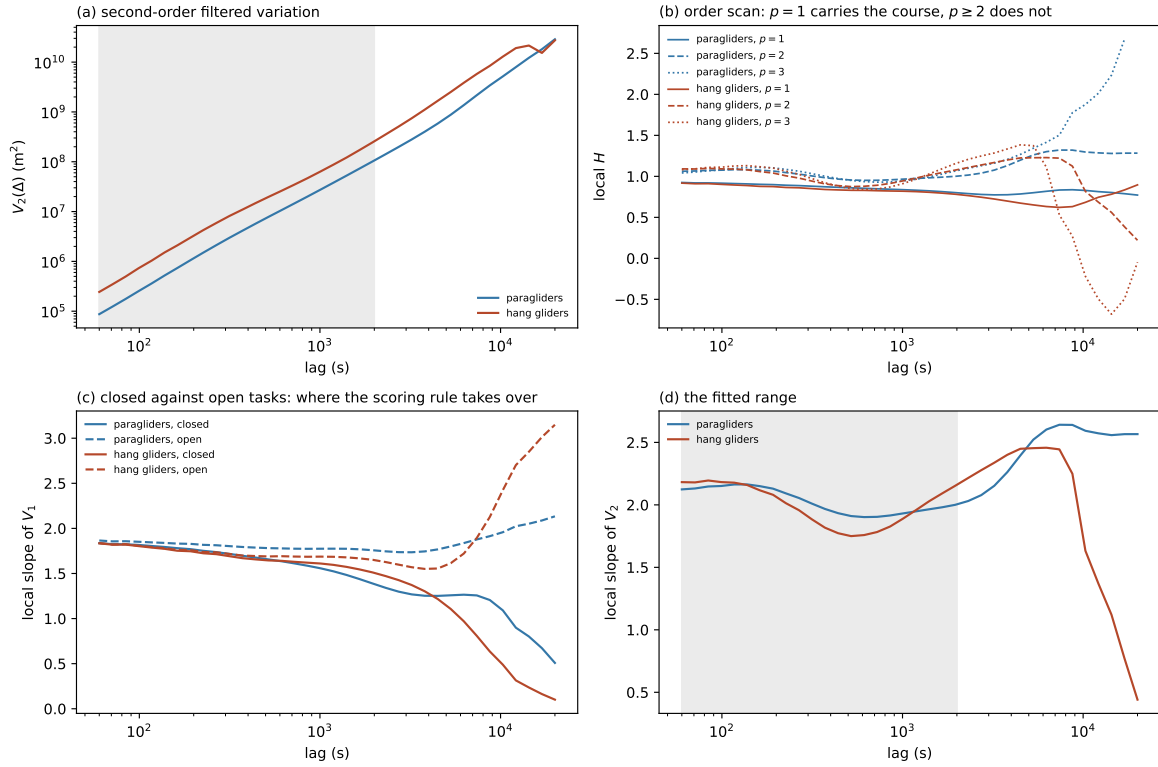
**What an exponent near two does not mean.**  $\alpha_2 = 2.02$  is  $2H$  with  $H = 1.009$ , at the upper boundary of the self-similar family and marginally past it. The reading that suggests itself is that the flight has gone locally straight over the range, and the estimator settles it against two calibrations of its own.

A plain increment cannot report above the ballistic value on any process with stationary increments: the displacement over  $n$  steps is a sum of  $n$  increments, Minkowski's inequality bounds its norm by the sum of theirs, and  $V_1(n\Delta) \leq n^2 V_1(\Delta)$  follows, so  $\alpha_1 \leq 2$  whatever the motion does. An estimator sitting at its own ceiling says nothing, and the second-order variation has no ceiling at that value. On a differentiable trajectory  $A_2 \mathbf{r} \simeq \ddot{\mathbf{r}} \Delta^2$ , so  $V_2 \sim \Delta^4$ : the second-order variation has room to report twice the ballistic value, and it reports it when the motion is smooth. Run on a persistent random walk over lags well inside its own correlation time, where the trajectory is differentiable at every lag it is read at, the same code returns an exponent between the ballistic value and that limit; run on a straight line it vanishes identically, the second difference annihilating a linear path exactly.

Read against that pair, 2.02 and 1.94 say that the increments stay correlated at every lag of the range and that the trajectory is rough at every lag of it. The wing turns at every scale the estimator reads, with a bias in the turning that has not decayed by the top of the

range; it is persistent, and it is not straight. A Hurst exponent above unity has no meaning for an exactly self-similar process with stationary increments, and the figure is quoted as an effective exponent over a declared window for that reason: how far the archive is from exact self-similarity is measured later in the chapter, on the distribution of the increments rather than on its second moment.

What the interval attached to each exponent is made of, and what the exponent does when the ensemble is cut by cadence, by wing class, by season and by declared task, is measured next.



**Figure 3.2.** The filtered variation, on which the exponent is read. (a)  $V_2(\Delta)$  against lag, log-log, with the fitted range shaded. (b) The local  $H = \frac{1}{2} d \log V_p / d \log \Delta$  at the three filter orders: inside the fitted range  $p = 1$  runs below  $p = 2$  and  $p = 3$ , which is the flown course read as a lower exponent. (c) The same local slope at  $p = 1$  with the ensemble split by declared task; closed and open courses track each other to about 3,000s and part company after it. (d) The local slope of  $V_2$ , the quantity the fit averages over the shaded range.

### 3.3 The uncertainty on the exponent

The archive holds 155,788 paraglider and 6,132 hang-glider flights, and 84,132 and 4,331 occasions on which one site was flown on one day. An interval on  $\alpha_2$  built on the first pair of counts is not an interval built on the second, so which of them carries the independent measurements has to be settled before either is quoted.

**The resampling unit.** The quantity clustered is one number per flight: the logarithm of that flight's own first-order variation at a lag in the middle of the grid, which is the per-flight input every fit in this chapter averages over. Its intraclass correlation at a given level is the share of its variance sitting *between* clusters rather than within them. Near zero the clustering



is irrelevant and flights may be resampled directly; near one a cluster is effectively a single measurement however many flights it holds.

For paragliders the ladder reads 0.00 at the flight, 0.15 at the calendar day, 0.29 at the pilot and 0.57 at the day and site; for hang gliders 0.26, 0.50 and 0.64. The first entry is not a measurement. Taking each record as its own cluster gives zero by construction, and it is in the ladder because it is the assumption a per-flight bootstrap makes. The ordering of the other three is the physical one. A date on its own pools every site in the country flying that day, and most of what two flights share washes out of it. A pilot carries a wing and a home site, which is worth more, and worth more again on the smaller archive, where a pilot returns to it more often. The air over one site on one day is what two flights actually have in common, and it is the level at which the coefficient is largest in both disciplines. Every bootstrap in this chapter therefore resamples days and sites, of which the archive holds 84,132 and 4,331.

The correction it buys is in the expected direction and is small. Resampling flights returns 0.000 on the paraglider  $\alpha_1$  against 0.001 from the clustered resample, and 0.002 against 0.003 on the hang-glider  $\alpha_2$ . Both figures sit below the last digit of the exponent they qualify. The ladder settles which of the two applies; neither of them is what the interval is made of.

**The sampling term and the range term.** The two are reported separately. On  $\alpha_2$  the day-and-site bootstrap returns 0.001 for paragliders and 0.003 for hang gliders. Refitting each half of the fitted window and halving the difference between the two exponents returns 0.104 and 0.128. The sampling term is negligible against the range term in both disciplines, and the quoted  $\pm 0.10$  and  $\pm 0.13$  are the two combined in quadrature, which is the range term with the sampling term invisible inside it.

The same construction at the other orders gives 0.062 at order one and 0.140 at order three, so the systematic grows with the order of the filter, as a higher difference of a noisier curve should. Nothing here turns on the third decimal of any of them. They are quoted to say that the second decimal of the exponent is not determined, and a number carrying the bootstrap alone would claim four significant figures on a curve that does not have two.

**What the least-squares error measures.** A third figure is available and measures something else. The standard error that least squares returns on the fitted slope is 0.006 on the paraglider time-averaged displacement, 0.017 on the paraglider ensemble fit and 0.031 on the hang-glider one. It asks how well a *straight line* describes the fitted points, and it treats each residual as an independent draw from a common noise. The residuals of these curves are neither. They are not independent, because every lag is an average over the same flights, which is what the effective count below puts at 2.0 points; and they are not noise, because the local slope drifts across the fitted decades. So the least-squares figure is dominated by the curve's departure from a power law, which is why it is larger on the ensemble fit than on the time-averaged one and tracks the residuals rather than the sample size. Which of it and the bootstrap comes out larger depends on the size of the ensemble, and a synthetic ensemble separates the two (implementation details: Sec. 3).

Neither candidate interval is an interval on the displacement exponents quoted earlier in this chapter. The bootstrap over flights treats 155,788 records as independent, which the ladder above refuses, and returns 0.001 for a curve that is not a power law; the least-squares figure measures the arch rather than the sampling. On a curve whose local slope covers 0.91 across the range it is fitted on there is no exponent for either to be an error *of*, and attaching a  $\pm$  would assert that there is one.

**What sets the two ends of the window.** Both ends come from the system rather than from the fit. Below, the smoothing of Sec. 2.7.7 has a window floor of 5 samples, so the



scale over which a trajectory has been made smooth is that many intervals of the logger that recorded it, and it is longest for the slowest loggers retained. The floor of the fitted range, 60 s, sits above that scale for every cadence in the archive; nothing under it is read as motion.

Above, the declared task takes over. The CFD scores closed courses, and 56 % of retained paraglider flights and 63 % of hang-glider flights fly one; a course that comes home suppresses any displacement statistic at lags set by the task rather than by the air. Splitting the ensemble on it and comparing local slopes measures where that begins to matter (Figure 3.2). Closed and open courses separate by 0.03 in the local slope at 60 s and by 1.12 at 12,156 s, the hang gliders from 0.00 to 2.39 over the same lags. The separation grows smoothly, so there is no lag above which the measurement becomes clean, and any threshold placed on it picks a point on a smooth curve at a place that moves with the sample size. The ceiling of the fitted range, 2,000 s, is set where the separation is still a small fraction of what it reaches, and the window that survives both ends is 1.5 decades wide.

**Four cuts of the ensemble.** Four stratifications ask whether the exponent belongs to the motion or to who is in the ensemble. The pass keeps one curve per flight, so each is a row selection on a matrix already in memory rather than another traversal of the fix table (implementation details: Sec. 3).

**Table 3.2.** What  $\alpha_2$  does when the paraglider ensemble is cut four ways, over the same range. The comparison that matters is with Sec. 2.8, where the same wing classes departed from the pooled ensemble MSD by 40 % and the orographic groups by 62 %. The declared task is entered as the open-minus-closed difference rather than as a spread, and it is the largest of the four.

| cut            | strata       | $\alpha_2$ range | spread |
|----------------|--------------|------------------|--------|
| native cadence | 6 classes    | 2.01–2.03        | 0.02   |
| wing class     | 6 classes    | 1.97–2.03        | 0.06   |
| season         | 17 seasons   | 2.00–2.03        | 0.03   |
| declared task  | closed, open | —                | −0.10  |

The cadence cut is a check on the pipeline. The smoothing scale is proportional to the logger’s own interval, so the 6 classes span the whole range of scales below which this archive has been made smooth, and across them the exponent spreads by 0.02. Whatever the estimator is reading, it is not the filter.

The wing-class and season cuts are the heterogeneity confound, and they are where the filtered variation separates itself from the displacement statistic it replaced. Across 6 wing classes the exponent spreads by 0.06 and across 17 seasons by 0.03. On the ensemble MSD of Sec. 2.8 the same wing classes departed from the pooled curve by 40 % and the orographic groups by 62 %, against 12 % for the seasons, which is the one stratification that section found compatible. A mixture of populations with different speeds and different courses is a mixture of trends, and a trend is what the second difference removes. The heterogeneity that makes the ensemble estimator unusable does not reach this one.

The declared task is the largest of the four, at −0.10 for paragliders and −0.07 for hang gliders, and it is the one cut of the four that divides flights by something other than who was flying. No entry in the table exceeds the interval the exponent already carries.

**The task stratum and the range dependence are one term.** The task gap at order two, −0.10, and the range dependence, 0.104, are the same size. They are plausibly the same effect, and therefore one term rather than two to be added.

Both measure how much of the fitted exponent is fixed by the long end of the window. The range term is what the exponent does when the window is halved and each half fitted alone. The task gap is what it does when the ensemble is split on the one attribute whose influence is confined to the long end: the local slopes of the two task populations are separated by 0.03 at the bottom of the fitted range and diverge monotonically above it. A systematic sourced where the scoring rule governs the trajectory is seen by a shift of the window and by the split alike, and its coming out the same size in the two is what a single term predicts. The signs are not comparable, one figure being a difference between two populations and the other a half-difference between two halves of a window; only the magnitudes are set against each other.

The quoted  $\pm 0.10$  already carries this systematic through its range half, so entering the task stratum as a fifth robustness term and adding it in quadrature would inflate the interval by counting the scoring rule twice. For hang gliders the same two figures are  $-0.07$  and  $0.128$ , of the same order and not of the same size, so the identification is weaker on that archive and the interval there is left as the range term alone.

One measurement would settle it and is not made: the range dependence inside the open-course population by itself. If the scoring rule is the whole of the systematic, that population's exponent is the one that should not move when the window does. It is a row selection on curves already stored.

**How many exponents the range supports.** The same question is put at its limit by fitting a continuous piecewise-linear curve to  $\log V_2$  against  $\log \Delta$ , with the number of breaks chosen by an information criterion. The positions are searched exhaustively on the lag grid, so there is no local optimum to fall into, and the penalty counts the effective rather than the nominal number of points, so a curve whose residuals are correlated cannot buy a break with points it does not independently have.

Selection returns 1 break for paragliders, at 267s, with a slope of 2.16 below it and 1.93 above; the two bracket the single-exponent value  $\alpha_2 = 2.02$ , and their separation is of the order of the range dependence, which is the same curvature seen as a shape instead of as an interval. Hang gliders take 2 breaks. A break here would be a genuine change of regime in the transport rather than the crossover of the ensemble curve under another name, that one being fixed at an elapsed time by a common origin this estimator does not have.

The archive cannot declare one. Run on surrogates that are a single power law plus the measured sampling noise — 0.0005 dex per lag for paragliders and 0.0027 for hang gliders, carrying the measured lag-to-lag correlation — the same procedure returns a break 16 % and 34 % of the time (implementation details: Sec. 3). A rule that invents a break that often on a curve that has none cannot be quoted as having found one, and neither count is established.

The reason is the resolution of the data rather than the fitting rule. The residuals of any fit to these curves are correlated across lags, because every lag is an average over the same flights, and the usual first-order correction puts 22 lags at no more than 2.0 independent points, which is the floor the estimator will report and is reached in both disciplines. On that budget one break sits at the edge of what is estimable and two are past it. What the range supports is one effective exponent over a declared window, carrying an interval set by where that window ends.

### 3.4 The exponent from the increment quantiles

Every exponent quoted so far is a second moment of a displacement, and this archive has already shown what a second moment can measure instead of the motion: averaged about a common origin, it returned the geometry of the launch site. What is wanted is a route to the

exponent that uses neither a moment nor a common origin. The first-passage time was tried for that role and withdrawn.

**First-passage times, tried and withdrawn.** The first-passage time  $T(L)$  is the first time a flight is a distance  $L$  from take-off, and for a self-similar process  $\langle T(L) \rangle \sim L^{1/H}$ , so the slope of  $\log \langle T \rangle$  against  $\log L$  gives the exponent in the space domain with no moment fitted over lags [21]. The audit arrays hold each flight's position at every lag, so it costs no further traversal of the fix table, and it was computed as a cross-check on  $\alpha_2$ .

Read naively it returns an exponent above the ballistic value, which no process with stationary increments can give. The cause is right censoring: at the larger radii only a minority of paraglider flights arrive, the ones that do not are the slow ones, so a median taken over arrivals alone is biased down and the fitted slope with it. Treating non-arrival as censored and taking a Kaplan–Meier median instead removes the bias, and returns values for both disciplines inside the paraglider interval  $\alpha_2 = 2.02 \pm 0.10$  they were meant to test. Refitting over sub-ranges of radius then moves the paraglider value across a spread several times that interval (implementation details: Sec. 3).

The diagnosis is the ensemble average's, transposed.  $T(L)$  is measured from take-off, so it carries the same origin into the space domain: at small  $L$  the wing is still climbing out inside its launch area, and at large  $L$  the censored median extrapolates past the duration of most flights. It is the crossover of Figure 3.1 read along the space axis instead of the time axis, and it is not an independent check on anything. It is withdrawn. The failure survives the change of axis, which places it on the common origin the two estimators share rather than on the elapsed-time axis only one of them uses.

**The propagator of the increments.** Write  $\Delta x(t_0, \Delta) = x(t_0 + \Delta) - x(t_0)$  for the increment of one horizontal component over a lag  $\Delta$ , taken inside a retained segment and never across the gap that ends one (Sec. 2.7.6). Pooled over starting times and over flights, its distribution is the propagator of the increments,

$$P(x, \Delta) dx = \Pr(\Delta x \in [x, x + dx]), \quad (3.6)$$

and for a self-similar process with stationary increments it takes the scaling form

$$P(x, \Delta) = \Delta^{-H} F\left(\frac{x}{\Delta^H}\right), \quad (3.7)$$

where  $F$  is a fixed density of unit integral, independent of the lag. It is defined and estimated on one component: pooling the two would presuppose isotropy, which Sec. 2.8 rejects on this ensemble.

Equation (3.7) makes every quantile of  $|\Delta x|$  grow as  $\Delta^H$ . Writing  $q_p(\Delta)$  for the  $p$ th quantile, the scaling form gives  $q_p(\Delta) = \Delta^H q_p(1)$ , so  $\log q_p$  against  $\log \Delta$  is a straight line of slope  $H$  at every  $p$ , and the four lines are parallel. A quantile is an order statistic: it is fixed by the middle of the distribution, where every flight contributes, and a heavy tail moves it not at all. Neither of the two defects of the earlier estimates enters here: there is no moment, so no small set of large increments can set it, and no absolute position, so the launch site cannot.

The four levels are the lower quartile, the median, the upper quartile and the ninetieth percentile. The last is far enough into the flank to move if the shape of  $F$  changes with the lag and far enough from the tail that the largest increments do not set it. The estimator is one streaming pass keeping a histogram of  $|\Delta x|$  per lag, per component and per native cadence, from which the quantiles are read (implementation details: Sec. 3).

**What it returns.**  $H = 0.901$  for paragliders and  $0.885$  for hang gliders, over 60 s to 2,000 s, from 489,204,233 and 11,631,258 increments in 6 and 2 native cadence classes. Doubled into the units the filtered variation is quoted in,  $\alpha = 1.80$  and  $1.77$ .

The smoothing window has a floor of 5 *samples* (Sec. 2.7.7), so the scale below which the pipeline has made the trajectory smooth is set by the logger’s own cadence, and the paraglider classes span a factor of ten in it. Across them the estimate moves between  $0.883$  and  $0.918$ . The smoothing scale therefore does not set the estimate.

**Table 3.3.** Three routes to  $H$ , on the same ensemble and the same window, 60 s to 2,000 s. The first and the third are second moments and differ in the frame they are taken in: the plain increment retains each flight’s course, the second-order variation annihilates it. The second is not a moment. The first is the quantity a published exponent for human soaring flight measures; the third is the route the chapter’s  $\alpha_2$  comes from, at  $\alpha_2 = 2H$ .

| route                        | what it uses                   | $H$ paragliders | $H$ hang gliders |
|------------------------------|--------------------------------|-----------------|------------------|
| plain increment $V_1$        | second moment, course retained | 0.876           | 0.852            |
| increment quantiles          | four order statistics          | 0.901           | 0.885            |
| second-order variation $V_2$ | second moment, course filtered | 1.009           | 0.972            |

**Three routes, and what they bracket.** Table 3.3 sets the quantile exponent beside the two second moments. They bracket it:  $H$  lies between  $0.876$  and  $1.009$  for paragliders and between  $0.852$  and  $0.972$  for hang gliders, and in both disciplines the bulk of the increment distribution sits inside that bracket, nearer the plain increment.

The gap between the two moments is the flown course, measured by the order scan and read here from the other side. It also fixes which row the published value is to be compared against. Vilpellet et al. [7] report a global  $H$  for human soaring flight on a different sample, obtained from a displacement statistic with no trend removed—which is the plain increment of the first row and not the second-order variation. The paraglider value in that row rounds to the reported one, on an independent archive and at the precision the reported value is quoted to. The second-order variation sits above it, and the sign of the gap is the one the order scan predicts: the archive’s closed tasks suppress displacement rather than inflate it, because a triangle comes home, so an exponent measured without removing the course is expected to fall below one measured with it. Which of the two answers a given question depends on the question, and the reference value is reproduced by the route that measures the same thing.

The bracket is the result rather than any of its decimals: two second moments taken in different frames and one statistic that is no moment at all, none of them measured from take-off.

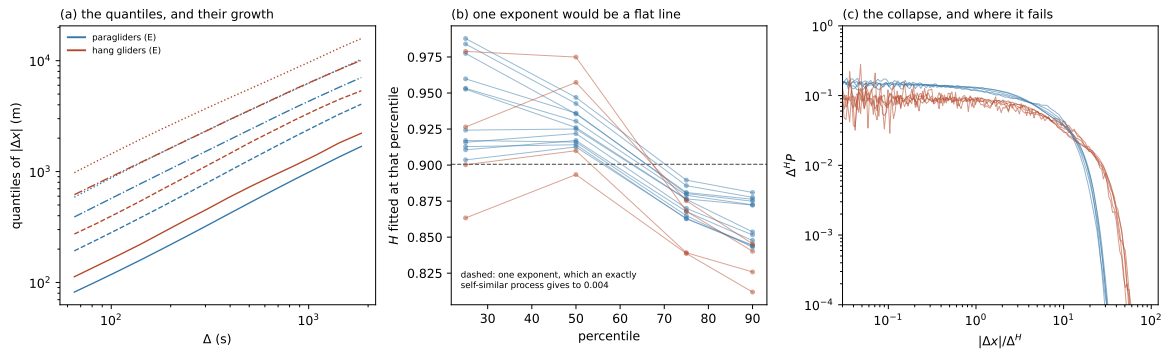
**What the quantiles cost the scaling form.** Equation (3.7) claims more than a single exponent, and the same histograms test the rest of it: the four quantiles must give the *same* slope. They do not. Fitted separately,  $H$  falls from  $0.926$  at the lower quartile and the median to  $0.872$  at the upper quartile and the ninetieth percentile for paragliders, and from  $0.918$  to  $0.840$  for hang gliders (Figure 3.3b). The fall runs in the same direction in both disciplines, on both components and at every cadence, which is what makes it a property of the increments rather than of one histogram.

Its size is read against a calibration measured the same way. The spread across the four quantiles is  $0.077$  and  $0.098$ , against  $0.004$  on an exactly self-similar process of  $H = 0.903$  fitted over the same window with less data (implementation details: Sec. 3). The collapse says the same in the other panel: rescaling the histograms by  $\Delta^H$  leaves  $0.031$  and  $0.044$  dex of scatter through the bulk, with visible separation where the flanks fall away (Figure 3.3c).

So the scaling form is not exactly satisfied over the range this chapter reads. The centre of the increment distribution spreads faster than its flanks, which is a distribution changing shape with the scale and not a single self-similar one.

That qualifies the exponent and does not refute it. The distance from the bulk value 0.926 to the flank value 0.872 is small beside the distance of either from the Brownian value, and no reading of these curves puts the transport near it. It is a second reason, independent of the movement of  $\alpha_2$  when the fitted range is halved, why a single exponent quoted to three figures would claim more than this archive supports.

**The anisotropy reaches the exponent.** Section 2.8 measures the two horizontal components as incompatible in amplitude. They differ in exponent as well: 0.890 east against 0.908 north for paragliders, and 0.860 against 0.911 for hang gliders, where the separation is the wider. Pooling the components would average that away. It is why they are kept apart here and in Sec. 2.8, and why Eq. (3.7) is written for one component directly rather than derived from a two-dimensional propagator through an isotropy substitution this ensemble does not support.



**Figure 3.3.** The exponent read from the distribution of the increments rather than from a moment of it. (a) The 25th, 50th, 75th and 90th percentiles of  $|\Delta x|$  against the lag, log-log, at the finest cadence of each discipline; a self-similar process gives four parallel lines. (b) The exponent fitted at each percentile separately, one curve per cadence and component, against the dashed single exponent an exactly self-similar process would give. Every curve falls from the bulk to the flank. (c) The same histograms rescaled by  $\Delta^H$ : they coincide through the bulk and part company where the flanks fall away.

### 3.5 The archive is not a Lévy walk

An exponent fixes one moment of one distribution, and two processes can agree on it and be different physics. A correlated Gaussian motion and a Lévy walk both give  $\langle |\Delta \mathbf{r}|^2 \rangle \sim \Delta^{2H}$ ; what separates them is the rest of the moment spectrum and the distribution of the durations over which the motion holds a heading. Both are measured on the same unsegmented trajectories, with no estimator in common, and both refuse the Lévy walk.

**The moment spectrum.** Write

$$\langle |\Delta \mathbf{r}(\Delta)|^q \rangle \sim \Delta^{q\nu(q)} \quad (3.8)$$

and read  $q\nu(q)$  as a function of  $q$ . A self-similar monofractal process gives a straight line through the origin with  $\nu(q)$  constant: one exponent governs every moment. A Lévy walk—flight durations drawn from a power-law tail, each traversed at finite speed—gives a bilinear

one, with a knee at its tail index and a right-hand branch of slope one, the ballistic front carried by the longest flights. The discrimination is between two shapes rather than between two fitted values, and it is computed on the whole archive rather than on a subsample of it.

Over 60 s to 8,000 s, on 24 lags and for  $q$  from 0.50 to 4, the spectrum is straight (Figure 3.4a). For paragliders  $\nu(q)$  varies by 0.019 across the whole range of  $q$ , between 0.847 and 0.866, about a single fitted  $\nu = 0.860$ ; for hang gliders it varies by 0.009 about  $\nu = 0.830$ . The rms departure of  $q\nu(q)$  from a straight line through the origin is 0.0109 and 0.0057. Neither archive shows strong anomalous diffusion.

Those departures are read against two references, and the two are not the same quantity. The first is what a Lévy walk leaves on this lag grid and this range of  $q$  when its spectrum is fitted the same way: its knee is a gross feature of the shape, and the measured departures sit an order of magnitude below what it produces. The second is the gate at which the fitting procedure declares a knee at all, set low so that a weak bend is not missed; the measured departures sit below that as well, by a smaller margin. Both references are stated in (implementation details: Sec. 3), and it is the first that carries the refusal: the archive is not merely below the threshold of declaration, it is far from the shape the model predicts.

The moments are estimable rather than merely computed. The share of each moment carried by the largest hundredth of its increments reaches 28 % at worst for paragliders and 17 % for hang gliders, and 16 % and 15 % at the top of the range,  $q = 4$  (Figure 3.4b). A moment more than half of which is carried by that hundredth is a statement about a few flights rather than an estimate; none of these is near it.

The Lévy walk is the model this thesis was framed around. On the whole trajectory, unsegmented, the archive does not support it, and what the spectrum leaves standing is a monofractal process: a single exponent governing every moment over the range read.

**The spectrum against the quantile spread.** The previous section fitted the quantiles of  $|\Delta x|$  separately and found that they do not share a slope:  $H$  falls from 0.926 at the lower quartile and the median to 0.872 at the upper quartile and the ninetieth percentile, and the same fall appears for hang gliders. The scaling form is therefore only approximately satisfied, while Eq. (3.8) has just been reported as straight. The two are compatible, and what reconciles them is what each averages over. A moment of order  $q$  integrates the whole increment distribution, so a local exponent drifting between the centre and the flanks enters  $\nu(q)$  weighted by where that moment takes its mass, and the drift partly cancels in the integral. A quantile fit reads the same collapse at four fixed probability levels and averages nothing.

The measured sizes are ordered as that argument requires. The spread across the four quantiles is 0.077 for paragliders and 0.098 for hang gliders, against spreads of  $\nu(q)$  across the whole range of  $q$  of 0.019 and 0.009: the quantity that averages is the smaller in both disciplines. The two are read over ranges sharing a lower end and differing at the top, 60 s to 2,000 s against 60 s to 8,000 s. What is not established is the bound: the drift measured at four probability levels has not been propagated through the moment integral to a predicted variation of  $\nu(q)$ , so the claim is the ordering and not that the one accounts for the other. The refusal above is untouched either way, since the departure a knee would leave exceeds both.

**The persistence runs.** The second route uses no moment. Each flight is decomposed into non-overlapping stretches whose sinuosity—arc length over chord—stays below a threshold, and the durations of those stretches are the geometric counterpart of a waiting time, the quantity whose power-law tail Appendix 3.A takes as the input of a continuous-time random walk. A Lévy walk ties the tail index  $\beta$  of that distribution to the transport exponent, so the description needs a run length that exists independently of how a turn is declared.

Scanning the sinuosity threshold across three values, from nearly straight to loosely sinuous,



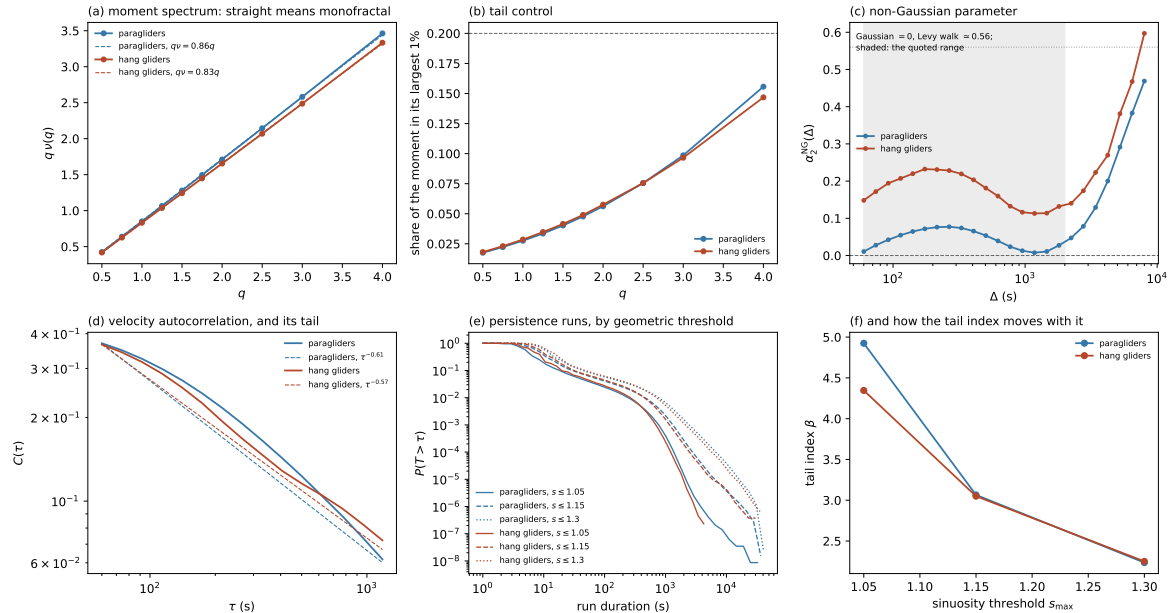
moves  $\beta$  from 4.92 through 3.07 to 2.24 for paragliders, a spread of 2.69, and from 4.35 through 3.05 to 2.25 for hang gliders, a spread of 2.10 (Figure 3.4f). The decomposition moves with it: 115,263,620 paraglider runs of median 5 s at the tightest threshold against 39,766,017 of median 11 s at the loosest. The survival curves at the three thresholds are not parallel (Figure 3.4e), which is the same statement before any tail is fitted.

A run length whose tail index moves by 2.69 with the angle at which a turn is declared has no scale of its own. That is the moment spectrum's result reached through the geometry instead of through the moments: there is no distinguished flight-leg duration for a Lévy description to rest on.

The runs are decomposed without overlap, and the alternative changes the answer rather than refining it. Sampling the run that contains each instant returns the length-biased distribution, whose tail index is one below the truth (implementation details: Sec. 3); at the values measured here that is not a bias to correct but a different regime.

The scan is also a prediction about Chapter 4. A segmentation into climbs and glides declares a threshold of the same kind on the same trajectories, and a tail index that moves by 2.69 across three sinuosity thresholds is the measured warning that the phase statistics will move with theirs. That threshold must be scanned and the movement reported rather than chosen once.

The two routes share no estimator, and neither supports a Lévy walk on the unsegmented trajectory. What survives is a monofractal process, one exponent for every moment over the range read, which fixes how the increment distribution scales and says nothing about its shape.



**Figure 3.4.** What the second moment cannot say. (a) The moment spectrum  $q\nu(q)$  with the straight line dashed: straight means one exponent governs every moment, and a Lévy walk would bend with a knee at its tail index. (b) The share of each moment carried by the largest 1% of its increments. (c) The non-Gaussian parameter against lag, with the Gaussian value and the Lévy-walk calibration marked and the quoted range shaded. (d) The velocity autocorrelation, log-log with the fitted tail dashed. (e) Survival of the persistence-run durations at three geometric thresholds. (f) The fitted tail index against the threshold that defines a run.

### 3.6 The propagator is not Gaussian

A straight moment spectrum says that one exponent governs every moment of the increment distribution. It fixes how that distribution *scales* and says nothing about its *shape*, so the natural next step—calling the process a correlated Gaussian motion—does not follow from it. The fourth and second moments already accumulated settle the shape directly, through the non-Gaussian parameter

$$\alpha_2^{\text{NG}}(\Delta) = \frac{\langle |\Delta \mathbf{r}(\Delta)|^4 \rangle}{2 \langle |\Delta \mathbf{r}(\Delta)|^2 \rangle^2} - 1, \quad (3.9)$$

evaluated on the increment over a lag  $\Delta$  rather than on the displacement from take-off, in keeping with how the propagator is defined above. The normalisation is the two-dimensional Gaussian value of  $\langle |\Delta \mathbf{r}|^4 \rangle / \langle |\Delta \mathbf{r}|^2 \rangle^2$ , so  $\alpha_2^{\text{NG}}$  vanishes for a Gaussian propagator and is positive for heavier-than-Gaussian tails. The superscript separates it from the filtered-variation exponents  $\alpha_p$ : the literature writes the non-Gaussian parameter  $\alpha_2$  as well, which at filter order  $p = 2$  is the symbol this chapter's exponent already carries. The two are unrelated quantities.

Over 60 s to 1,805 s the median value is +0.042 for paragliders and +0.181 for hang gliders. Neither is flat across it. Both rise to an interior maximum and fall back: +0.077 at 266 s for paragliders, against +0.011 at the lower end of the window and +0.027 at the upper, a factor of 7.2; and +0.232 at 174 s for hang gliders. The rise and the fall are each monotone over seven or more consecutive lags, so this is a shape and not scatter.

A maximum at two to four hundred seconds is a scale, and the chapter has otherwise found none: the moment spectrum is straight, the run-length tail index has no stable value, and the exponent is read as a single effective number over the whole window. What the maximum says is narrower than a scale in the transport — it is the lag at which the *shape* of the increment distribution departs furthest from Gaussian, with the departure smaller both below it and above it. Below, the increments are averages over a smoothing window and a few heading changes; above, a central-limit Gaussianisation begins to act. Reporting the two ends of that curve as a difference, which an earlier version of this section did, is the chord of an arch — the error diagnosed for the ensemble MSD in Sec. 3.1 and repeated here.

**What those values are worth.** A number of this size is readable only against what the same estimator returns on processes whose shape is known, and the reference has to be read at the sample size the archive is measured at. On an exact fractional Brownian motion the estimator returns a small positive value over the modest sample the calibration uses, and that value collapses towards zero, sign included, as the sample is grown; the offset is finite-sample bias and not a floor of the estimator. On a Lévy walk it returns a figure far above either archive's and does not move with the sample (implementation details: Sec. 3). Against that pair both archives are non-Gaussian and neither is close to Lévy. The hang-glider figure sits well clear of the fractional-Brownian offset; the paraglider figure is a small multiple of it.

**The disciplines are not equally non-Gaussian.** They are not averaged either. Over the window the hang-glider parameter never falls below +0.113 and the paraglider parameter never rises above +0.077, so the two do not overlap at any lag in it, and the separation between them is larger than the movement either curve shows from one end of the window to the other. Whatever makes these increments heavier-tailed than Gaussian acts more strongly on the smaller and slower discipline, by the ratio of +0.181 to +0.042. That asymmetry is a property of the two archives rather than scatter between them, and any mechanism Chapter 4 proposes for the shape of the increment distribution has to reproduce it.

**Beyond the fitted range.**  $\alpha_2^{\text{NG}}$  climbs, to +0.47 and +0.60 at the longest lags of panel (c) of Figure 3.4. Those values are reported and not quoted, for the two reasons that bound every fitted range in this chapter. They lie where the declared task governs the trajectory rather than the air, and a fourth moment of a displacement the scoring rule bounds is a statement about the scoring rule; and they lie where that moment is carried by the fewest flights, since the ensemble contributing to a lag thins as the lag approaches the flown durations.

The process is therefore monofractal and its propagator is not Gaussian. One exponent for every moment fixes how the increment distribution scales, and leaves the shape it scales as heavy as it is measured to be.

### 3.7 The memory of the heading

Writing  $\delta\mathbf{v}$  for the horizontal velocity fluctuation about a record's own mean, the velocity autocorrelation is

$$C(\tau) = \frac{\langle \delta\mathbf{v}(t_0) \cdot \delta\mathbf{v}(t_0 + \tau) \rangle_{t_0}}{\langle |\delta\mathbf{v}(t_0)|^2 \rangle_{t_0}}, \quad (3.10)$$

normalised so that  $C(0) = 1$ , with the flown course, which would otherwise hold every lag up, removed before the average is taken. It is computed within a retained segment and never across the gap that ends one (Sec. 2.7.6), by the Fourier identity rather than by a double loop, and averaged across records at each lag of the grid the moments use (implementation details: Sec. 3).

**Positive at every lag, and small at the first.** Two things come off the measured curve directly (Figure 3.4). It is positive throughout and changes sign nowhere on the grid:  $C = +0.37$  for paragliders at the floor and +0.37 for hang gliders, decaying from there without a zero crossing. A sign change is what a circling wing produces, since half a turn later its velocity points the other way, and there is none. At the ensemble level the circling does not survive the average, which is the first constraint on Chapter 4: a phase decomposition has to *produce* the anti-persistence of a climb, because no statistic on the unsegmented trajectory carries it.

The second is that  $C$  is already small at the bottom of the grid. It falls below  $1/e$  by 74 s for paragliders and 60 s for hang gliders, within the first lags of it. That figure is a property of the pipeline as much as of the flying: the grid starts at 60 s because that is the smoothing scale of the slowest logger retained (Sec. 2.7.7), so the decay of  $C$  over the first decade is not measured here. What survives above the floor is small.

**Integrability rather than size.** That last statement and the exponent look incompatible. A memory spent by 60 s should leave the motion diffusive over a range reaching to 2,000 s, and the filtered variation says the increments are still correlated across all of it, at  $\alpha_2 = 2.02$  and 1.94. Both cannot hold if the size of  $C$  is what sets the growth law, and it is not.

For a stationary velocity the Green–Kubo relation

$$\langle |\Delta\mathbf{r}(\Delta)|^2 \rangle = 2 \int_0^\Delta (\Delta - \tau) C(\tau) d\tau \quad (3.11)$$

makes the growth law a statement about the convergence of an integral. A tail  $C(\tau) \sim \tau^{-\gamma}$  with  $\gamma < 1$  has no finite integral, and Eq. (3.11) then gives  $\langle |\Delta\mathbf{r}|^2 \rangle \sim \Delta^{2-\gamma}$  across the window however small  $C$  has already become; with  $\gamma > 1$  the integral converges and the motion is diffusive beyond the correlation time whatever  $C$  is worth at the floor. Size and integrability are independent properties of the same curve, and it is the second that Green–Kubo turns on: multiplying  $C$  by a constant moves every value on it and leaves  $\gamma$  where it was.

Over 60 s to 1,179 s, which is 15 lags and the same lags in both disciplines, the measured tail is a power law of  $\gamma = 0.61$  for paragliders and 0.57 for hang gliders. Neither is integrable. The memory is not spent at the bottom of the grid, only small there, and the objection does not stand.

**The comparison is one-sided.** Two features of the estimate push  $\gamma$  up, and both are known in sign. The correlation is taken about each record's own mean velocity, and a mean estimated from the same record pulls every lag down by about the record-mean of  $C$ ; that is negligible where  $C$  is large and is most of it where  $C$  is small, which is the tail, so the fitted decay is steeper than the true one. And the first decade of the integral in Eq. (3.11) lies below the grid, so the fit reads the tail without the stretch that carries the largest part of the weight. The exponent  $2 - \gamma$  implied for the displacement is therefore a floor rather than an estimate: 1.39 and 1.43.

The quantity that has to clear it is the plain increment, which is the displacement statistic Eq. (3.11) predicts, with no trend removed:  $\alpha_1 = 1.75$  for paragliders and 1.70 for hang gliders. Both clear their floors. That is a consistency between two channels rather than a second determination of the exponent, since a bound cleared from above fixes nothing. What it excludes is a velocity memory integrable over this window, which would have contradicted the position statistic outright.

**The turning angles.** The persistence has a microscopic form and it is measured, in the same units a walker model would use: the angle between successive horizontal displacement vectors, taken at a stride of 5 s so that it is the flying and not the smoothing that is read. Its median is  $11^\circ$  for paragliders, over 272,503,728 angles, and  $19^\circ$  for hang gliders over 6,893,892; 63 % and 56 % of them fall within  $\pi/6$  of the previous heading. Memoryless reorientation is the reference, and it is nowhere near: its median is a right angle, and the share it leaves within  $\pi/6$  is the fraction of the half-turn that cone spans.

A wing that changes heading by  $11^\circ$  at a time is a correlated walk and not a memoryless one. A positive  $C(\tau)$ , a tail too slow to be integrable, and increments still correlated at 2,000 s are the ensemble consequences of the same forward peak. It is also the warrant for the persistent random walk of Sec. 4.4, which needs the angle distribution as an input rather than as a check.

The distribution is not that peak alone, and the remainder is the second constraint on Chapter 4. A wing turns when it climbs, and Sec. 2.8 measures a paraglider climbing 45 % of its time and a hang glider 47 %; the angles outside the forward cone are a minority of the same order. Pooled, the two components give one forward-peaked distribution and a  $C(\tau)$  with no sign change in it, which is why the circling is invisible to both statistics here and not absent from the flying. The segmentation of Sec. 4.1 separates them, and the turning angles conditioned on phase are where this result has to reappear.

## 3.8 Speed and vertical velocity

The horizontal speed and the vertical velocity are marginals of the smoothed kinematics of Sec. 2.7.7, taken over every retained in-segment fix. They need no lag grid and no scaling argument, and they are accumulated in the same streaming pass that builds the increment histograms of Sec. 3.4 (implementation details: Sec. 3). Each is read against a reference fixed outside the archive: the bounds under which the cleaning admitted the data, and the altitude a flight of several hours must return to.

**Horizontal speed.** The median is  $9.6 \text{ m s}^{-1}$  for paragliders and  $15.6 \text{ m s}^{-1}$  for hang gliders, with ninetieth percentiles of  $13.4 \text{ m s}^{-1}$  and  $22.0 \text{ m s}^{-1}$  and ninety-ninth percentiles of  $17.1 \text{ m s}^{-1}$  and  $28.6 \text{ m s}^{-1}$ . The disciplines separate at every percentile quoted, and the hang-glider median lies above the paraglider ninetieth percentile.

Both ninety-ninth percentiles sit well inside the fix-level speed bounds of Sec. 2.7.2,  $45 \text{ m s}^{-1}$  for paragliders and  $55 \text{ m s}^{-1}$  for hang gliders. That margin is a check on the cleaning: a retained population pressed against the threshold that admitted it would mean the bound had cut into the fast tail of the flying rather than into instrument artefacts. The comparison is one-sided, since the bound acts on the inter-fix speed and the percentiles are of the smoothed velocity, which is the smaller of the two.

**Vertical velocity.** The vertical velocity is centred just below zero, at  $-0.16 \text{ m s}^{-1}$  and  $-0.18 \text{ m s}^{-1}$ , with a 10 % to 90 % band of  $-2.08 \text{ m s}^{-1}$  to  $2.04 \text{ m s}^{-1}$  for paragliders and  $-2.51 \text{ m s}^{-1}$  to  $2.49 \text{ m s}^{-1}$  for hang gliders, inside the fix-level bound of  $13 \text{ m s}^{-1}$  at both ends — though a decile band says less about the cleaning than the ninety-ninth percentile of the speed does.

Paragliders climb 45 % of the time and hang gliders 47 %. A flight that ends near the altitude it began at divides its time between climbing and sinking in the inverse ratio of the two rates, and the band above makes those rates comparable, so over the median flight of 2.63 h and 3.07 h recorded in Sec. 2.8 the climbing share is required to be near a half, which both measured shares are. Both fall short of a half, which is the same fact as the negative median: the wing spends more of its time sinking, so its mean rate of climb exceeds its mean rate of sink.

The sign of the vertical velocity is the coarsest climb–glide rule available, so these two shares are what a segmentation in Chapter 4 must return as its threshold goes to zero, and this is the distribution any such threshold cuts.

**The return probability.** The return probability is the increment propagator at its centre,  $P_0(\Delta) \equiv P(\mathbf{0}, \Delta)$ ; the scaling form of Sec. 3.4 makes it decay as  $\Delta^{-dH}$  in  $d$  dimensions and as  $\Delta^{-H}$  on one component, so its slope is another reading of  $H$ . It is defined here and not measured.

That reading would not repeat the one already made. Sec. 3.4 reads  $H$  at four fixed probability levels and finds that they do not share a slope, by 0.077 and 0.098; the peak reads the same  $F$  at a fifth level, inside the innermost of the four. The ordering measured there predicts an exponent at the peak at or above 0.926, and a value below it would place the drift in the estimator rather than in the distribution. It is also the cheapest check remaining on the headline exponent: the streaming histograms of  $|\Delta x|$  per lag, per component and per native cadence that supply those quantiles hold their own centre as well, so  $P_0$  costs a fit and no further traversal of the fix table.

First-passage times are not on this list: they were measured and withdrawn, in Sec. 3.4, because a time to reach a radius from take-off carries the launch geometry into the space domain.

### 3.9 What this establishes, and what the segmentation receives

Over 60 s to 2,000 s, measured within a retained segment by an estimator that annihilates a polynomial trend rather than estimating one, cross-country soaring gives  $\alpha_2 = 2.02 \pm 0.10$  for paragliders and  $1.94 \pm 0.13$  for hang gliders, on 155,788 and 6,132 flights. The wing classes that depart from the pooled ensemble MSD by 40 % (Sec. 2.8) move this exponent by 0.06; native cadence, across a factor of ten in the scale below which the trajectory has been smoothed,

moves it by 0.02, season by 0.03, and the declared task by  $-0.10$ , the largest of the four. The two disciplines agree within their uncertainties, which is not guaranteed: they differ in speed and in glide ratio.

**The class of process.** Every moment scales with that one exponent. Over the fitted range and  $q$  from 0.50 to 4,  $\nu(q)$  varies by 0.019 for paragliders and 0.009 for hang gliders, and the moment spectrum departs from a line through the origin by 0.0109 and 0.0057 in rms, an order of magnitude below the departure a Lévy walk’s knee would leave on this grid. The propagator is nevertheless not Gaussian:  $\alpha^{\text{NG}} = +0.042$  and  $+0.181$ , flat across the same window. The velocity autocorrelation is positive at every lag of the grid, and its tail is a power law of  $\gamma = 0.61$  and  $0.57$ , non-integrable in both, which is what sustains correlated increments to 2,000 s from a correlation already below  $1/e$  by 74 s. No leg duration is distinguished: the tail index of the persistence-run durations moves from 4.92 to 2.24 as the sinuosity threshold that defines a run is scanned, a spread of 2.69. The quantiles of the increment distribution, which are neither a moment nor a statistic about a common origin, put  $H$  at 0.901 and 0.885, between the two moment routes. Over this window the archive is a monofractal, non-Gaussian, directionally persistent motion with a non-integrable velocity memory — neither the Lévy walk this thesis was framed around nor a Gaussian one.

**The limits.** The window is 1.5 decades wide, and neither end is set by the size of the archive. Below it the trajectory has been smoothed over a window whose floor is 5 samples and which therefore scales with the logger’s cadence (Sec. 2.7.7); nothing below that scale is read as motion. Above it the declared task governs the displacement, the separation between closed and open courses in the local slope growing from 0.03 at 60 s to 1.12 at 12,156 s. Inside the window the scaling form is only approximate:  $H$  falls from 0.926 at the bulk of the increment distribution to 0.872 at its flank, a spread across the quantiles of 0.077 and 0.098 against 0.004 on an exactly self-similar process measured the same way. The exponent itself moves by 0.104 when the fitted range is halved, against 0.001 from resampling at the day and site, and the 22 lags of the fit carry no more than 2.0 effective degrees of freedom, which is a budget for one exponent and not for a count of regimes. The filtered variation has an effective exponent of 2.02 over 60 s to 2,000 s; that the transport is characterised by  $\alpha = 2.02$  is a different statement, and the range dependence and the quantile spread are each enough to withhold it. The window will not widen with more data: a soaring day ends, so the upper cutoff follows from the system rather than from the method.

**What was withdrawn.** Four measurements are withdrawn. The ensemble MSD about take-off is withdrawn as a source of exponents and yields a timescale instead: displacement from the launch site crosses over where the population leaves its launch area, at a median of 1,725 s for paragliders and 1,530 s for hang gliders on the first crossing of 5 km. The first-passage time was brought in as a route to  $H$  in the space domain and withdrawn when its exponent moved across sub-ranges of radius by more than the uncertainty it was meant to check; what it bought is the sharpest evidence that the ensemble curve’s failure belongs to the common origin rather than to the time axis, since exchanging the time axis for a radius reproduces it (implementation details: Sec. 3). The sign-run count on the MSD residuals is kept as a description of how those residuals are arranged and withdrawn as a test, every lag of that curve being an average over the same flights; what carries the claim instead is the size of the residual against the standard error on each lag. And the reading of a straight moment spectrum as a Gaussian propagator is withdrawn, one exponent for every moment being a statement about how the distribution scales and none about its shape; what it bought is the non-Gaussian parameter, computed from moments the spectrum had already accumulated.



**The hand-over.** Chapter 4 inherits a class of transport and six constraints on how it may be decomposed.

- (i) Any statistic measured from take-off inherits the launch geometry. On the displacement statistic that contamination governs the curve out to the departure times above, 1,725 s and 1,530 s, and it is not confined to the observable that exposed it: any phase statistic binned by elapsed time since take-off carries it below those times. The remedy is the one used throughout this chapter, to measure within a segment and never about a common origin (Sec. 2.7.6).
- (ii) The estimator is additive over any partition of the sample.  $V_2(\Delta)$  is a mean over the placements of a second difference; labelling those placements by phase and recombining the per-phase means with the weights actually observed returns the pooled curve identically, and with it  $\alpha_2 = 2.02$  and  $1.94$  over 60 s to 2,000 s. That is a property of the estimator rather than of the transport: it asserts no decomposition of an exponent into phase exponents, only that the arithmetic closes. A recombination that misses the pooled value is therefore evidence about the segmentation — placements dropped at a phase boundary, counted twice, or straddling one and assigned to neither — and is to be read that way before it is read as physics.
- (iii) A climb–glide threshold is declared in the same way as the sinuosity threshold of the persistence runs, and that threshold moved the run-length tail index by 2.69 across its scan. Every phase statistic is to be reported against a scan of the threshold that produced it, with the movement quoted.
- (iv) The two horizontal components are analysed separately and never pooled. They are incompatible in amplitude (Sec. 2.8) and they differ in exponent:  $H = 0.890$  east against 0.908 north for paragliders, and 0.860 against 0.911 for hang gliders. Pooling presupposes an isotropy this ensemble does not have.
- (v) The non-Gaussian parameter is +0.042 for paragliders and +0.181 for hang gliders over the same window: whatever makes these increments heavier-tailed than Gaussian acts about fourfold more strongly on the smaller, slower wing. Any mechanism proposed for the phase structure has to produce that gap rather than average it away.
- (vi) The ergodicity question is handed on in the two forms that need no power law. The first is the drift of the exponent with the time already flown, obtained by running the same within-segment estimator on successive stretches of a record: an ageing process shows such a drift and a launch-geometry artefact does not (Sec. 4.1). The second is the ergodicity-breaking parameter of Appendix 3.A, which compares the distribution of the per-flight time average across flights rather than two fitted slopes. The usual comparison of  $\alpha_{EA}$  with  $\alpha_{TA}$  requires both curves to be power laws, and this archive supplies one of the two.

The class of transport is fixed over one window; which flight behaviour produces it is not. Every number above mixes climbing, gliding and searching into one, and the persistent random walk of Sec. 4.4 takes its warrant from the memory measured here rather than from any single one of them. Chapter 4 separates the phases, and is answerable to the six constraints.

## Chapter 4

# Flight phases, and what they explain

Chapter 3 determines which class of transport the archive shows. It cannot say which flight behaviour generates it, because every observable there is an average over climbs, glides and searches together. Separating them needs a segmentation of each flight into its phases, and that is the work this chapter plans.

The distinction is worth stating plainly, because it sets what a segmentation is allowed to claim. Pre-segmentation determines *which* class of transport is present; segmentation explains *which mechanism* produces it. A segmentation explains an exponent, it does not measure one — and if the two disagree, the disagreement is evidence about the segmentation rather than about the transport.

None of this chapter is implemented. Each section states what is to be built and what it would settle.

### 4.1 Segmentation into flight phases

Building on the previous work of Vilpellet et al. [7], each flight is segmented into the three phases of the soaring cycle:

**Transition (glide).** The inter-thermal cruise: a near-straight, energy-efficient glide toward the next source of lift (sinking, large directed horizontal displacement). This is the directed “step”.

**Search.** The exploratory phase entered when no thermal is found at a comfortable altitude: gentle turns and probing manoeuvres to sample the air mass and locate the next thermal while limiting altitude loss. It mediates the passage from one step to the next trap.

**Climb (thermallng).** Circling within rising air to regain altitude: persistent turning with small net horizontal displacement. This is the localized “trap”.

That work performs the segmentation with a hidden Markov model [22] whose observations are a small set of *binary* trajectory-derived features — the sign of the vertical speed, the persistence of the curvature angle, and indicators of trajectory straightness.

A first task of this thesis is to scrutinize this segmentation rather than take it for granted: to assess whether those binary variables are adequate and robust on the larger, more heterogeneous dataset assembled here, and whether the feature set or the model should be refined, extended, or replaced. The segmentation method is therefore itself an object of study, and the specific variables are deliberately left open at this stage. This three-state description refines the two-state caricature underlying continuous-time random walks and Lévy walks.

## 4.2 Phase-resolved observables

The segmentation gives access to a second group of observables, defined per phase occurrence and then as ensemble distributions. These are the building blocks of an intermittent-transport description and, unlike those of Section 3.1, cannot be obtained without first segmenting the flight.

**The waiting phase: climb and search.** Between two consecutive glides the flier advances little horizontally: it searches for a thermal and climbs in it. For a transport model these two phases together form the *waiting time* between directed steps, so we treat them on an equal footing and build three distributions – the climb duration  $\tau_c$ , the search duration  $\tau_s$ , and their sum  $\tau_w = \tau_s + \tau_c$ , the total time spent essentially stationary between glides. The distinction matters: Vilpellet et al. [7] report the search times to be broadly *power-law* distributed while the climb times are closer to *exponential*, so that the heavy tail of the combined waiting time  $\psi(\tau_w)$ , the quantity that determines whether trapping drives sub-diffusion, would be inherited from the search phase. Whether this still holds on the larger dataset is itself one of the things to re-check once the segmentation is in place. We therefore report  $\psi(\tau_w)$  together with the separate  $\psi(\tau_c)$  and  $\psi(\tau_s)$ . Alongside the durations we record the climb kinematics (altitude gained  $\Delta z$ , mean climb rate, thermalling radius and period) and the search geometry (path length, tortuosity, net displacement) – the search being the phase in which prior work locates the signature of pilot skill.

**The directed step: glide (transition).** Each glide contributes a horizontal step of length  $\ell$  and duration  $\tau_g$ , with ground speed, glide ratio (horizontal distance per unit altitude lost) and sink rate. Because a glide is close to ballistic,  $\ell \approx v_g \tau_g$ , step length and duration are expected to be nearly proportional; we test this directly through the joint law of  $(\ell, \tau_g)$  – the  $\ell$ – $\tau_g$  scatter, the conditional mean  $\langle \ell \mid \tau_g \rangle$ , and the spread of the ratio  $\ell/\tau_g$  (the glide speed). Both marginals matter: the tail of the step-length distribution  $\lambda(\ell)$  controls super-diffusive contributions, so we compute the distributions of  $\ell$  and  $\tau_g$  separately as well as their joint law.

**Anatomy of a flight.** Number of thermals (climbs) per flight, inter-thermal distances, and the fraction of both time and along-track distance spent in each phase: the coarse composition that sets the weight of each phase in the global transport.

**Angular diffusion between glides.** The turning angle between the heading of one transition segment and the next – the segment-level counterpart of the turning angle of Section 3.1. How fast successive glide directions decorrelate is the key ingredient of the minimal model below.

Two couplings then decide which transport model applies, and are stated here as explicit objects of study. The first is the *step–waiting coupling*: whether a long glide tends to be followed by a long wait, i.e. whether  $\ell$  and  $\tau_w$  are correlated. A genuine coupling is the defining feature of a Lévy walk and is what separates it from a decoupled CTRW, in which jumps and waiting times are drawn independently.

The second is the *memory in the phase sequence*. A flight is a discrete sequence over the three states {transition, search, climb}, and memory can hide in two places. (i) The *order* of the phases, summarised by the transition matrix  $P_{ij} = \text{Pr}(\text{next} = j \mid \text{current} = i)$ : a perfectly cyclic flight ( $T \rightarrow S \rightarrow C \rightarrow T$ ) gives a near-deterministic matrix, whereas aborted climbs, repeated searches or skipped phases appear as off-cycle probabilities and reveal how stereotyped the soaring cycle actually is. (ii) The *durations*, through correlations between consecutive phase durations – for instance whether a long climb tends to precede a long glide.

Both probe the *renewal* assumption underlying the textbook CTRW and Lévy-walk results, in which each step and wait is independent of the history: if the transition matrix is structured or the durations are correlated, the process is non-renewal and those formulas must be amended. These measurements determine whether a memoryless model is admissible.

### 4.3 Reproducing the reference results

A first objective of the analysis, undertaken once the segmentation is in place, is to check whether the enlarged—and eventually multi-source—dataset reproduces the quantitative and graphical results of Vilpelle et al. [7]. That study covers paragliders, hang gliders and sailplanes; the present data cover the first two (Chapter 2), so both the paraglider and the hang-glider comparisons are within reach, while the sailplane figures become reachable once that source is added. Two figures are the main targets.

The first is the per-phase conditional MSD (their Fig. 3): the transition segments should stay ballistic ( $\alpha = 2$ , i.e.  $H = 1$ ) across time lags; the search phase should be strongly sub-diffusive ( $H \approx 0.3$  for paragliders and hang gliders,  $\approx 0.2$  for sailplanes); and the climb phase should be quasi-ballistic at long lags but with an effective velocity about two orders of magnitude smaller, including the transient anti-correlation near the  $\sim 30$  s thermalling period.

The second is the set of per-phase descriptive statistics (their Fig. 4): the glide ratios ( $\approx 8.5$ ,  $10$ , and  $25$  for paragliders, hang gliders, and sailplanes), the horizontal-velocity statistics, the heavy-tailed distribution of transition times, the search-time fraction and the effective search radius, the vertical climb velocities, the thermalling radii, and the thin, near-exponential distribution of climb times—together with the global time budget ( $\approx 60\%$  transition,  $\approx 35\%$  climb, the remainder searching). The global exponent is not on this list: it needs no segmentation, and Sec. 3.9 has already put this archive’s value beside the reported  $H \approx 0.88$ .

Recovering these results would validate the acquisition pipeline and the segmentation and would test the universality of the reported behaviour on a substantially larger sample; systematic departures would be equally informative, pointing to dataset- or method-dependent effects to investigate. This comparison is therefore the bridge between the present data-acquisition stage and the original analysis that follows.

### 4.4 Modelling and simulation

The aim is a *minimal* stochastic model that reproduces the asymptotic anomalous scaling reported for soaring (a Hurst exponent  $H \approx 0.88$ ) with as few ingredients as possible. A preliminary working hypothesis—to be made precise, and possibly revised, once the empirical phase statistics are available—is that the transport is carried mainly by the transition (glide) phases. The model would therefore resolve the transition segments explicitly while merging search and climb into a single effective non-transition state, so that the soaring cycle reduces to a sequence of directed glides interrupted by reorientations. Anomalous transport would then emerge from the directed transition segments combined with an *angular diffusion* of the heading from one transition to the next: successive glide directions decorrelate gradually rather than abruptly, producing a correlated random walk whose persistence controls the long-time exponent. Simulated ensembles would be compared with the data to test whether these ingredients suffice. These are baseline ideas, to be refined—or changed—as the analysis progresses.

## **4.5 Tooling**

Mirror the acquisition package with an analysis package, and let the figures and tables of this document be generated automatically from the analysis outputs – as the dataset statistics already are – so that this document stays in step with the work and remains a reliable basis for the thesis and for presentations.

# Bibliography

- [1] Ralf Metzler and Joseph Klafter. The random walk’s guide to anomalous diffusion: a fractional dynamics approach. *Physics Reports*, 339(1):1–77, 2000.
- [2] V. Zaburdaev, S. Denisov, and J. Klafter. Lévy walks. *Reviews of Modern Physics*, 87(2):483–530, 2015.
- [3] G. M. Viswanathan, Sergey V. Buldyrev, Shlomo Havlin, M. G. E. da Luz, E. P. Raposo, and H. Eugene Stanley. Optimizing the success of random searches. *Nature*, 401(6756):911–914, 1999.
- [4] Jean-Philippe Bouchaud and Antoine Georges. Anomalous diffusion in disordered media: Statistical mechanisms, models and physical applications. *Physics Reports*, 195(4–5):127–293, 1990.
- [5] T. H. Solomon, Eric R. Weeks, and Harry L. Swinney. Observation of anomalous diffusion and lévy flights in a two-dimensional rotating flow. *Physical Review Letters*, 71(24):3975–3978, 1993.
- [6] Gautam Reddy, Antonio Celani, Terrence J. Sejnowski, and Massimo Vergassola. Learning to soar in turbulent environments. *Proceedings of the National Academy of Sciences*, 113(33):E4877–E4884, 2016.
- [7] Jérémie Vilpelle, Alexandre Darmon, and Michael Benzaquen. From random walks to thermal rides: Universal anomalous transport in soaring flights. *arXiv:2601.01293 [cond-mat.stat-mech]*, 2026.
- [8] Fédération Aéronautique Internationale, International Gliding Commission. Technical specification for IGC-approved GNSS flight recorders. Second Edition, with Amendment 8, 2023. Defines the IGC flight-data file format. [https://www.fai.org/sites/default/files/igc\\_fr\\_specification\\_with\\_al8\\_2023-2-1\\_0.pdf](https://www.fai.org/sites/default/files/igc_fr_specification_with_al8_2023-2-1_0.pdf).
- [9] Fédération Aéronautique Internationale. FAI sporting code, common section 7: Hang gliders and paragliders. 2024 edition, 2024. Defines the FAI hang-glider classes and their sub-class ceilings (§§1.2, 1.4.1). <https://www.fai.org/civl-documents>.
- [10] CEN/TC 136/WG 6. Paragliding equipment — paragliders — part 2: Requirements and test methods for classifying flight safety characteristics. Technical Report Draft of FprEN 926-2, document N 0054, submitted to formal vote, European Committee for Standardization, 2012. Final draft of the standard published as EN 926-2:2013. <http://www.xcmag.com/wp-content/uploads/2012/08/DraftEN926-2.pdf>.
- [11] United States Department of Defense. Global positioning system standard positioning service performance standard. Technical Report 5th Edition, Department of Defense, Washington, DC, 2020. Table 3.8-3. <https://www.gps.gov/technical/ps/2020-SPS-performance-standard.pdf>.



- [12] Omar García Crespillo, Rafael Oliva González, and Stefano Caizzzone. Airborne time-correlated GNSS multipath error modeling of carrier-phase smoothed code. In *Proceedings of the 2024 International Technical Meeting of the Institute of Navigation (ION ITM)*, 2024. German Aerospace Center (DLR).
- [13] scikit-learn developers. `sklearn.metrics.pairwise.haversine_distances`. scikit-learn API reference. States the great-circle formula used here. [https://scikit-learn.org/stable/modules/generated/sklearn.metrics.pairwise.haversine\\_distances.html](https://scikit-learn.org/stable/modules/generated/sklearn.metrics.pairwise.haversine_distances.html).
- [14] National Geospatial-Intelligence Agency. Department of defense world geodetic system 1984: Its definition and relationships with local geodetic systems. Technical Report NGA.STND.0036, Version 1.0.0, National Geospatial-Intelligence Agency (NGA), 2014. <https://earth-info.nga.mil/php/download.php?file=coord-wgs84>.
- [15] Frank R. Hampel. The influence curve and its role in robust estimation. *Journal of the American Statistical Association*, 69(346):383–393, 1974.
- [16] Laurie Davies and Ursula Gather. The identification of multiple outliers. *Journal of the American Statistical Association*, 88(423):782–792, 1993.
- [17] Peter J. Rousseeuw and Christophe Croux. Alternatives to the median absolute deviation. *Journal of the American Statistical Association*, 88(424):1273–1283, 1993.
- [18] European Space Agency. Ellipsoidal and Cartesian coordinates conversion. Navipedia. [https://gssc.esa.int/navipedia/index.php/Ellipsoidal\\_and\\_Cartesian\\_Coordinates\\_Conversion](https://gssc.esa.int/navipedia/index.php/Ellipsoidal_and_Cartesian_Coordinates_Conversion).
- [19] Abraham Savitzky and M. J. E. Golay. Smoothing and differentiation of data by simplified least squares procedures. *Analytical Chemistry*, 36(8):1627–1639, 1964.
- [20] Y. He, S. Burov, R. Metzler, and E. Barkai. Random time-scale invariant diffusion and transport coefficients. *Physical Review Letters*, 101(5):058101, 2008.
- [21] Sidney Redner. *A Guide to First-Passage Processes*. Cambridge University Press, Cambridge, 2001.
- [22] L. R. Rabiner. A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2):257–286, 1989.
- [23] P. Welch. The use of fast Fourier transform for the estimation of power spectra: a method based on time averaging over short, modified periodograms. *IEEE Transactions on Audio and Electroacoustics*, 15(2):70–73, 1967.

# Appendices

Each appendix is numbered after the body chapter that first cites it: Appendices 2.A and 2.B belong to Chapter 2, Appendix 3.A to Chapter 3. They follow one another in the order in which the text first cites them.

## Appendix 2.A

# Power spectral density and Welch's method

This appendix explains the tool used in Chapter 2 to compare the two altitude channels of an IGC file, namely the *power spectral density* (PSD) estimated with *Welch's method* [23]. No prior familiarity with spectral estimation is assumed.

### The power spectral density

Consider a signal sampled at a fixed rate, here an altitude series  $z_n = z(n \Delta t)$ ,  $n = 0, \dots, N - 1$ , with sampling interval  $\Delta t$  (so the sampling frequency is  $f_s = 1/\Delta t$ ). Any such finite signal can be written as a sum of sinusoids of different frequencies (its discrete Fourier transform). The **power spectral density** describes how the power (the variance) of the signal is distributed across frequencies. A slow drift puts its power at low frequency; rapid jitter puts its power at high frequency. Formally the PSD  $S(f)$  is the Fourier transform of the autocovariance of the signal, and it is normalised so that the total area under it equals the variance of the signal,

$$\int_0^{f_s/2} S(f) df = \text{Var}(z), \quad (2.A.1)$$

with units of (metres)<sup>2</sup> per hertz. The upper limit  $f_s/2$  is the *Nyquist frequency*: a series sampled every  $\Delta t$  cannot resolve anything faster than one oscillation per two samples. This separates a slowly drifting but locally clean channel (power concentrated at low  $f$ ) from one with a high-frequency noise floor (power that stays large up to the Nyquist frequency).

### The periodogram

The naive estimator of  $S(f)$  is the **periodogram**: take the discrete Fourier transform  $\hat{z}(f) = \sum_n z_n e^{-2\pi i f n \Delta t}$  and form  $|\hat{z}(f)|^2$  (suitably normalised). It is asymptotically unbiased but *not consistent*: its variance does not shrink as the record gets longer, because a longer signal gives more frequency points, each as noisy as before. Plotted directly it is a spiky curve, too noisy for a reliable comparison of two channels.

### Welch's method

Welch's method turns the noisy periodogram into a usable estimate by *averaging*. It proceeds in three steps:

- (i) **Segment.** Split the record into several shorter, overlapping segments.

---

(ii) **Window.** Multiply each segment by a smooth taper (a Hann window) that falls to zero at its ends. This suppresses the spurious high-frequency power (*spectral leakage*) that an abrupt segment boundary would otherwise inject.

(iii) **Average.** Compute the periodogram of each windowed segment and average them.

Averaging  $K$  segments cuts the variance of the estimate by roughly a factor  $K$ , at the cost of coarser frequency resolution (set by the segment length rather than the whole record). The result is a smooth PSD estimate whose *level* can be compared meaningfully between two signals; it is the standard, robust spectral estimator. Each segment is typically detrended before transforming (e.g. by removing a linear fit), so that the estimate reflects the fluctuations about the trend, where the sensor noise lives, rather than the trend itself (implementation details: Sec. 2.7.1).

The spectra of the two altitude channels, and the choice of channel they motivate, are discussed in Sec. 2.7.1; this appendix covers only the method.

## Appendix 2.B

# Geodetic transforms: from $(\varphi, \lambda, h)$ to ECEF to ENU

This appendix derives the two formulas that Sec. 2.7.5 uses as given: the geodetic-to-ECEF transform, Eq. (2.6), and the ECEF-to-ENU rotation matrix. It closes with the size of the tangent-plane error quoted there. Notation as in the body: WGS84 semi-major axis  $a$ , flattening  $f$ ,  $b = a(1 - f)$  the semi-minor axis,  $e^2 = f(2 - f) = 1 - b^2/a^2$  the first eccentricity squared [14].

### 2.B.1 Geodetic coordinates to ECEF

The ECEF frame is Cartesian, centred at the Earth's centre  $O$ : the  $Z$  axis along the polar axis, the  $X$  axis through the intersection of the equator and the prime meridian,  $Y$  completing the right-handed triad. By symmetry, a point at longitude  $\lambda$  lies in the meridian plane obtained by rotating the  $X$ - $Z$  plane by  $\lambda$ ; writing  $p$  for the distance from the polar axis, its ECEF coordinates are

$$X = p \cos \lambda, \quad Y = p \sin \lambda, \quad Z = Z. \quad (2.B.1)$$

The problem therefore reduces to the meridian ellipse: find  $(p, Z)$  of a point at geodetic latitude  $\varphi$  and height  $h$ .

**The surface point.** The meridian section of the ellipsoid is the ellipse

$$\frac{p^2}{a^2} + \frac{Z^2}{b^2} = 1. \quad (2.B.2)$$

The geodetic latitude is defined through the surface *normal* (Figure 2.4). The gradient of the left-hand side of (2.B.2) is proportional to  $(p/a^2, Z/b^2)$ , so the outward normal at a surface point  $(p_s, Z_s)$  makes an angle  $\varphi$  with the equatorial plane given by

$$\tan \varphi = \frac{Z_s/b^2}{p_s/a^2} \quad \implies \quad Z_s = \frac{b^2}{a^2} p_s \tan \varphi = (1 - e^2) p_s \tan \varphi. \quad (2.B.3)$$

Substituting (2.B.3) into (2.B.2), with  $b^2 = a^2(1 - e^2)$ ,

$$\frac{p_s^2}{a^2} (1 + (1 - e^2) \tan^2 \varphi) = 1 \quad \implies \quad p_s^2 = \frac{a^2 \cos^2 \varphi}{\cos^2 \varphi + (1 - e^2) \sin^2 \varphi} = \frac{a^2 \cos^2 \varphi}{1 - e^2 \sin^2 \varphi}, \quad (2.B.4)$$

that is,

$$p_s = N(\varphi) \cos \varphi, \quad N(\varphi) \equiv \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}}, \quad (2.B.5)$$

and, back through (2.B.3),

$$Z_s = (1 - e^2) N(\varphi) \sin \varphi. \quad (2.B.6)$$

The quantity  $N(\varphi)$  so defined is the *prime-vertical radius of curvature*; a short computation shows it is precisely the distance, along the normal, from the surface point to the polar axis (the segment  $Q-P$  of Figure 2.4): the normal direction in the meridian plane is  $(\cos \varphi, \sin \varphi)$ , and following it inward from  $(p_s, Z_s)$  a length  $N$  lands on the axis,  $p_s - N \cos \varphi = 0$  by (2.B.5).

**Adding the height.** The height  $h$  of a fix is measured along the same normal, whose unit vector in ECEF components is

$$\hat{\mathbf{u}} = (\cos \varphi \cos \lambda, \cos \varphi \sin \lambda, \sin \varphi). \quad (2.B.7)$$

Adding  $h \hat{\mathbf{u}}$  to the surface point (2.B.5)–(2.B.6) in the meridian plane and restoring the longitude via (2.B.1):

$$X = (N + h) \cos \varphi \cos \lambda, \quad Y = (N + h) \cos \varphi \sin \lambda, \quad Z = (N(1 - e^2) + h) \sin \varphi, \quad (2.B.8)$$

which is Eq. (2.6). The asymmetry between the horizontal factor  $(N + h)$  and the vertical one  $(N(1 - e^2) + h)$  is the ellipsoid’s flattening at work: on a sphere ( $e = 0$ ) both reduce to  $R + h$  and the geodetic and geocentric latitudes coincide.

**How far the two latitudes differ.** The geocentric latitude  $\psi$  of a point on the surface is the angle its position vector makes with the equatorial plane, so from (2.B.5)–(2.B.6), at  $h = 0$ ,

$$\tan \psi = \frac{Z_s}{p_s} = \frac{N(1 - e^2) \sin \varphi}{N \cos \varphi} = (1 - e^2) \tan \varphi, \quad (2.B.9)$$

the classic relation: the two agree only where  $e = 0$ , or at the equator and the poles, where  $\tan$  is 0 or infinite. To see the size of the gap, write  $\psi = \varphi - \Delta$  and expand for small  $e^2$ . Using  $\tan(\varphi - \Delta) \simeq \tan \varphi - \Delta (1 + \tan^2 \varphi)$  in (2.B.9),

$$\tan \varphi - \Delta (1 + \tan^2 \varphi) \simeq (1 - e^2) \tan \varphi \implies \Delta \simeq \frac{e^2 \tan \varphi}{1 + \tan^2 \varphi} = e^2 \sin \varphi \cos \varphi, \quad (2.B.10)$$

that is

$$\varphi - \psi \simeq \frac{e^2}{2} \sin 2\varphi, \quad (2.B.11)$$

which is the expression quoted in Sec. 2.7.5. It is largest where  $\sin 2\varphi = 1$ , i.e. exactly at  $\varphi = 45^\circ$ , and there equals  $e^2/2 = 3.347 \times 10^{-3}$  radian, or  $0.192^\circ \approx 11.5'$ . Mistaking one latitude for the other therefore displaces a fix along the meridian by  $R_\oplus (e^2/2) \approx 21$  km at mid-latitudes—the  $\sim 20$  km quoted in the body. The error vanishes at the equator and at the poles and is symmetric about  $45^\circ$ .

## 2.B.2 ECEF to local ENU

The local frame at the origin fix  $(\varphi_0, \lambda_0, h_0)$  is spanned by three unit vectors: East, tangent to the parallel; North, tangent to the meridian, pointing to the pole; Up, along the ellipsoid normal (2.B.7). East is the direction in which the position (2.B.1) changes under  $\lambda$  alone,

$$\hat{\mathbf{e}}_E = \frac{\partial \mathbf{X} / \partial \lambda}{\|\partial \mathbf{X} / \partial \lambda\|} = (-\sin \lambda_0, \cos \lambda_0, 0), \quad (2.B.12)$$



Up is  $\hat{\mathbf{u}}$  of (2.B.7) evaluated at  $(\varphi_0, \lambda_0)$ , and North completes the right-handed triad,

$$\hat{\mathbf{e}}_N = \hat{\mathbf{e}}_U \times \hat{\mathbf{e}}_E = (-\sin \varphi_0 \cos \lambda_0, -\sin \varphi_0 \sin \lambda_0, \cos \varphi_0). \quad (2.B.13)$$

The ENU components of a displacement  $\Delta \mathbf{X}$  are its projections on the triad,  $E = \hat{\mathbf{e}}_E \cdot \Delta \mathbf{X}$  and likewise for  $N$  and  $U$ ; stacking the three row vectors gives exactly the matrix of Sec. 2.7.5,

$$\begin{pmatrix} E \\ N \\ U \end{pmatrix} = \underbrace{\begin{pmatrix} -\sin \lambda_0 & \cos \lambda_0 & 0 \\ -\sin \varphi_0 \cos \lambda_0 & -\sin \varphi_0 \sin \lambda_0 & \cos \varphi_0 \\ \cos \varphi_0 \cos \lambda_0 & \cos \varphi_0 \sin \lambda_0 & \sin \varphi_0 \end{pmatrix}}_R \Delta \mathbf{X}, \quad (2.B.14)$$

whose rows are orthonormal by construction, so  $R$  is a rotation ( $R^T R = \mathbb{I}$ ) [18]. Two remarks used in the body follow directly. First, the rows depend on the *geodetic* latitude because Up is the ellipsoid normal — the same  $\varphi$  that enters Eq. (2.6), which is why no latitude conversion appears anywhere in the pipeline. Second, the matrix is evaluated once per flight, at the origin fix: the frame is a fixed tangent frame, not one that travels with the wing.

### 2.B.3 Size of the tangent-plane error

Working in a fixed tangent frame replaces distances measured *along* the curved surface by straight-line (chord) distances in the plane. For two surface points separated by an arc  $d$  on a sphere of radius  $R_\oplus$ , the chord is

$$c = 2R_\oplus \sin \frac{d}{2R_\oplus} = d - \frac{d^3}{24 R_\oplus^2} + O(d^5), \quad (2.B.15)$$

so the horizontal distance error of the tangent-plane picture is  $d^3/(24R_\oplus^2)$ : about 1 cm at  $d = 20$  km and 13 cm at  $d = 50$  km — far below the metre-scale GPS noise on the same displacement, which is the comparison made in Sec. 2.7.5. The *vertical* coordinate of a distant surface point is a different matter: the surface falls below the tangent plane by the sagitta  $\approx d^2/(2R_\oplus)$ , which reaches tens of metres already at  $d = 20$  km. This is why the analysis never takes its working altitude from the plane geometry: the vertical coordinate is the barometric altitude itself (Sec. 2.7.5), and the  $U$  component serves only the geometric construction.

## Appendix 3.A

# CTRW with power-law waiting times: EA-MSD vs TA-MSD

This appendix derives the ensemble-averaged MSD (EA-MSD) and the time-averaged MSD (TA-MSD) for the continuous-time random walk (CTRW) with heavy-tailed waiting times, and shows explicitly that the two differ and that the TA-MSD depends on the observation time  $T$ .

### Model

A walker takes steps of length  $l_i$  (iid, symmetric,  $\langle l^2 \rangle = \sigma^2 < \infty$ ) separated by waiting times  $\tau_i$  (iid) drawn from

$$\psi(\tau) \sim \frac{\alpha \tau_0^\alpha}{\tau^{1+\alpha}}, \quad \tau \rightarrow \infty, \quad 0 < \alpha < 1. \quad (3.A.1)$$

The mean waiting time diverges ( $\langle \tau \rangle = \infty$ ), so the walker is anomalously slow. Let  $n(t)$  be the number of steps completed by time  $t$ ,  $T_n = \sum_{i=1}^n \tau_i$  the time of the  $n$ -th step, and

$$x(t) = \sum_{i=1}^{n(t)} l_i \quad (3.A.2)$$

the position at time  $t$ .

### Ensemble-averaged MSD

Since steps and waiting times are independent,

$$\langle x^2(t) \rangle = \sigma^2 \langle n(t) \rangle. \quad (3.A.3)$$

For large  $t$  the mean number of renewals in  $[0, t]$  follows from the renewal equation and the Tauberian theorem applied to the Laplace transform of  $\psi$ . Because  $\hat{\psi}(s) \equiv \int_0^\infty e^{-s\tau} \psi(\tau) d\tau \simeq 1 - \Gamma(1-\alpha) (\tau_0 s)^\alpha + \dots$  for  $s \rightarrow 0$  (a consequence of  $\psi$  having a power-law tail with exponent  $1+\alpha$  and amplitude  $\alpha\tau_0^\alpha$ ), the renewal theorem gives

$$\langle n(t) \rangle \sim \frac{t^\alpha}{\Gamma(1-\alpha) \Gamma(1+\alpha) \tau_0^\alpha} = \frac{\sin \pi \alpha}{\pi \alpha} \left( \frac{t}{\tau_0} \right)^\alpha, \quad t \rightarrow \infty. \quad (3.A.4)$$

Hence the **EA-MSD** grows sub-linearly,

$$\text{MSD}(t) = \langle x^2(t) \rangle \sim \frac{\sigma^2}{\Gamma(1-\alpha) \Gamma(1+\alpha) \tau_0^\alpha} t^\alpha \equiv K_\alpha t^\alpha, \quad 0 < \alpha < 1. \quad (3.A.5)$$

The transport is sub-diffusive with Hurst exponent  $H = \alpha/2 < 1/2$ .

## Time-averaged MSD and its mean

For a trajectory of duration  $T$ , the time-averaged MSD at lag  $t \ll T$  is

$$\overline{\delta^2}(t; T) = \frac{1}{T-t} \int_0^{T-t} [x(t_0+t) - x(t_0)]^2 dt_0. \quad (3.A.6)$$

Since  $\langle [x(t_0+t) - x(t_0)]^2 \rangle = \sigma^2 \langle n(t_0+t) - n(t_0) \rangle$ , the mean of the integrand equals  $\sigma^2$  times the mean number of steps in any window of width  $t$ . The integral  $\int_0^{T-t} [n(t_0+t) - n(t_0)] dt_0$  counts, for each step  $k$  that lands at time  $T_k \in [0, T]$ , the length of the  $t$ -wide sliding window that contains  $T_k$ . For  $T_k$  deep in the interior ( $t \leq T_k \leq T-t$ ) this length is exactly  $t$ ; the contributions from the two boundary layers of width  $t$  are negligible for  $t \ll T$ . Therefore

$$\int_0^{T-t} [n(t_0+t) - n(t_0)] dt_0 \approx t n(T), \quad t \ll T, \quad (3.A.7)$$

and

$$\overline{\delta^2}(t; T) \approx \frac{\sigma^2 t n(T)}{T}. \quad (3.A.8)$$

Taking the ensemble average and using  $\langle n(T) \rangle \sim K_\alpha T^\alpha / \sigma^2$ ,

$$\text{TAMSD}(t, T) \equiv \langle \overline{\delta^2}(t; T) \rangle \approx \sigma^2 \frac{\langle n(T) \rangle}{T} t \sim K_\alpha T^{\alpha-1} t. \quad (3.A.9)$$

Comparing (3.A.5) and (3.A.9):

- The EA-MSD scales as  $t^\alpha$  (sub-diffusive), whereas the TA-MSD scales as  $t^1$  (linear in the lag) – a qualitatively different dependence on  $t$ .
- The TA-MSD carries an explicit  $T^{\alpha-1}$  prefactor: since  $\alpha < 1$  this is a decreasing function of  $T$ , so longer trajectories yield a smaller TA-MSD at the same lag. This  $T$ -dependence is the signature of *aging*: the increment statistics depend on how long the process has already been observed.
- The ratio  $\text{TAMSD}(t, T)/\text{MSD}(t) \sim (t/T)^{1-\alpha} \rightarrow 0$  as  $T \rightarrow \infty$  at fixed  $t$ .

## Distribution of the TA-MSD: non-self-averaging

Equation (3.A.8) shows that  $\overline{\delta^2}(t; T) \approx \sigma^2 t n(T)/T$ , so the randomness in the TA-MSD comes entirely from  $n(T)$ , the total number of steps in  $[0, T]$ . For a power-law renewal process the rescaled count

$$\xi \equiv \frac{n(T)}{\langle n(T) \rangle} \quad (3.A.10)$$

converges in distribution as  $T \rightarrow \infty$  to a non-degenerate limit law: the *Mittag-Leffler distribution*  $M_\alpha$ , whose moments are  $\langle \xi^n \rangle = n! \Gamma(1+\alpha)^n / \Gamma(1+n\alpha)$  [20]. This is a broad, non-Gaussian distribution on  $(0, \infty)$  that does *not* concentrate around  $\xi = 1$  for  $\alpha < 1$ : different trajectories give systematically different values of  $\overline{\delta^2}(t; T)$  even in the limit  $T \rightarrow \infty$ . Hence the TA-MSD is *non-self-averaging*: the value from a single trajectory is not a reliable estimate of any ensemble quantity, however long the trajectory.

Quantitatively, the ergodicity-breaking parameter

$$\text{EB}(t, T) \equiv \frac{\langle [\overline{\delta^2}(t; T)]^2 \rangle - \langle \overline{\delta^2}(t; T) \rangle^2}{\langle \overline{\delta^2}(t; T) \rangle^2} = \text{Var}(\xi) \xrightarrow{T \rightarrow \infty} \frac{2 \Gamma(1+\alpha)^2}{\Gamma(1+2\alpha)} - 1 > 0 \quad (3.A.11)$$

---

tends to a finite positive constant rather than to zero [20]: the scatter of  $\overline{\delta^2}$  across trajectories survives even in the limit of infinite observation time (for  $\alpha = 1$  the constant is 0 and ergodicity is recovered). Monitoring the full distribution of  $\overline{\delta^2}^{(k)}(t)$  across trajectories, and not only its mean, therefore diagnoses non-ergodicity directly: the distribution collapses with growing  $T$  for a self-averaging ergodic process and converges to a non-trivial limit for a non-ergodic one.

For comparison, in an ordinary random walk ( $\alpha = 1$ , finite mean waiting time) the ratio  $n(T)/\langle n(T) \rangle$  tends to 1 in probability as  $T \rightarrow \infty$ , so the distribution of the TA-MSD collapses to a delta function,  $\overline{\delta^2}(t; T) \rightarrow \text{MSD}(t)$  almost surely, and the process is ergodic.

# Implementation and Computational Details

# Chapter 1

## Overview

This part collects the implementation- and computation-level detail behind the choices made in the main text: the exact code module and script that produced each figure or table, the configuration values in force, sample sizes and computation times, and the status of each piece (implemented and tested, designed but not yet built, or planned). It is organised as a document within the document: its own chapters, **numbered independently of the appendices**, mirror the body chapter by chapter. It is not meant to be read start to end – the body links to the relevant place at the point where a detail was trimmed out of the text, via a cross-reference such as “(implementation details: Sec. 3.5.2)”.

### Contents of this part

|          |                                                          |            |
|----------|----------------------------------------------------------|------------|
| <b>1</b> | <b>Overview</b>                                          | <b>89</b>  |
| <b>2</b> | <b>The dataset</b>                                       | <b>92</b>  |
| 2.1      | Data acquisition . . . . .                               | 92         |
| 2.1.1    | Pipeline and tooling . . . . .                           | 92         |
| 2.1.2    | Reproducibility . . . . .                                | 93         |
| 2.1.3    | Integrity . . . . .                                      | 93         |
| 2.2      | Where the data lives . . . . .                           | 93         |
| 2.3      | The IGC tracklog format . . . . .                        | 97         |
| 2.4      | Flight metadata and the catalog . . . . .                | 97         |
| 2.5      | Glider categories . . . . .                              | 97         |
| 2.6      | Coverage and statistics . . . . .                        | 97         |
| 2.7      | Trajectory pre-processing . . . . .                      | 97         |
| 2.7.1    | Choice of altitude channel . . . . .                     | 98         |
| 2.7.2    | Fix-level cleaning . . . . .                             | 101        |
| 2.7.3    | Trimming of ground phases . . . . .                      | 106        |
| 2.7.4    | Flight-level filtering . . . . .                         | 106        |
| 2.7.5    | From geographic to local Cartesian coordinates . . . . . | 107        |
| 2.7.6    | Uniform sampling within a flight . . . . .               | 107        |
| 2.7.7    | Smoothing and differentiation . . . . .                  | 108        |
| 2.8      | Preliminary characterization . . . . .                   | 109        |
| <b>3</b> | <b>Global transport</b>                                  | <b>111</b> |
| 3.1      | Observables on the whole trajectory . . . . .            | 111        |
| 3.1.1    | The two estimators . . . . .                             | 111        |
| 3.1.2    | The fit, and what is quoted from it . . . . .            | 112        |

|          |                                                           |            |
|----------|-----------------------------------------------------------|------------|
| 3.1.3    | The audit pass . . . . .                                  | 113        |
| 3.1.4    | The shape observables, and what calibrates them . . . . . | 114        |
| 3.1.5    | Known limitations, with their size . . . . .              | 116        |
| <b>4</b> | <b>Flight phases</b>                                      | <b>117</b> |
| 4.1      | Segmentation into flight phases . . . . .                 | 117        |
| 4.2      | Phase-resolved observables . . . . .                      | 117        |
| 4.3      | Reproducing the reference results . . . . .               | 117        |
| 4.4      | Modelling and simulation . . . . .                        | 117        |
| 4.5      | Tooling . . . . .                                         | 117        |

Table 1.1 is a quick index of which script reproduces which auto-generated figure or table (all paths relative to `scripts/reporting/`); the chapters that follow go into the reasoning and numbers behind each one.



| Figure/Table                                                                 | Script                                   | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stats macros;<br>Tables 2.3, 2.4                                             | <code>generate_stats.py</code>           | Reads committed <code>data/*/seasons_index.csv</code> ; instant; part of the pre-commit hook.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| The three processed tables (Sec. 2.2)                                        | <code>../preprocess.py</code>            | Runs stages (i)–(vii) over an archive; ~80 min for both disciplines on 8 workers. <code>../verify_dataset.py</code> checks the result against the invariants of Ch. 2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Figure 3.1;<br><code>\StatMsd*</code>                                        | <code>generate_msd_figure.py</code>      | Streams <code>fixes.parquet</code> once, through <code>soaring.analysis.derived.stream_flights</code> , and accumulates in that single pass both estimators and the three fixed-duration cohorts (Sec. 3.1); ~20 min on the paraglider archive. The cohorts cost nothing beyond three more accumulators, because the expensive step is the read. <code>--redraw</code> re-renders the figure from the written <code>msd_curve.csv</code> in a second, without touching the archive: nothing there recomputes an estimator, so a change to the drawing need not cost another pass over $10^9$ rows.                                                                                                                                                             |
| <code>\StatPipe*</code> , the per-stage shares quoted around Table 2.7       | <code>generate_pipeline_census.py</code> | Queries <code>flights_meta.parquet</code> ; instant.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Table 2.7 itself;<br><code>\StatData*</code> ,<br><code>\DataCascade*</code> | <code>generate_dataset_stats.py</code>   | Reads <code>flights_meta.parquet</code> and the catalogue; seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Census macros<br>( <code>\StatScan...</code> )                               | <code>generate_census_stats.py</code>    | Reads the cached track scans on the SSD (never rescans); instant; best-effort in the pre-commit hook, so the committed macros refresh whenever the SSD is mounted. Also emits the flight-level filtering census ( <code>LongEnough/Overlong/ShortPath</code> families, Sec. 2.7.4) and the missing-barometric-fix statistics ( <code>BaroMiss</code> , Sec. 2.7.1), against the thresholds in <code>configs/preprocessing.yaml</code> , and re-exports those thresholds as the <code>\Preproc*</code> macros quoted by the pipeline map (Table 2.8).                                                                                                                                                                                                           |
| — (all of the above, in order)                                               | <code>../regenerate.sh</code>            | Runs every row of this table that depends on the processed dataset, in the one order that is correct, and refuses to start while <code>preprocess.py</code> is still writing. The ordering is a real constraint rather than a convenience: the driver rewrites <code>&lt;data_root&gt;/derived/</code> in place and completes <code>flights_meta</code> only after its last flight, so a generator run against a live archive reads a partial fix table beside the previous run’s metadata and produces numbers that are wrong and look right. That happened once. The macro contract below is checked before the build, because an undefined generated macro inside a <code>\SI</code> is a fatal L <sup>A</sup> T <sub>E</sub> X error rather than a warning |

# Chapter 2

## The dataset

*Mirrors Chapter 2 section by section, opening with the data acquisition (Sec. 2.1) and including the pre-processing and preliminary- characterization sections (Secs. 2.7, 2.8) – the cleaning and first-look steps are part of describing the dataset, not of the planned analysis in Chapter 3.*

Table 2.1 collects the implementation status of every step discussed below; the sections point to it instead of repeating status remarks.

| Pipeline step                                           | Status                           | Where               |
|---------------------------------------------------------|----------------------------------|---------------------|
| Data acquisition (fetch, download, verify)              | implemented                      | Sec. 2.1            |
| IGC parsing and format validity                         | implemented, unit-tested         | Sec. 2.3            |
| Catalog and season index ( <code>build-catalog</code> ) | implemented                      | Sec. 2.4            |
| Census scan, diagnostic figures and macros              | implemented                      | Sec. 2.7, Table 1.1 |
| Glider-label normalisation                              | designed                         | Sec. 2.5            |
| Altitude-channel choice ( <code>alt_source</code> )     | implemented, unit-tested         | Sec. 2.7.1          |
| Fix-level cleaning (three detectors)                    | implemented, unit-tested         | Sec. 2.7.2          |
| Validation battery for the cleaning                     | planned, follows the cleaning    | Sec. 2.7.2          |
| Trimming of ground phases, interior guard               | implemented, unit-tested         | Sec. 2.7.3          |
| Flight-level filtering                                  | implemented, unit-tested         | Sec. 2.7.4          |
| ENU conversion                                          | implemented, unit-tested         | Sec. 2.7.5          |
| Uniform sampling and segmentation                       | implemented, unit-tested         | Sec. 2.7.6          |
| Smoothing and differentiation                           | implemented; timescales measured | Sec. 2.7.7          |
| Preliminary characterization                            | implemented                      | Sec. 2.8            |
| MSD audit (residuals, controls, isotropy)               | implemented                      | Sec. 3.1            |
| Ensemble MSD (Sec. 3.1)                                 | implemented, unit-tested         | Sec. 3.1            |

**Table 2.1.** Implementation status of the pipeline steps, collecting the status remarks of this appendix in one place. *Implemented* code is committed and in use; *designed* steps have their rules and thresholds fixed (and, where noted, their parameters already in the configuration) but no running routine yet; *planned* steps follow the designed ones.

### 2.1 Data acquisition

*Mirrors Sec. 2.1.*

#### 2.1.1 Pipeline and tooling

Package `soaring.acquisition.ffv1`, exposed as two CLI entry points, `soaring-para` and `soaring-delta`, sharing subcommands: `fetch-xml` (archive season XMLs), `download` (fetch IGC files), `build-catalog` (rebuild catalog/index), `status` (per-season coverage), `verify`

(post-hoc integrity scan, Sec. 2.1.3). One package serves both disciplines because the two FFVL sites expose an identical URL structure and the same machine-readable XML export format; only the host and the data root differ between the two CLIs.

**Resumable downloads.** Files already present on disk are skipped, so an interrupted job resumes freely without re-fetching.

**Atomic writes.** Each file is written to a temporary `.part` and then renamed, so a file under its final name is always complete – important on the exFAT external disk, which has no journaling.

**Content validation.** A response is accepted only if it looks like a real IGC file (an A header record followed by at least one B fix record); HTML error pages and truncated responses are rejected and retried.

**Rate-limited networking.** A small bounded thread pool, per-request timeouts, exponential backoff, and a minimum delay between requests keep the load on the server low; a browser TLS fingerprint is impersonated to pass the site’s Cloudflare challenge.

### 2.1.2 Reproducibility

Destination directory (`data_root`) set per machine via an environment variable – `SOARING_PARA_DATA_ROOT` for paragliders, `SOARING_DELTA_DATA_ROOT` for hang gliders – or a config file shipping a placeholder (`configs/para_download.yaml/ configs/delta_download.yaml`); every command validates the setting at start-up and aborts with an explicit message if the target is missing.

### 2.1.3 Integrity

The `verify` subcommand re-scans every downloaded file on disk, reusing the same IGC validation applied at download time, and flags truncated or invalid files, leftover `.part` fragments, and read errors.

## 2.2 Where the data lives

Everything is on one external disk and nothing is in the repository, so a reader who has the disk can locate any number this thesis quotes. There is one root per discipline, identically laid out, found through an environment variable rather than a hard-coded path (`SOARING_PARA_DATA_ROOT`, `SOARING_DELTA_DATA_ROOT`); the repository keeps only the small versioned `data/<discipline>/seasons_index.csv`, so the season tables of Sec. 2.5 build without the disk mounted.

| Path                                                                           | Size          | What it is                                                                                 |
|--------------------------------------------------------------------------------|---------------|--------------------------------------------------------------------------------------------|
| <i>raw/</i> — immutable, never modified after download                         |               |                                                                                            |
| <i>raw/igc/&lt;season&gt;/</i>                                                 | 186,052 files | the track files, one directory per season, named <i>&lt;date&gt;_&lt;flight_id&gt;.igc</i> |
| <i>raw/raw_xml/&lt;year&gt;.xml</i>                                            | 27 files      | the season listings exactly as the site served them: the provenance of the catalog         |
| <i>catalog/</i> — regenerable from <i>raw/</i> , metadata only                 |               |                                                                                            |
| <i>catalog/catalog.csv</i>                                                     | 87 MB         | one row per listed flight, 23 columns (Sec. 2.3)                                           |
| <i>catalog/seasons_index.csv</i>                                               | 2 kB          | one row per season: listed, with a track, downloaded                                       |
| <i>logs/</i> — what the acquisition did                                        |               |                                                                                            |
| <i>logs/download.log</i> ,<br><i>logs/failures.csv</i>                         | 80 kB         | the run log and the retry registry, one row per track that could not be fetched            |
| <i>derived/</i> — regenerable from <i>raw/</i> , everything the analysis makes |               |                                                                                            |
| <i>derived/track_scan.parquet</i>                                              | 8.5 MB        | the census cache: one scalar row per parsed flight (Sec. 2.7)                              |
| <i>derived/fixes.parquet</i>                                                   | 43 GB         | the processed trajectories: $1.36 \times 10^9$ rows for paragliders                        |
| <i>derived/segments.parquet</i>                                                | 4 MB          | one row per segment the splitting produced, retained or not                                |
| <i>derived/flights_meta.parquet</i>                                            | 10 MB         | one row per flight <i>attempted</i> , dropped ones included                                |

The three levels are a rule and not a habit: *raw/* is never written to after the download, *catalog/* is a function of *raw/*, and *derived/* is a function of both. Deleting *derived/* costs computation and never data. It also means the two census families cannot be confused: *track\_scan.parquet* describes *parsed* tracks and is what the `\StatScan*` macros query, because a cut must be audited on the population it acts on; *flights\_meta.parquet* describes what the pipeline *did*, and is what the `\StatPipe*` macros query.

**The trajectory table.** *fixes.parquet* carries one row per grid point of every retained segment, keyed (*source, flight\_id, segment\_id*). Printed as it is on disk (the hang-glider root; `scripts/reporting/show_dataset.py` produces this view):

34,389,245 rows in 43 row groups, 1,156.0 MB on disk, ZSTD

dtypes:

|            |       |   |         |     |         |                 |      |
|------------|-------|---|---------|-----|---------|-----------------|------|
| source     | str   | t | float32 | v_E | float32 | interpolated    | bool |
| flight_id  | str   | E | float32 | v_N | float32 | edge            | bool |
| segment_id | int16 | N | float32 | v_z | float32 | hampel_flagged  | bool |
|            |       | z | float32 | a_* | float32 | alt_invalidated | bool |

head(4), columns 1-9:

| source     | flight_id | segment_id | t    | E          | N           | z           | v_E       | v_N       |
|------------|-----------|------------|------|------------|-------------|-------------|-----------|-----------|
| hangglider | 975       | 0          | 0.0  | -2.644085  | -4.948205   | 1458.585693 | 13.275144 | 1.215108  |
| hangglider | 975       | 0          | 10.0 | 111.172691 | -33.921207  | 1460.657104 | 9.778760  | -5.975531 |
| hangglider | 975       | 0          | 20.0 | 198.742096 | -103.775276 | 1468.514282 | 8.025670  | -6.961107 |
| hangglider | 975       | 0          | 30.0 | 279.220947 | -138.911942 | 1463.914307 | 7.008325  | -2.623670 |

head(4), columns 10-17:

| v_z       | a_E       | a_N       | a_z       | interpolated | edge  | hampel_flagged | alt_invalidated |
|-----------|-----------|-----------|-----------|--------------|-------|----------------|-----------------|
| -0.532143 | -0.436803 | -1.029317 | 0.192857  | False        | True  | False          | False           |
| 0.721429  | -0.262474 | -0.408811 | 0.057857  | False        | True  | False          | False           |
| 0.625000  | -0.088144 | 0.211696  | -0.077143 | False        | False | False          | False           |
| -1.466667 | 0.072790  | 0.463077  | -0.047143 | False        | False | True           | False           |

The four rows are the first four grid points of a 10 s logger: *t* starts at zero, *E* and *N* within metres of it, *z* at its measured 1,458 m. Coordinates are `float32`, a part in  $10^7$  on a coordinate

of tens of kilometres and orders below the GPS noise on it. The file is written in batches of 400 flights, and an analysis streams it rather than loading 43 GB—which is how the mean-squared displacement of Sec. 3.1 is computed.

A row group is not a batch, and the difference cost a measurement. The writer hands Parquet 400 flights at a time; Parquet then cuts what it is handed into row groups of its own default size, which is why the header above reads 43 row groups for an archive written in 16 batches. A reader that takes a row group for a unit of flights therefore meets, at each boundary, one flight arriving as two fragments—and processes it as two flights. The ensemble MSD counted 157,134 paraglider flights where the pipeline had written 156,017, 0.7% too many; worse, the time-averaged MSD is computed *within* a segment and cut at half its length, so a straddling segment stopped answering at half of its longer fragment instead of half of itself, and the longest segments—the ones that carry the long-lag tail, and the ones likeliest to straddle—were the ones truncated. The reading was thus biased by the writer’s buffering rather than by the data. Every consumer now goes through one reader, `soaring.analysis.derived.stream_flights`, which holds back the last flight of each row group and prepends it to the next, so what it yields is always a whole flight; it raises rather than continues if the writer’s ordering ever stops holding. The dataset verifier reads through it too, since it had the same blind spot in a worse place: the step *joining* two fragments was never differenced, so it was never checked, and a check with a blind spot reports success over it.

The three timing conventions a reader of this table has to know are argued elsewhere in the chapter and worth restating together: *t* is zero at the first fix of free flight of the parent flight, so a segment never restarts it (Sec. 2.7.6); *E* and *N* are zero at that same fix (Sec. 2.7.5); and *z* is the adopted altitude at its measured value, never re-zeroed. The only times in another clock are `ground_phase_start_s` and `ground_phase_end_s` in `flights_meta`, which are in the recorded clock of the IGC file because the integrity gate counts its removals inside that window.

**The segment table.** One row per segment the splitting produced, retained or not, and the reason for each drop:

| source         | flight_id    | segment_id | t_start                     | t_end   | n_fix | n_fix_raw | frac_interpolated |
|----------------|--------------|------------|-----------------------------|---------|-------|-----------|-------------------|
| hangglider     | 975          | 0          | 0.0                         | 20640.0 | 2065  | 2064      | 0.001453          |
| hangglider     | 1032         | 0          | 0.0                         | 14175.0 | 0     | 691       | 0.001479          |
| censored_start | censored_end | kept       | drop_reason                 |         |       |           |                   |
| False          | False        | True       | <NA>                        |         |       |           |                   |
| False          | False        | False      | channel_not_reconstructable |         |       |           |                   |

`n_fix` counts the rows the segment contributed to `fixes` (zero when dropped) against `n_fix_raw`, its measured fixes. `segment_id` is assigned at split time and stays stable, so a gap in the numbering is itself the record of a drop.

**The flight table.** One row per flight *attempted*, forty-one columns, read down rather than across—one retained flight beside one the flight-level filter rejected:

|                  | a retained flight | a dropped one          |
|------------------|-------------------|------------------------|
| source           | hangglider        | hangglider             |
| flight_id        | 975               | 830                    |
| pipeline_version | 1.1.0             | 1.1.0                  |
| drop_stage       | NaN               | flight_filter          |
| drop_reason      | NaN               | duration_below_minimum |
| alt_source       | baro              | gnss                   |
| n_fix_raw        | 2099              | 349                    |

|                                   |            |          |
|-----------------------------------|------------|----------|
| <code>n_fix_clean</code>          | 2069       | 348      |
| <code>n_removed_frozen</code>     | 30         | 0        |
| <code>n_flagged_kept</code>       | 40         | 8        |
| <code>n_boundaried</code>         | 0          | 0        |
| <code>integrity_fraction</code>   | 0.0        | 0.004587 |
| <code>ground_phase_start_s</code> | 300.0      | 485.0    |
| <code>duration_flight_s</code>    | 20645.0    | 1085.0   |
| <code>path_km</code>              | 283.408578 | 6.044969 |
| <code>lat0</code>                 | 43.812717  | NaN      |
| <code>dt_native_s</code>          | 10.0       | NaN      |
| <code>n_segments_kept</code>      | 1.0        | NaN      |
| <code>savgol_window_horiz</code>  | 5.0        | NaN      |

(abridged: the table carries 47 columns in all.) A dropped flight keeps everything the stages *before* the verdict had already measured—flight 830 was judged on a duration of 1,085 s, and its cleaning counters remain auditable even though none of its fixes reached the trajectory table. That is what makes the removals a census rather than an absence.

**What is not stored.** Anything that is pure algebra of the stored columns—the speed  $|\mathbf{v}|$ , the heading, the angular velocity, the turn radius, the glide ratio, the mechanical energy—is computed at analysis time and never materialised. Only the outputs of a non-trivial parametric step are kept, which is what makes the table’s size a function of the archive rather than of the questions asked of it.

**Checking it.** `scripts/verify_dataset.py` reads the three tables back and tests, on the data itself, the properties this chapter claims: completeness within a segment, a strictly increasing and uniform time base, no impossible step between two interior samples of a segment (Sec. 2.7.2), the origin at  $r(0) = 0$ , reachability— $|\mathbf{r}(t)| \leq v_{xy}^{\max} t$  for every sample—and referential integrity between the tables. Each check names the section it comes from, so a failure says which claim is false rather than only that something is wrong.

Reachability is the one check stated without reference to anything the pipeline did: it follows from the frame and the speed bound alone, because the flight began at the origin. That is what makes it able to catch a defect no per-step test can see. A flight whose *first* fix is corrupt has every step plausible—it simply sits, in the archive, 4,500 km from where it says it began, with the whole trajectory referred to that point (Sec. 2.7.2).

The qualification in the third check is not a hedge but the statement being precise. The bound is a claim about *measured* motion, and two kinds of sample in the table are not measurements. At a segment edge the smoothing polynomial is evaluated off-centre (Sec. 2.7.7), so its position carries more variance than a centred one and on a slow logger can imply a step over the bound where the underlying fixes do not. Across a bridged gap the monotone cubic is a reconstruction, and its slope may exceed the secant it spans—a hole crossed at  $42 \text{ m s}^{-1}$ , itself under the bound and so legitimately bridged, leaves interpolated points implying nearly  $60 \text{ m s}^{-1}$ . Both kinds are flagged in the table for exactly this reason (**edge**, **interpolated**), so the check reads the invariant between samples that are measured and centred, and *reports* the excess on the others. That reported number is one of the acceptance criteria Sec. 2.7.7 asks for: measured, rather than assumed away.

The script found two defects this text would otherwise have asserted without warrant. A randomised robustness suite (`test_pipeline_robustness.py`, which combines injected spikes, null-island runs, frozen locks, time defects, gaps and offsets at random and asserts only these invariants) found two more, and both were faults in the *statement* rather than in the pipeline: the invariant had been written without either qualification above. A third, the corrupt opening block of Sec. 2.7.2, was found by neither: it was found by the ensemble MSD of Chapter 3 being wrong, which is the expensive way to find a bug, and the reachability check

above is what makes it cheap next time. It also found one in the cleaning—a track crossing the antimeridian is read as motionless, because the great-circle distance is correctly aware of the wrap and therefore reads the two numerical extremes of a straddling bounding box as metres apart. No flight in the present archive reaches within ten degrees of that meridian; the further sources of Sec. 2.1 may.

## 2.3 The IGC tracklog format

The parser (`soaring.analysis.igc.parse_igc`) checks, at decode time and independently of any flight-dynamics judgement: that the time is a valid 24-hour `hh:mm:ss`; that the arc-minutes field of latitude and longitude is  $< 60$  (the `DDMM.mmm` encoding is undefined otherwise); that the hemisphere letter is one of the two valid characters; and that the decoded latitude/longitude fall within  $[-90, 90]/[-180, 180]$  degrees. A record failing any of these is dropped as unparseable, on the same footing as a record with the wrong length or a non-numeric field. *This format-validity check and the fix-level plausibility checks are distinct stages, and both are implemented and unit-tested (Sec. 2.7.2, Table 2.1).*

## 2.4 Flight metadata and the catalog

Both `catalog.csv` and `seasons_index.csv` are regenerated on demand (Sec. 2.3) by the acquisition CLI’s `build-catalog` subcommand (Sec. 2.1.1) – never hand-edited.

## 2.5 Glider categories

The placeholder class is the literal “0” in `aile_class`; where possible the true class is recovered from the raw `aile` (wing-model) field. The normalisation runs at the fix-level/pre-processing stage (Sec. 2.7.2) and is defined in Sec. 2.4. Status: Table 2.1.

## 2.6 Coverage and statistics

The season tables of Sec. 2.5 are typeset from the versioned `seasons_index.csv` (Sec. 2.4); the trajectory-level census numbers quoted from Sec. 2.7 onward come from a different pipeline, the cached track scan described at the head of Sec. 2.7.

## 2.7 Trajectory pre-processing

*Mirrors Sec. 2.7 and its subsections below.* Parsing: each IGC file is decoded by the `soaring.analysis.igc` parser, which reads the B records at their fixed character positions (Sec. 2.3) and returns both altitude channels alongside time and geodetic coordinates. The Welch/PSD procedure used to compare altitude channels is stated once, in Sec. 2.7.1; the smoothing calibration (Sec. 2.7.7) reuses it, applied to the horizontal components and to the flight’s adopted altitude channel.

**The census scan and its cache.** One full parse of the archive backs every trajectory-census number and figure in this appendix and in Chapter 2: the `load_or_scan_tracks` scan reads every parsed track (186,025 paraglider, 6,716 hang-glider) once and caches a per-flight summary table at `<data_root>/derived/track_scan.parquet`, one cache per discipline.



That cache is worth describing, since every census number in the chapter is a query against it. It is a table with *one row per parsed flight* and one column per summary quantity, all produced by `track_stats` in `soaring.analysis.census` as it walks a flight's fixes once:

- `duration_s`, `n_fix`, `dt_s` — the recorded span, the number of B records, and the median inter-fix interval;
- `path_km`, `extent_km` — the summed great-circle steps and the farthest any fix lies from the first;
- `max_gap_ratio`, `missing_fraction` — the largest inter-fix gap in units of `dt_s`, and the share of a uniform grid at that interval with no fix;
- `max_vxy_mps`, `max_vz_mps` — the extreme inter-fix speeds;
- `baro_alt_min_m`, `baro_alt_max_m`, `baro_present_frac` — the barometric range and how much of the flight the channel covers.

Nothing in it is a trajectory: it is deliberately one scalar row per flight, which is what makes it small enough to hold in memory for the whole archive and to re-query in seconds. Its purpose is exactly that—to decouple the cost of *reading* the archive (hours, once) from the cost of *asking* it a question (instant, as often as a threshold changes). Every `\StatScan*` macro, and the flight-level and sampling diagnostics of Figs. 2.3 and 2.6, are functions of these columns alone.

*What the census is not.* It is measured on the **raw** archive, one parse of the B records with no cleaning, no trimming and no filtering—which is the only thing it could be, since its purpose is to choose the thresholds that cleaning, trimming and filtering then apply, and a threshold chosen on the population that survives it is chosen circularly. So every `\StatScan*` number in this thesis is a statement about the archive *as recorded*: a duration is the recorded span and not the airborne one, a path includes whatever a corrupt fix contributed, and a retained-fraction curve is what a cut would keep if applied alone to the raw population. The retained ensemble has its own numbers and its own family, `\StatPipe*`, computed from `flights_meta.parquet` after all seven stages have run (Table 2.7); where a quantity appears in both, the two are not expected to agree, and the difference is precisely what the pipeline did. The two families are never mixed in one statement.

`generate_census_stats.py` recomputes the quoted macros from that cache against the thresholds currently in `configs/preprocessing.yaml`, so a threshold change propagates to every quoted count by re-running one script, with no rescan. The cache is refreshed when the archive changes—which for the FFVL collections means once a season closes, and more substantially when a new source is added (the sailplanes of Sec. 2.1 above all); between refreshes the cached counts can trail the season tables' download counts, and the two are not expected to agree to the flight. The same scan feeds Figures 2.3, 2.6 and 2.7, regenerated together by one run of `generate_preproc_figure.py` (Table 1.1); the fix-level diagnostics of Figure 2.2 are the one exception, drawn from a separate sampled pass (Sec. 2.7.2).

### 2.7.1 Choice of altitude channel

**The `alt_source` field.** The channel a flight uses is stored as an `alt_source` column (`baro` or `gnss`)—two-valued and categorical, so its storage cost is negligible—in the processed dataset's per-flight table, alongside the duration and the raw fix count—the number of B records the IGC file yields at parse time, before any cleaning, since the channel choice precedes the cleaning in the pipeline order.

**Welch/PSD procedure and its sample (Figure 2.1a–c).** The sample is a seeded random subsample of 3,000 flights per discipline. The size is set by convergence of the ensemble spectrum, not by the spectral shape: with a few hundred flights the high-frequency part of the median barometric curve is still visibly ragged (under-averaged), and it smooths out only with a few

thousand flights, so 3,000 buys a stable band at modest cost. This is an assertion the reader cannot presently check, and it should not stay one: the convergence figure—the same median spectrum drawn at a few hundred, a thousand and 3,000 flights, on one pair of axes—is to be produced and shown here, so that the choice of sample size rests on a picture rather than on a claim. Panel (a)/(b) use one representative flight (the one with the most fixes among the 3,000 sampled flights of that discipline); its measured offset (panel b) is annotated on the panel itself, not restated here.

IGC loggers record one fix per second nominally, but timestamps are not guaranteed to be perfectly regular (a missed fix, a logger writing at 1.05 s instead of 1.00 s); Welch’s method (Appendix 2.A) requires a uniformly sampled signal, so the following runs on every flight before the transform.

- (i) **Modal sampling period.** For each flight the median inter-fix interval is computed and rounded to the nearest integer second. The value that appears most often across all flights in the subsample (the *mode*) is taken as the common target  $\Delta t$  (in practice  $\Delta t = 1$  s for both disciplines, giving  $f_s = 1$  Hz and  $f_{\text{Nyq}} = 0.5$  Hz).
- (ii) **Flight selection.** Only flights whose own median period lies within 0.25 s of the modal  $\Delta t$  are retained, so pooled spectra share one frequency axis. Mixing cadences here would not be admissible: a spectrum is defined on a frequency axis that runs to  $1/(2\Delta t)$ , so a 1 s and a 9 s logger do not even share a Nyquist frequency, and averaging their periodograms per frequency bin would compare different bands. That is the whole reason for this cut—and it is a cut on the *diagnostic* sample only. The pipeline proper keeps every flight at its native cadence (Sec. 2.7.6); it is only this spectral estimate, which pools flights, that needs them commensurate. Flights with fewer than  $N_{\text{seg}} = 256$  fixes ( $\sim 4$  min at  $\Delta t = 1$  s) are also excluded (a series shorter than one Welch block of  $N_{\text{seg}} = 256$  samples cannot contribute a periodogram, Appendix 2.A). Of the 3,000 sampled flights per discipline, this leaves 1,641 paragliders but 727 hang gliders: paragliders are predominantly 1 s loggers (69 %, Sec. 2.7.6), so the population mode is a good match for most of them, whereas hang gliders split across several native rates (1, 2, 5,  $\geq 8$  s, Figure 2.7) and only their 1 s-logger minority survives this cut.
- (iii) **Uniform resampling.** The altitude series  $z(t)$  is linearly interpolated onto a grid  $t_0, t_0 + \Delta t, t_0 + 2\Delta t, \dots$  spanning the flight. Linear and not cubic, because the resampling must not disturb the band the estimate is about to read. Interpolation acts on a spectrum as a filter: linear interpolation attenuates high frequencies smoothly and monotonically, whereas a higher-order interpolant has a transfer function that can exceed unity near the Nyquist frequency, adding power that was never in the signal. Since the quantity being measured is the flat floor at high frequency, an interpolant that can lift it would bias the very number in question; one that only attenuates cannot manufacture a floor, and the timestamps are near-uniform to begin with, so the attenuation it does apply is slight.
- (iv) **Welch on the resampled series.** `scipy.signal.welch` is called with  $f_s = 1/\Delta t$ ,  $N_{\text{seg}} = 256$ , 50% overlap, a Hann window of the same  $N_{\text{seg}} = 256$  samples (the window spans one Welch segment by construction—256 s at the 1 s cadence, giving a frequency resolution of about 4 mHz), and linear detrending per segment.
- (v) **Robust ensemble estimator.** The per-flight PSDs are kept individually (per discipline and channel) and reduced across flights by the per-frequency *median*, with a shaded 10 to 90 percentile band; panel (c) then pools the two disciplines, whose medians coincide. The median, not the arithmetic mean, because the per-flight spectra are heavy-tailed: a mean swings by one to two orders of magnitude between samples – for the hang-glider barometric floor, from 0.8 to 93 m<sup>2</sup> Hz<sup>−1</sup> between an  $n = 114$  and an  $n = 727$  draw

- whereas the median is stable. Because  $\Delta t$  is the same for every retained flight by construction, the figure’s Nyquist dotted line is a single fixed value.

### Reading the band (panel c).

**Why the medians coincide.** For the typical flight both channels are dominated at high frequency by the metre resolution to which the IGC format rounds every altitude: the white-noise floor of a 1 m quantiser,  $\frac{1}{12}/f_{\text{Nyq}} \approx 0.17 \text{ m}^2 \text{ Hz}^{-1}$ ,<sup>1</sup> sits just below where the measured medians level off ( $\approx 0.3 \text{ m}^2 \text{ Hz}^{-1}$ ).

**Where the channels part.** In the tail: at the 90th percentile the high-frequency GNSS floor is  $4\times$  the barometric one for paragliders and  $108\times$  for hang gliders, and 13 % to 20 % of flights carry a GNSS floor above three times their own barometric one. What sits in that tail is receiver tracking noise scaled by the dilution of precision, since a raised floor is by definition a white contribution; multipath, being correlated over minutes (Sec. 2.7.1), does not show up there. Why the hang-glider fleet is worse in this respect is not settled by these spectra alone—older or lower-grade receivers are the obvious candidate—and the figure is not claimed to answer it.

**Consequence.** This heterogeneity, rather than a uniform gap, motivates the choice of the barometric channel (Sec. 2.7.1).

**Census (panel d).** Panel (d) is a full census, not a sample: the barometric-presence fraction is read off the shared track scan (Sec. 2.7), already paid for by the flight-level diagnostics.

“Present” needs a number, and the number is a decision. The scan records `baro_present_frac`, the share of a flight’s fixes carrying a non-zero pressure altitude, and a flight counts as barometric when that share reaches `BARO_PRESENT_MIN = 95 %` (defined once, in `soaring.analysis.altitude_noise`, and imported by both the cleaning and the census so the three cannot drift). The threshold is high deliberately. Presence is bimodal—the census below shows the channel is either essentially absent or essentially complete—so any value in the middle of the range separates the same two populations; but a mid-scale cut would also admit a flight missing, say, a third of its pressure altitudes, and such a flight is not one the vertical analysis should treat as barometric: its  $v_z$  would rest on a channel with holes, reconstructed by interpolation over stretches long enough to erase transitions. Setting the cut at 95 % sends those flights to the GNSS fallback, where their altitude is at least uniformly sourced. The number quoted in the body (Sec. 2.7.1) is the same one.

Presence is whole-file, checked on the same census: when the channel is absent it reads exactly zero for 99.3 % (paragliders) / 100 % (hang gliders) of those flights, and when present it covers every fix for 96 % / 98 % of them; the rest miss individual fixes, dropped one by one (Sec. 2.7.2), and no more than 5 % of the flight by construction, since below that the channel is not counted as present at all.

The bimodality asserted above is what these numbers measure. Where the channel is absent it is *exactly* zero for 99.3 % of those flights, and where it is present it is complete for 96 %; the population in between is thin. Raising the threshold from 0.5 to 0.95 reclassified 241 paraglider flights and no hang-glider ones—0.13 % of the archive—which is the direct measurement of how thin. Those flights are also where the worst partial coverage sat: the largest number of missing barometric fixes on a flight still counted as barometric fell from 14,878 to 1,397 when the threshold moved, so most of what looked like a pathological tail was

<sup>1</sup>Rounding to a step  $q$  adds an error that is well modelled as uniform on  $[-q/2, q/2]$  and independent between samples, of variance  $q^2/12$ . A white sequence of variance  $\sigma^2$  has a flat one-sided power spectral density  $\sigma^2/f_{\text{Nyq}}$ , since integrating it over  $[0, f_{\text{Nyq}}]$  must return the variance. With  $q = 1 \text{ m}$  and  $f_{\text{Nyq}} = 0.5 \text{ Hz}$  this is  $(1/12)/0.5 \approx 0.17 \text{ m}^2 \text{ Hz}^{-1}$ .

a classification artefact rather than a sensor one. The liveness threshold is `baro_min_range_m` (key `alt_channel`; value in Table 2.8); rule and rationale in Sec. 2.7.1.

### 2.7.2 Fix-level cleaning

This section gives the working definitions the narrative (Sec. 2.7.2) states in words: the three detectors, their parameters, the order they run in, and how the whole step is validated. In the order of Table 2.6, the three families map onto a robust local test (a short window of context), a structural rule (a whole run), and an absolute bound (no context). Status: Table 2.1.

The detectors run in a fixed order, which the subsections below follow:

- (1) format validity, at decode time (Sec. 2.3);
- (2) time defects: backward steps, then duplicates;
- (3) position outliers (the impossibility gate; the Hampel flag is recorded, not required—Sec. 2.7.2);
- (4) frozen-lock runs;
- (5) altitude: absolute bounds and altitude outliers.

The order matters because a speed or a residual is only meaningful once the time base is monotone and free of duplicates. Merging duplicates is componentwise and unweighted: fixes sharing one UTC second collapse to a single fix at that second, carrying the mean of their latitudes and longitudes and the mean of each altitude channel over the members that have one; the V flag of the merged fix is the logical *or* of the members', so a degraded member cannot be averaged away. The position-outlier and frozen-lock passes are one left-to-right scan that carries the last accepted fix as an anchor: it never re-walks the whole track, and it absorbs a run by extending the anchor across the flagged block – tracking the block's running bounding box to test its bounding diameter against  $\delta_{xy}$  – so the cost is  $O(N)$  in the number of fixes.

#### Time-base defects

The parser unwraps the UTC-midnight roll-over — a drop of  $\sim 86,400$  s advances the day count — and leaves residual backward jitter in place (Sec. 2.3). That is deliberate, and it is the one behaviour the redesign changed: the parser used to flatten the jitter with a running maximum, which is a repair, and repairing it there destroys the evidence this stage has to act on. A backward step that is *not* a roll-over is now removed rather than absorbed – the fix with the corrupt time is deleted as a node, just as a fix with a corrupt position is. A corrupted timestamp is recorded as a defect rather than silently repaired. Which fix carries the corrupt time is decided by minimal removal: delete the smallest set whose removal leaves a strictly increasing clock (the complement of the longest increasing subsequence), so a single fix with a forward-jumped clock is itself removed, rather than condemning every later fix against a running maximum. This requires moving the clamp out of the parser: `parse_igc` keeps the roll-over unwrap (a format concern, needing the raw time of day) but stops flattening backward jitter with the running maximum – flattened, the evidence the cleaning must act on no longer exists downstream.

Fixes sharing one UTC second are collapsed to a single representative at that second's centroid position. This replaces the provisional “keep the first” of the earlier design: keeping the first privileges an arbitrary member and, for a genuine  $\geq 2$  Hz logger, would silently halve the cadence, whereas the centroid is order-independent and the sub-second resolution it discards is discarded anyway once the flight is resampled to a common  $\Delta t$  (Sec. 2.7.6). One dilution case is accepted: if one member of a duplicate pair is itself a position spike, the centroid halves its amplitude before the outlier pass (which runs after the time defects) sees it;

a large spike is still flagged at half size, and one small enough to slip under  $k\sigma_k$  after halving is within what the smoother absorbs.

### Robust local-outlier test and action

The test is Eqs. (2.2)–(2.3) of Sec. 2.7.2, evaluated on a *time* window of half-width  $w$  (not a sample count), so it adapts to the native cadence and to irregular sampling; working defaults in Table 2.2. The window must hold at least a handful of fixes (order five); a sparser neighbourhood falls back to the absolute bounds alone.<sup>2</sup>

Three properties distinguish the test from the bare speed bound:

- *The floor  $\varepsilon_{\min}$  is not cosmetic.* On a very smooth stretch  $\sigma_k \rightarrow 0$ , so without it any sub-metre wiggle would exceed  $k\sigma_k$  and the test would chase GPS noise where there is no outlier.
- *A burst is one block.* A short burst of consecutive spikes inflates the windowed MAD only mildly, so it is still flagged, and it is removed as the block whose bracketing neighbours rejoin within scale (the attribution argument is the removal-policy paragraph of Sec. 2.7.2).
- *It does not clip real manoeuvres.* A genuine sharp turn is drawn by consecutive, mutually consistent fixes, each well predicted by its neighbours and so of small residual; only an isolated point off the local trend scores large. *The one failure mode, the under-sampled tight turn (Sec. 2.7.2), is measured directly by the injected-defect test (check (1) below).*

The flag itself is context-sensitive – a  $30\text{ m s}^{-1}$  step out of a  $12\text{ m s}^{-1}$  glide is flagged while the same step passes unremarked where it is ordinary – but a flag alone does not delete (next paragraph): a flagged-but-physically-possible fix is *kept*, its flag recorded as a per-flight anomaly count alongside the integrity-gate counters. *The test runs on the horizontal position alone.* An earlier draft of this appendix had it running on the adopted altitude channel as well, which the three-part argument of Sec. 2.7.2 supersedes: on  $z$  an absolute bound is sufficient where on  $xy$  it is not, a local test on  $z$  would flag the thermal entries the segmentation is built on, and a false vertical flag is expensive exactly where it is most likely. The vertical is therefore cleaned by its absolute bounds and by nothing else.

**From flag to action.** The identifier only *flags*; it never repairs. A flagged horizontal fix is *deleted* as a node – not overwritten with the local median – *with all reconstruction deferred to the single resampling step (Sec. 2.7.6; the one-policy rationale is Sec. 2.7.2).* This is the *Hampel identifier*, the detection half of the eponymous filter, not the median-substituting filter itself: the local median enters only as the reference for the residual  $r_k$ , never as an output value.

**Position.** Deletion is the impossibility gate: a fix goes when it is unreachable from the last accepted fix at the absolute  $v_{xy}$  bound *and* the removal of the block it belongs to lets the track rejoin within that bound. The flag is neither necessary nor sufficient—why it cannot be required is Sec. 2.7.2—and the genuine under-sampled tight turn is kept because it flies at an ordinary speed, not because anything exempts it.

**Byte-identical, on the parsed table.** The frozen-lock witness of Eq. (2.4) is specified on the raw B records, but the parser returns decoded floats and keeps no raw text. No raw text is needed: the `DDMM.mmm` field is a fixed-width integer count of thousandths of

---

<sup>2</sup>The residual and run-diameter distances are the haversine great-circle distance of Eq. (2.1); a vectorised off-the-shelf implementation exists as `sklearn.metrics.pairwise.haversine_distances`, should the hand-rolled one need cross-checking.



an arc-minute, so the decode is injective and two records are byte-identical in latitude and longitude *iff* their decoded values compare exactly equal. The test is therefore `lat[i]==lat[i+1]andlon[i]==lon[i+1]` on the parsed columns, with exact equality and no tolerance — a tolerance would turn the conclusive signature into the jittering-freeze case it is meant to exclude.

**Discontinuity.** When no bounded block restores continuity – a position discontinuity, such as a re-acquisition offset, where every removal still leaves an impossible step – the scan stops deleting and marks a split at the step, handled like a long gap (Sec. 2.7.6): both sides are genuine, only the transition is unknown.

**How long a block may run.** “Bounded” needs a number, and it is the Hampel half-width  $w$ : a block that has not rejoined within 20s is called a discontinuity instead. The value is reused rather than introduced, and it also fixes the scale at which the two cases part—an offset smaller than  $v_{xy}^{\max}w \approx 900$  m becomes reachable from the anchor before the block expires and is removed as an excursion rather than split. That is a real limit of the rule and a bounded one: at most one block of genuine fixes, at a discontinuity, against the alternative of splitting a flight in two for every isolated spike.

**The postcondition, and its counter.** After the scan, any step between two *surviving* fixes that still breaks the bound is made a segment boundary (Sec. 2.7.2). The number of boundaries this adds is recorded per flight as `n_boundaried` in `flights_meta`, alongside the other cleaning counters: it is the audit of a rule that would otherwise repair without saying so, and it is expected to be small—a large value would indict the scan rather than the data.

**Altitude.** No flag reaches it: the vertical is cleaned by the absolute bounds of Table 2.6 alone, which mark the value missing and never touch the fix (a deferral: Sec. 2.7.2). Both bounds act on the channel the flight *adopted*, whichever it is. The  $|v_z|$  bound was placed on barometric distributions, but what it expresses is a climb/sink envelope of the aircraft, which does not change with the instrument reading it; leaving a GNSS-fallback flight with no vertical cleaner at all — the consequence of dropping the local test — would be the worse reading. It fires more often there, and the rate is recorded per `alt_source` so the audit can see by how much.

**Isolated, not sustained.** The  $|v_z|$  bound marks an altitude missing when the record jumps away from it and back — both adjoining steps past the bound, with opposite signs. This is check (5) below made operational: a *run* of super-threshold steps of one sign is a sustained manoeuvre, a spiral dive holding  $15 \text{ m s}^{-1}$  to  $20 \text{ m s}^{-1}$  of real sink, and it is counted and left alone rather than censored.

**The first fix.** An anchor-carrying scan never questions its own first anchor, and no rule here supplies one: the discontinuity is left as a discontinuity, for the reason Sec. 2.7.2 gives. The guarantee is enforced one stage later, by the reach criterion of Sec. 2.7.4, and checked on the written table (Sec. 2.2).

This is the *default to keep* rule of Sec. 2.7.2: removal requires positive, corroborated evidence.

### Frozen-lock detection

The rule, its witnesses and its margins are Eq. (2.4) and Sec. 2.7.2; parameter values in Table 2.2. Three points are implementation-level.

- *Diameter, not step.* The run’s bounding diameter is the discriminating statistic because a step-based test would fire on every 1 Hz climb: the fix-to-fix step of a circling wing,  $v_{xy}\Delta t \approx 10\text{ m}$ , is of the order of  $\delta_{xy}$  itself.
- *Robust witness.*  $\Delta z_{\text{baro}}$  is evaluated between medians of the run’s ends, so an altitude spike inside the run – not yet cleaned, since the altitude pass comes last – cannot fake a climb.
- *Bookkeeping.* On a barometric flight the V flag and the zeroed GNSS altitude are recorded as corroborating diagnostics only; they fill the witness slot on GNSS-fallback flights alone (Eq. (2.4)).

A run the rule condemns is marked as a data gap; the split it induces is realised at the uniform-sampling stage (Sec. 2.7.6), on the same footing as a native long gap.

### Absolute bounds and gates

The six numbers of Table 2.6 are the `FixLevelThresholds` dataclass, which `load_preproc_config` reads from `configs/preprocessing.yaml` under the key `fix_level`; nothing is hard-coded in Python. They are the context-free floor of the design, and are placed on the real per-fix distributions (Figure 2.2) rather than guessed. The horizontal-speed bound is set where the bin-to-bin ratio of a finely binned ( $2.5\text{ m s}^{-1}$ ) histogram settles at  $\approx 1$ , i.e. where the decay has actually stopped, not where it first visibly slows ( $\approx 40\text{ m s}^{-1}$  for hang gliders, versus the adopted  $55\text{ m s}^{-1}$ ), since a shallow but real decreasing stretch can still be genuine rare dynamics. The Hampel and frozen-lock parameters live under the same key; Table 2.2 collects them.

**Flight-level integrity gate.** The pass records how much it changed each flight (or segment) – `n_fix_raw`, `n_fix_clean` and `frac_interpolated` in `flights_meta`. A flight whose deleted-plus-reconstructed fraction exceeds  $f$  (10 %, Table 2.2) is dropped whole; the counting window (airborne only) and the rationale are Sec. 2.7.2, whose threshold sweep validates the adopted value rather than setting it. This is the cleaning counterpart of the sampling-regularity cut (Sec. 2.7.6).

### Diagnostics and validation

**Diagnostic sample (Figure 2.2).** Computed by `fix_level_distributions` and rendered by `make_fixlevel_diagnostics_figure` (Table 1.1). Panels (a)–(c) pool a seeded random sample of 15,000 paraglider flights (126,723,499 consecutive-fix pairs,  $\sim 73\text{ s}$  to compute) and the complete 6,716-flight hang-glider population (a full census at this size, 37,915,334 pairs,  $\sim 25\text{ s}$ ) – under two minutes total. The large sample is deliberate: resolving the sparse artefact tail well enough to tell it from real dynamics needs far more points than a headline statistic would. The  $x$ -range in (a)/(b) extends to each discipline’s own 99.99th percentile rather than cropping just past the bound; panel (c) keeps a narrower range, its cut sitting far out in an otherwise tight, unimodal distribution. Panel (b) bins one integer  $\text{m s}^{-1}$  per bin because barometric altitude is logged to the metre: differencing at a 1 s cadence returns integer rates, and a finer grid would alias them into a spurious comb.<sup>3</sup>

**Validating the cleaning.** The primary validation is the two-stage procedure of Sec. 2.7.2: freeze the working values a priori, then audit the per-rule removal fractions over the archive

<sup>3</sup>Not every flight samples at 1 s (Figure 2.7), and differencing over a longer native step returns finer values (multiples of  $0.2\text{ m s}^{-1}$  at 5 s) that partly fill the gaps once cadences are pooled: among 1 s-native fixes about 99% land within  $0.05\text{ m s}^{-1}$  of an integer, against only 34–42% at every other cadence.



| Parameter                                 | Role, and the scale that sets it                                                                                                                    |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Robust local-outlier test (Hampel)</i> |                                                                                                                                                     |
| $w$                                       | half-width of the local time window; a few native intervals                                                                                         |
| $k$                                       | flag threshold, in robust standard deviations $\sigma_k$                                                                                            |
| $\varepsilon_{\min}$                      | absolute residual floor; a few times the horizontal GPS noise, so a smooth stretch ( $\sigma_k \rightarrow 0$ ) is not chased                       |
| <i>Frozen-lock rule (Eq. 2.4)</i>         |                                                                                                                                                     |
| $\delta_{xy}$                             | run-diameter tolerance; a few times the GPS noise, below the $2R \approx 30$ m to 100 m of a circling climb                                         |
| $\delta_z$                                | barometric-flatness witness; a few times the sensor noise and quantisation, $\ll$ the $\gtrsim 60$ m a real climb gains over $\tau_{\text{freeze}}$ |
| $\tau_{\text{freeze}}$                    | minimum span of a cut run; two thermalling periods, so a slow, tight climb is never mistaken for a freeze                                           |
| <i>Flight-level integrity gate</i>        |                                                                                                                                                     |
| $f$                                       | maximum deleted-plus-reconstructed fraction; past it the flight is being rebuilt, not cleaned, and is dropped whole                                 |

**Table 2.2.** Fix-level cleaning: what each parameter of the robust and structural detectors does, and the physical scale that fixes it. The *values* are deliberately not repeated here—they are in Table 2.5, with the rest of the pipeline’s working values, so that there is one place to read a number and one place to change it. They are fixed a priori and checked a posteriori on the per-rule removal fractions (Sec. 2.7.2), with the injected-defect calibration as the direct measurement of the error rates. All live as config keys under `fix_level` in `configs/preprocessing.yaml`, alongside the absolute bounds of Table 2.6.

and sweep each threshold around its working value. The checks below are the fuller battery behind it, run as the implementation matures.

- (1) *Injected defects.* The synthetic-defect procedure of Sec. 2.7.2, item (d), gives the detection floor – the smallest spike caught – and, where the removal-fraction audit flags a rule, recalibrates  $k$ ,  $w$ ,  $\delta_{xy}$  and  $\tau_{\text{freeze}}$  to a target rate: the only direct measurement of the cleaning’s false negatives.
- (2) *Downstream invariance.* Recompute the transport estimators – the displacement-scaling exponent and the waiting-time-tail exponent – over a grid of cleaning strengths, i.e. re-running the cleaning with all thresholds tightened or loosened together by a common factor around the working point, and take the operating point on the plateau where they stop moving; movement under stricter cleaning is the signature of a cut biting into signal.
- (3) *Detector nesting.* The fixes flagged by the absolute bound should nest inside those flagged by the local test (the impossible is a subset of the locally anomalous); a large disagreement points to a defect class neither was designed for – a cheap guard.
- (4) *Per-phase removal rates.* The error this pipeline most wants to avoid is losing a slow-phase fix. Once the segmentation is in place (Ch. 3), the removal and split rates are measured *per phase* and required to be flat across transition, search and climb rather than enriched in the two slow phases; and synthetic tight climbs – circles at flying speed, over a range of radii and wind drift – planted in otherwise clean tracks confirm directly that the frozen-lock detector never splits a real climb. This is the false-*positive* complement of check (1), and its natural readout is the tail of  $\psi(\tau_w)$ : a cut biting into climb or search would thin that tail before it moved anything else.
- (5) *Isolation of the vertical cuts.* The fixes cut by  $|v_z| > 13 \text{ m s}^{-1}$  must be *isolated*, not

consecutive: a sustained spiral dive holds  $15\text{ ms}^{-1}$  to  $20\text{ ms}^{-1}$  of real sink, so a run-shaped excess is the signature of a genuine manoeuvre – calling for a raised bound or a coherent-run exemption rather than censoring – whereas an isolated super-threshold difference, bracketed by ordinary rates, is a sensor error and safe to cut.

### 2.7.3 Trimming of ground phases

The two thresholds of Eq. (2.5) ( $v_0, T_0$ ) are the `TrimmingThresholds` dataclass, loaded from `configs/preprocessing.yaml` (key `trimming`); values in Table 2.8.

The interior-ground guard of Sec. 2.7.3 keeps `interior_ground_s` ( $T_{\text{ground}}$ , 600s) and `ground_flatness_m` (5 m) in the same group. The flatness is tested on the *detrended* altitude, as Sec. 2.7.3 requires: fit a straight line to the adopted channel over the candidate stint by least squares, subtract it, and compare the residual’s central span—its 5th to 95th percentile—against `ground_flatness_m`. The linear term is the barometric drift, which over a stint of  $T_{\text{ground}}$  is already of the order of the tolerance; the residual is what is left of genuine vertical motion, and the central span rather than the full range because a range grows with the length of the stint and a central span does not (Sec. 2.7.3). No new config key is needed.

Building it made one thing explicit that the rule as stated leaves open. Detrending cannot by itself separate a pressure drift from a steady climb, because both are linear and the fit removes either completely: a wing thermalling at  $1\text{ ms}^{-1}$  in still air holds a ground speed near zero, gains 600 m over a  $T_{\text{ground}}$  stint, and leaves a detrended residual of *zero* — so the flatness condition alone would excise it as a mid-flight landing, which is precisely the failure this guard exists to avoid. What separates the two is the *rate*: drift runs at tens of metres per hour, a soaring climb two orders of magnitude faster. The fitted slope is therefore bounded as well, at  $\delta_z/\tau_{\text{freeze}}$  — the frozen-lock rule’s own statement of what counts as motionless, read as a rate, so the guard borrows a number the configuration already fixes instead of introducing one. Below  $T_{\text{ground}}$  the same predicate is stored as a list of suspect intervals per flight, which the segmentation later maps onto the phases they touch (consumption of the flag: Sec. 2.7.3).

### 2.7.4 Flight-level filtering

The three cuts of Sec. 2.7.4 are the `FlightLevelThresholds` dataclass (key `flight_level` in `configs/preprocessing.yaml`); values in Table 2.8. The census measurements behind the path floor and the upper duration bound, and the totals of Figure 2.3, are quoted through macros computed from the shared track scan (Sec. 2.7); the headline duration and path medians are quoted in Sec. 2.8.

The plausibility and reach bounds of Sec. 2.7.4 are not further `flight_level` keys: both read `max_horizontal_speed_mps` from the `fix_level` block, so the same number governs a single step, a whole-flight average and a whole-flight displacement, and the three cannot drift apart. Both are evaluated on the trimmed track, after the per-block sums of the duration and the path, and they are the last of the whole-flight cuts because each is a statement about a record the earlier ones have already accepted. The extent they need is returned by `flight_quantities` alongside the duration and the path, from the same haversine call, so it costs nothing: the maximum of a vector the function already forms. Unlike the duration and the path it is *not* summed per block—a displacement from a fixed origin has nothing to sum—which is why a split does not weaken it.

The chain behind every number quoted in Sec. 2.7.4 is short enough to name in full, so that a count can be traced to the code that produced it:

- `soaring.analysis.igc.parse_igc` decodes one IGC file into a table of fixes;
- `soaring.analysis.census.track_stats` reduces those fixes to the one summary row described in Sec. 2.7, and `scan_tracks` / `load_or_scan_tracks` do that for every flight

- of a discipline and cache the result;
- `fraction_retained` and `retention_curve` in the same module evaluate a single cut against a column of that table—the first at one threshold, the second along a grid of them, which is what the bottom row of Fig. 2.3 plots;
- `scripts/reporting/generate_census_stats.py` calls the same functions with the thresholds read from `configs/preprocessing.yaml` and writes the `\StatScan*` macros the text quotes.

Two consequences follow, and both matter for reading the numbers. The counts are *marginal*: each is the effect of one criterion applied alone to the whole parsed population, not of the cascade. And they are pre-cleaning: the scan runs on parsed tracks, so a count in Sec. 2.7.4 is what the criterion removes from the parsed archive and not from the cleaned one. That is deliberate and is argued for above (*What the census is not*): a threshold chosen on the population the cleaning has already reshaped would be chosen against its own effect. The post-cleaning counts are the cascade of Sec. 2.9, and the two answer different questions.

### 2.7.5 From geographic to local Cartesian coordinates

No implementation content: this step is pure coordinate geometry with no configurable parameters, and both of its figures (Figure 2.4, Figure 2.5) are hand-drawn TIKZ diagrams, not data plots. The mathematics behind Eqs. (2.6) and the ENU rotation is derived in Appendix 2.B.

### 2.7.6 Uniform sampling within a flight

The sampling thresholds (key `sampling`, the `SamplingThresholds` dataclass; values in Table 2.8) act as follows.

**Split bound.**  $g_{\max} = \min(\max\_gap\_factor \times \Delta t, \max(\max\_gap\_seconds, 2 \Delta t))$ , pinned by the constraint chain of Sec. 2.7.6. Its audit is a per-flight count – the census scan records `max_gap_ratio` and `dt_s`, so the absolute gap is their product – and joins Figure 2.6 when regenerated.

**Missing fraction.** Acts per *segment* (rule: Sec. 2.7.6); the per-flight value in the census scan is the pre-split diagnostic, not the operational gate.

**Segment gate.** `min_segment_duration_s` (90 s). It only needs to support the smoothing window and the within-segment statistics: a few smoothing timescales, never less than  $w \Delta t$ ; the only gate applied at segment level (Sec. 2.7.6).

**Measured vs interpolated.** A grid point counts as measured when a fix lies within half a native step of it, interpolated otherwise; the same tolerance decides whether a flight is uniform as recorded (no interpolated points) or was resampled (`was_resampled`).

The bound is on the *time base* and is the same for every channel: an order-of-magnitude smaller  $v_z$  buys no wider gap for the altitude, since the interpolation error scales with curvature, not speed, and a wide altitude gap would erase the climb–glide transitions that  $v_z$  is there to resolve (Sec. 2.7.6).

**Segment handling.**

**Origin and clock.** A split assigns a `segment_id` and nothing else (semantics: Sec. 2.7.6).

**Tables.** One row per fix in `fixes`, keyed (`flight_id`, `segment_id`); per-segment records (start/end, fix count, `frac_interpolated`, censoring flags) in a `segments` table beside `flights_meta`. No separate files; the ensemble MSD at lag  $t$  reads  $\mathbf{r}(t)$  wherever some segment covers  $t$ .

**Censoring.** Any phase duration truncated at a segment boundary is flagged censored, for the  $\psi(\tau)$  fits to exclude.

**Per-channel fill.** Across a short gap the horizontal position is interpolated by monotone piecewise cubics (`scipy.interpolate.PchipInterpolator`) and the adopted altitude channel linearly (its within-phase smoothness makes the choice among low-order methods immaterial), realising the completeness invariant of Secs. 2.7.2 and 2.7.6. Each filled grid point is flagged in an `interpolated` boolean column of `fixes`, and `frac_interpolated` records the per-segment fraction for the integrity gate.

Figures 2.6 and 2.7 are rendered by `make_gap_diagnostics_figure` and `make_sampling_figure` from the shared track scan (Sec. 2.7).

**Gap-ratio discreteness (Figure 2.6).** Checked directly: only 86%/63% (para/hang) of largest-gap-ratio values are exact integers, the rest a genuinely continuous scatter; the integer peaks visible for hang gliders sit around  $k = 3, 6, 10, 15$ . Panels (a)/(b) of the figure therefore use linear axes: a logarithmic axis would compress precisely this small-integer structure, on which the adopted cut sits.

**Native-rate discreteness (Figure 2.7).** Checked directly:  $\Delta t$  lands on an exact whole second for 99.9% of paragliders and 100% of hang gliders. **Per-rate breakdown:** paragliders sit mostly (69%) at 1 s; hang gliders spread across the rates below.

| Native rate  | 1 s  | 2 s  | 5 s  | 10 s | other     |
|--------------|------|------|------|------|-----------|
| Hang gliders | 30 % | 13 % | 27 % | 15 % | remainder |

### 2.7.7 Smoothing and differentiation

Three decisions are pinned for the implementation (status: Table 2.1):

- the filter runs per segment (never across a boundary) with `scipy.signal.savgol_filter` in `mode='interp'`, so the terminal half-windows are fitted on the edge window rather than padded, and edge samples inherit the segment-boundary caution of Sec. 2.7.6;
- the window is floored at  $w = 5$ , the smallest a cubic fit accepts (slow-logger consequences: Sec. 2.7.7, item iii);
- the polynomial order and the three smoothing timescales are the `SavgolParams` dataclass in `configs/preprocessing.yaml` (key `savgol`): `tau_c_horizontal_s` for  $(E, N)$ , `tau_c_vertical_baro_s` and `tau_c_vertical_gnss_s` for the vertical channel (the flight's adopted altitude, Sec. 2.7.1); values in Table 2.8.

**The measured timescales.** The knees were measured with `estimate_savgol_timescales.py`, which applies the Welch procedure of Sec. 2.7.1 to  $E$ ,  $N$  (local equirectangular metres) and to the altitude channel each flight actually uses, on a seeded sample (908 horizontal flight-components, 719 barometric and 189 GNSS-fallback flights at 1 Hz). The knee rule: the floor is the median PSD over 0.35 Hz to 0.5 Hz;  $f_c$  is the lowest frequency whose median PSD falls below three times it. The measured floors sit at  $\approx 0.60$  and  $\approx 0.26 \text{ m}^2 \text{ Hz}^{-1}$  against the theoretical quantisation floors  $q^2/12/f_{\text{Nyq}}$  of 0.57 and 0.17; the three knees coincide at

$f_c \approx 0.2$  Hz, hence  $\tau_c = 5$  s for every channel (Sec. 2.7.7, item (iv), for the reading), and even the 90th-percentile spectra move the knee by less than a second. The two vertical keys stay, conditioned on `alt_source`, for the noisy-minority refinement; the validation of Sec. 2.7.7 (flat residual spectrum, stability under one window step) is what confirms the values. The filter runs on the resampled grid, so the small interpolated fraction (`frac_interpolated`) is already in place when it reads its window.

## 2.8 Preliminary characterization

`scripts/reporting/generate_prelim_figure.py` produces both figures of Sec. 2.8 and the `\StatPrelim*` macros. It reads three things: the per-flight summary and the per-lag positions `audit_msd.py` wrote (Sec. 3.1), `flights_meta.parquet` for the take-off coordinates, and the season catalogue for `wing_class` and `season`. The join on `flight_id` resolves for 99.99 % of retained flights.

Reading the audit arrays rather than the fix table is what makes the stratification affordable: a stratified MSD is then a row selection on a matrix already in memory, so the three stratifications and the isotropy check together cost one pass that has already been paid, instead of one 43 GB scan each.

**The basemap.** The take-off panels are drawn on Natural Earth admin-0 country polygons (public domain), which give the coastline and the borders in one layer and carry the overseas departments inside the France feature—which is what puts La Réunion on the map at all. The geometry is not fetched at draw time and Cartopy is not a dependency: `scripts/reporting/build_basemap.py` downloads it once, keeps the rings that intersect each panel, decimates them ( $0.01^\circ$  for France,  $0.001^\circ$  for La Réunion,  $0.25^\circ$  for the world frame) and writes `data/basemap.json`, which is committed at 0.76 MB. The figure then needs nothing but Matplotlib and regenerates from a clean checkout with no network. Rings are kept whole rather than clipped to the frame—Matplotlib clips to the axes anyway, and clipping a polygon properly would need the geometry library the arrangement exists to avoid. Re-run the builder only to change a panel’s extent.

**The three frames.** Metropolitan France  $\lambda \in [-5.5, 10]$ ,  $\varphi \in [41, 51.5]$ ; La Réunion  $\lambda \in [55.15, 55.92]$ ,  $\varphi \in [-21.42, -20.82]$ ; the world panel for everything outside both. The split is the archive’s, not a choice: 4,939 paraglider flights launch on La Réunion and 3,359 outside France altogether, and a single frame that held them would render metropolitan France four pixels wide. Each panel sets its aspect to  $1/\cos \varphi$  at its own mid-latitude, so the coastline has its true shape rather than a stretched one.

**The orographic boxes.** The strata are longitude/latitude boxes rather than polygons traced from the map: a box that is written down can be checked against a gazetteer, and the test the strata support is whether the curves separate, not where exactly the Alps end. Alps  $\lambda \in [5.4, 10.0]$ ,  $\varphi \in [43.8, 46.6]$ ; Pyrenees  $\lambda \in [-1.9, 3.3]$ ,  $\varphi \in [42.0, 43.5]$ ; Massif Central  $\lambda \in [1.8, 4.6]$ ,  $\varphi \in [43.6, 46.2]$ ; anything else inside  $\lambda \in [-6, 10]$ ,  $\varphi \in [41, 52]$  is “lowland” and anything outside it is “abroad”. The boxes are coarse and overlap along their edges, with the first match winning in the order listed; they are a stratification, not a geography, and the test they support is whether the curves separate rather than where exactly the Alps end.

**The stratum floor.** A stratum below 2,000 flights is pooled into the residual group rather than drawn. Below that the curve’s own sampling error at the long-lag end is comparable to the separations being tested for, and a panel that draws it invites the reader to compare noise.

The floor is why the hang-glider archive, at 6,132 flights, contributes only to the isotropy panel: one wing class and one orographic group clear it, so the stratified panels would compare a stratum against itself.

**The isotropy test.** The variance ratio is computed on the same per-lag position matrix, column by column, over the flights covering that lag. The Kolmogorov–Smirnov comparison runs at four lags spanning the fit range, on the signed components rather than their magnitudes, so that a difference in *spread* and a difference in *shape* both register. Only the distance is quoted. At 155,788 flights the two-sample *p*-value underflows to zero for any departure worth naming, so it separates nothing; the distance is the effect size and is what the text reads.

**Stratum departure.** Each stratum’s curve is divided by the pooled curve over the fit range and the largest  $|\cdot - 1|$  recorded. Two summaries are generated: the maximum over strata (`\StatPrelim*MaxDeparturePct`) and the median over strata (`*TypicalDeparturePct`). The maximum alone reads as a statement about the smallest and noisiest stratum, which is what it usually is; the median is what distinguishes an axis along which the ensemble genuinely splits from one along which a single small group wanders. Both are quoted in Sec. 2.8.

## Chapter 3

# Global transport

This chapter mirrors Chapter 3. Built and recorded here: the two mean-squared-displacement estimators and their audit, the filtered-variation family that replaces them as a source of exponents, the moment spectrum, the non-Gaussian parameter, the velocity autocorrelation and its tail, and the persistence runs. One observable the body catalogues is defined and not measured, and Sec. 3.8 says which.

### 3.1 Observables on the whole trajectory

The mean-squared displacement of Sec. 3.1 is implemented and unit-tested, as is every other observable Chapter 3 defines except the return probability, which Sec. 3.8 names as the one gap. The estimators live in `soaring.analysis.observables.transport` and the figure, the curve table and the `\StatMsD*` macros come from one pass of `generate_msd_figure.py` (Table 1.1). A second pass, `audit_msd.py`, keeps what the first one averages away, and `audit_msd_report.py` reduces it into the `\StatAudit*` macros; both are described at the end of this section.

#### 3.1.1 The two estimators

`MSDAccumulator` builds the ensemble average and `TAMSDAccumulator` the time average. Both accumulate by streaming rather than by loading: the `fixes` table runs to  $\sim 1.4 \times 10^9$  rows over the two archives, and neither estimator needs more than one flight in memory at a time.

**Coverage.** A flight answers a lag  $t$  with the fix nearest to it, and only if two conditions hold: that fix lies within  $\max(0.5\text{ s}, \Delta t/2)$  of  $t$ , and  $t \geq \Delta t$ , where  $\Delta t$  is the median inter-fix time of the flight’s own concatenated segments. The second condition is the one that is easy to omit and expensive to omit. Without it a 5 s logger asked for a lag of 1 s answers with its fix at  $t = 0$ —a displacement of exactly zero—and a slow-cadence minority drags the short-lag MSD down until the grid crosses their step. The same rule appears in the time-averaged estimator, which refuses any lag below its segment’s own step. A lag falling inside a gap between two retained segments is answered by neither side, since the split bound (Sec. 2.7.6) makes every gap wider than the tolerance.

**The lag grid.** `log_lag_grid` spaces 90 points geometrically from 1 s to 43,200 s, the duration bound of the flight-level filter, and snaps them to the second. The snapping is what makes the short end usable: unsnapped, lags of 1.00, 1.13 and 1.27 seconds all resolve to the same fix at a 1 Hz cadence, and the curve there is a staircase of repeated values whose local slope alternates between zero and a spurious spike. Snapping merges them instead, which is why the grid



delivers 80 distinct lags rather than 90, and why the lags below 8 s are consecutive integers. At long lags the rounding is a part in  $10^4$ . The consecutive-integer run is also what lets the coverage artefact of Sec. 3.1 have a period; that is the cost of the snapping, it is measured in Sec. 3.1 by splitting the archive by cadence, and it sits an order of magnitude below the lower end of every fit range.

**The time average.** `time_averaged_msd` is computed within a segment, never across the gap that ends it. The direct double loop is  $O(n^2)$  and hopeless at  $10^4$  samples per segment; writing  $D_j = |\mathbf{r}_j|^2$ , the identity  $\delta^2(k) = \langle D_j + D_{j+k} \rangle_j - 2\langle \mathbf{r}_j \cdot \mathbf{r}_{j+k} \rangle_j$  turns the second term into an autocorrelation, which the FFT gives in  $O(n \log n)$ . The curve is cut at half a segment's length, beyond which the average over starting times has too few of them to be one.

### 3.1.2 The fit, and what is quoted from it

**The range.** The lower end is an argument, not a consequence: at 120 s it sits above the thermalling period, so what the slope measures is transport rather than the circling superimposed on it, and above the whole region the short-lag artefacts occupy. The upper end is a consequence. `coverage_limited_range` walks the count-per-lag curve and cuts where it has fallen to `min_coverage` = 0.25 of its value at the lower end; past that the curve is an average over a survivorship-selected minority. Applied to the four curves this gives the ranges quoted with each exponent.

**The two errors.** `fit_msd_exponent` fits unweighted in log space, which gives every decade the same say; a fit weighted by the much larger absolute errors at long lags would be dominated by the last decade alone. It returns the least-squares error on the slope, which is written out as `\StatMsd*AlphaErr0ls` and is *not* the quoted uncertainty. `bootstrap_alpha_error` resamples 200 ensembles of flights with replacement, refits each over the same range, and reports the standard deviation as `\StatMsd*AlphaErr`. Neither figure is attached to an exponent in the body: both MSD exponents are quoted bare, because the bootstrap treats clustered flights as independent records and because the range dependence is fifty times larger where both have been measured. Sec. 3.1 makes that argument and gives the synthetic check that establishes what each of the two errors measures; the exponents that do carry an interval are those of Sec. 3.3.

**The cohorts.** Three fixed-duration cohorts at  $T = 1, 2$  and 4 h. Within a cohort every member is present at every lag up to  $T$ , so the population is constant by construction and any remaining growth is motion; each is therefore fitted from the range's lower end up to its own  $T$ , which is as far as the guarantee reaches. The time-averaged cohorts are keyed on segment span and capped at  $T/2$ , for the same reason the time-averaged curve is. A cohort is compared against the *pooled curve refitted over the cohort's own lags*, never against the pooled exponent over the pooled range: the local slope is not constant across those two spans, so the naive comparison charges the shape of the curve to the change of population. Both readings are generated (`\StatMsd*CohortSpread` and `*CohortSpreadNaive`), and on the ensemble estimator they differ by a factor of five, which is the size of the mistake avoided.

**The local slope, and why it is a regression.** The panel that carries the most interpretation is the one whose estimator is easiest to get wrong. The local slope is  $d \log \text{MSD} / d \log t$ , and the obvious way to compute it—a centred difference between neighbouring lags—divides by the grid spacing, which on the logarithmic grid used here is about 0.05 decades. The division multiplies whatever scatter the curve carries by  $1/(2 \times 0.05) \approx 4$ : a point-to-point scatter of a

few tenths of a per cent in the MSD, which is about what the ensemble’s own standard error allows, becomes a slope that jitters by 0.05 and reads as noise. The panel then testifies against a curve that is not in fact noisy.

`soaring.analysis.observables.transport.local_slope` therefore works on a window: at each lag it takes the points within 0.15 decades on either side—five to six at this grid spacing—and reports the *median of the slopes of all pairs* of them, a Theil–Sen estimator. The choice is for resistance to a single displaced lag rather than to a heavy-tailed residual, which is the failure this curve actually has: a coverage artefact of a few per cent at one lag tilts a least-squares window and moves a median by nothing. The uncertainty reported with it, and drawn as a band, is the interquartile range of those pairwise slopes divided by 1.35 and by the square root of the window’s size—the window’s own scatter about its line, expressed as an error on the slope. Over the fitted range it is 0.017, against a point-to-point variation of 0.052: the curve moves more between neighbouring lags than the scatter within a window accounts for, which is the quantitative statement that what the panel shows is structure and not noise. The half-width is a stated compromise rather than a tuned one: narrower and the amplification returns, wider and a genuine bend is flattened into the straight line the panel exists to test for; 0.15 decades resolves a crossover of half a decade. It is not a smoother in disguise. The residual wiggle of this archive is correlated across some four grid points, so it survives the window—which is the right outcome for something that is not noise, and the reason the window is reported with the figure rather than chosen to make the curve look better.

### 3.1.3 The audit pass

The ensemble curve turned out not to be a power law (Sec. 3.1), and none of the questions that decides—is the shape the coverage rule, the survivorship, a shared heading, or the motion—can be asked of an averaged curve after the fact. Each is a different reduction of the same per-flight quantity, and that quantity is gone by the time the average exists.

`audit_msd.py` therefore makes a second streaming pass and writes what the first one discards: one row per flight and one column per lag, holding each flight’s  $E$  and  $N$  at that lag under the estimator’s own coverage rule, NaN where it does not cover it, plus a per-flight summary (duration, path, native step, mean-velocity components, largest in-segment step speed, coordinate extent). The coverage rule is *copied* into that script rather than imported, so that a change to the estimator surfaces as a disagreement between the figure and the audit instead of being tracked silently. The arrays run to a few hundred megabytes per discipline and are analysis products, not thesis ones: they live outside the repository, under `AUDIT_DIR`.

`audit_msd_report.py` reduces them, together with the curve table `msd_curve.csv` and the segment and flight tables, into `\StatAudit*`. Everything the audit asserts in the body reaches it through those macros. What it computes: the rms residual each estimator leaves against the power law fitted to it, in dex and as a percentage; the range of the local slope over the fitted decades; the mean vector’s share of the mean square,  $|\langle \mathbf{r} \rangle|^2 / \langle |\mathbf{r}|^2 \rangle$ ; the local slope recomputed on sub-ensembles of a single native cadence, where the population cannot change, and on fixed-duration cohorts, where it cannot be selected; the first-crossing time of 5 km from take-off, per flight; the per-component variance ratio and the Kolmogorov–Smirnov distance between the  $E$  and  $N$  marginals; and the pre-processing losses of Table 2.7 decomposed by the rule that caused them.

Two of its numbers are worth naming here because they are checks on the pipeline rather than on the estimator. The retained tables carry 0 non-finite coordinates and 0 non-monotone clocks, which is what Sec. 2.2 asserts and this pass confirms independently. And the flights whose first retained segment did not survive resampling start their record at  $t > 0$  rather than at the origin—1.4% of paraglider and 11.7% of hang-glider flights. Their position is

still measured from their own take-off, since the ENU frame is anchored there and a dropped segment does not move it; they simply do not answer the lags before their record starts. For the rest, the position at  $t = 0$  is the origin to within 0.09 m in the median, the smoothing's displacement of the first sample.

### 3.1.4 The shape observables, and what calibrates them

**The non-Gaussian parameter.** Computed from the moment spectrum already accumulated, as  $\alpha_2^{\text{NG}} = \langle |\Delta \mathbf{r}|^4 \rangle / 2 \langle |\Delta \mathbf{r}|^2 \rangle^2 - 1$  at each lag, on order-1 increments. Two calibrations make the archive's value readable and both are pinned by `tests/analysis/observables/test_moments.py`, at the sample size the test uses: an exact fractional Brownian motion returns 0.02 in the median over a ten-point grid spanning the quoted range, and a Lévy walk of tail index 1.5 returns 0.55. The Gaussian 0.02 is estimation bias and not shape — a fourth moment over  $10^4$  increments is not yet its own expectation — and sixteen times the data collapses it to within a hundredth of zero, sign included. The same test pins that too, since it is what makes the archive's own value a departure rather than an offset. It is quoted over 60 s to 2,000 s rather than over the whole grid; above that the declared task governs the trajectory and the fourth moment rests on the fewest flights, and the curve climbs to +0.47 there.

**The tail of the velocity autocorrelation.** Green-Kubo is used in its scaling form,  $C(\tau) \sim \tau^{-\gamma}$  with  $\gamma < 1$  giving  $\langle |\Delta \mathbf{r}|^2 \rangle \sim \Delta^{2-\gamma}$ , and not as the integral. Two reasons, both properties of the archive: the integral runs from zero and  $C$  is estimable only above the smoothing scale of the slowest logger, so its first decade is missing; and  $C$  is estimated per flight with that flight's own mean velocity removed, which pulls every lag down by about the record-mean of  $C$  and so steepens the tail. Both push  $\gamma$  up, so  $2 - \gamma$  is a floor and `vacf_tail_exponent` documents it as one. The one-sidedness is itself tested, on an Ornstein-Uhlenbeck velocity whose true correlation is known: the estimate with the mean removed is always the steeper.

**The filter kernel, against a closed form.** For any process with stationary increments and  $S(\Delta) = \sigma^2 \Delta^{2H}$ , the two orders are related exactly by  $V_2 = 4S(\Delta) - S(2\Delta) = S(\Delta)(4 - 2^{2H})$ . That identity pins the kernel in a way a second convolution cannot: the order-2 kernel is symmetric, so an independent `np.convolve` route performs the same three products in the same order and agrees to the last bit whether or not the kernel is right — which is what a recomputation of the paraglider exponent by that route in fact found. The estimator also has to be unbiased where the archive sits, at  $H$  near 1, which is the boundary of the family: on exact fractional Brownian motion of  $H = 0.99$  it returns 1.98.

**When a knee is declared.** A bilinear fit to  $q\nu(q)$  has two more parameters than a straight line through the origin, always reduces the residual, and on a grid of ten moment orders buys the reduction cheaply — fitted to a fractional Brownian motion, which has no knee, BIC prefers the bilinear form while the straight line already fits to an rms of 0.003. So BIC alone is not the rule. `bilinear_fit` declares a knee only when three conditions hold together: BIC prefers the bilinear form, *and* the straight line leaves a residual above `min_departure` = 0.02, *and* the knee sits inside the  $q$  grid rather than on its edge. On the synthetic pair those three give no knee for fractional Brownian motion and a knee at 1.57 with a right-hand slope of 0.96 for a Lévy walk of index 1.5. A Lévy walk of index 1.5 leaves a departure of about 0.084 on this grid; the archive leaves 0.0109 and 0.0057, which is roughly eight and fourteen times smaller, and about half and a third of the declaration threshold. The margin against a knee is therefore an order of magnitude; the margin against the gate is a factor of two, and the two should not be confused.

**Why the drift is filtered and never subtracted.** `centring_bias` measures the alternative rather than arguing against it: fractional Brownian motion of known exponent is generated, the mean-squared displacement is computed raw and after subtracting each realisation's own net velocity, and the two are compared lag by lag. Subtracting a velocity estimated from the record's endpoint turns the series into a bridge whose increments must sum to zero, so the centred curve is pulled down at every lag and hardest at the long ones — 2 % at a lag of 0.05 % of the record and 76 % at half of it, with no range in between where the distortion is negligible. On  $\alpha = 1.50$  the operation returns 1.35. The filtered variation of `soaring.analysis.observables.variations` estimates no drift at all, so none can be estimated badly, and the order scan is what reads off how much drift there was.

**The variation pass, and why stratifying is free.** `measure_variations.py` traverses the fix table once and keeps, per flight and per filter order, one curve over the lag grid together with the flight's cadence, wing class, season and declared task. Every stratification of Sec. 3.3 is then a row selection on that table rather than another pass over  $1.4 \times 10^9$  rows, which is what makes four cuts affordable where one would otherwise be a day's work.

**The persistence runs, and the bias that decomposition avoids.** `persistence_runs` walks the record greedily from its start, extending a run while the sinuosity — arc length over chord — stays under the threshold and beginning the next run where the last one ended, with a floor of five samples. The stopping rule is *first violation*: the condition is not monotone in the endpoint, since a stretch can come back inside the threshold after leaving it, so the rule is stated rather than implied. No sample belongs to two runs, and that is the point. Sampling the run length at every instant instead returns each run in proportion to its own length — the length-biased distribution, whose tail index is one below the truth. `inspection_biased_runs` implements the wrong way on purpose so the size of the error can be measured: on synthetic runs of known index 2.0 the two estimators return 2.0 and 1.1. At the values this archive shows that is not a bias to correct afterwards but a different answer. The first implementation of the biased estimator measured the residual life instead of the containing run, which is a third quantity again; a test caught it.

**First-passage times, and why they are not reported.** The audit arrays keep each flight's position at every lag, so the first-passage time  $T(L)$  — the first time a flight is  $L$  from take-off — costs no traversal, and  $\langle T(L) \rangle \sim L^{1/H}$  would give the exponent in the space domain, independently of any moment fitted over lags. It was computed and is not reported, for a reason worth recording so it is not attempted again. Read naively it returns  $\alpha = 2.46$ , which no process with stationary increments can give. The cause is right censoring: at 34 km only a quarter of paraglider flights arrive and the missing three quarters are the slow ones. Treating non-arrival as censored and taking a Kaplan–Meier median instead returns 1.95 for paragliders and 2.03 for hang gliders, both inside the paraglider interval — but refitting over sub-ranges of radius moves the paraglider value between 1.51 and 2.03, a spread five times the uncertainty it was meant to check. The reason is the one Sec. 3.1 gives for the ensemble MSD.  $T(L)$  is measured from take-off, so it shares that estimator's origin: at small  $L$  the wing is still climbing out inside its launch area, and at large  $L$  the censored median extrapolates past the duration of most flights. It is the same crossover seen along the space axis instead of the time axis, and it is not an independent check on anything.

**Why the runs test is not read against a normal table.** The Wald–Wolfowitz statistic is asymptotically normal under a null of *independent* signs, and on a curve every lag of which averages the same flights the signs are nothing of the kind: the measured lag-1 residual

autocorrelation is 0.89 to 0.95 on all four curves. Simulating an exact power law observed with AR(1) noise of that correlation on the same forty-lag grid, the rule  $|z| > 2$  rejects it in 84 % of trials at  $\rho = 0.8$  and 90 % at  $\rho = 0.9$ , and the true five-per-cent threshold is  $|z|$  near 5.1. The  $z$  values this archive returns,  $-4.9$  to  $-5.8$ , are therefore not significances, and `residual_runs` now says so in its own docstring.

Calibrating the null on the *observed* residual autocorrelation would be circular in the way this project has been caught by once already, since a genuinely bent curve has a high residual autocorrelation because of the bend. So the conclusion rests on a different comparison, `residual_against_noise`: the residual against the standard error of each lag’s own mean, taken from the stored 10 % to 90 % band and count on a normal equivalence, which understates a right-skewed spread and so returns a lower bound. It gives 22 to 51 across the four curves — a departure that many times the error on each point cannot be scatter, whatever its arrangement.

### 3.1.5 Known limitations, with their size

Four, each bounded rather than repaired.

*Edge samples enter.* The smoothing evaluates its polynomial off-centre at the ends of a segment and the table flags those samples `edge` (Sec. 2.7.7). `verify_dataset.py` qualifies its invariants by that flag—which is why its speed invariant holds—and the MSD estimators do not, and read every fix. Those samples are under 1 % of the table and account for 26,368 of the  $26,368 + 32$  in-segment steps above the fix-level speed bound, 0.001,9 % of all steps. The worst step between two *measured* samples is  $49 \text{ m s}^{-1}$  against a bound of  $45 \text{ m s}^{-1}$ : an excess of under a tenth, which the verifier reports and does not fail on, its threshold for a data defect being twice the bound. Excluding the flagged samples moves the ensemble MSD by up to 13 % below 10 s and by under 0.2 % above 60 s, so the fit ranges are unaffected; excluding them would also cost every flight its shortest lags, since the first samples of a record are exactly the flagged ones.

*Absolute position carries re-acquisition offsets.* Bounded in Sec. 3.1 at under a per cent of the mean over the fitted range, on 0.029 % of paraglider flights.

*The archive holds a few duplicate flights.* The same physical flight uploaded twice, detected by take-off coordinates, duration and path length agreeing to four decimal places, one second and ten metres: 186 paraglider flights in 93 groups, 0.119 % of the archive, and 29 hang-glider flights in 14 groups, 0.473 %. Ensemble averaging assumes independent draws, and at this rate the violation is immaterial; it is reported so that it is not assumed to be zero.

*The last lag of the grid is empty.* 43,200 s is the flight-level duration bound, so no retained flight reaches it and the row is written with a null MSD and a zero count. It is dropped by every fit and every plot through the finiteness mask, and appears in `msd_curve.csv` for grid completeness.

**Reading the table.** Through `soaring.analysis.derived.stream_flights`, never by iterating Parquet row groups: a row group is a unit of storage and cuts across flights, so the direct reading counted 0.7 % of flights twice and truncated the longest segments, which are precisely the ones the long-lag end of the time-averaged curve rests on.

# Chapter 4

## Flight phases

This chapter mirrors Chapter 4. None of that code exists yet, so each entry is a placeholder rather than a populated record; each will gain its own implementation content — module names, sample sizes, reproduction commands — as the code is written.

### 4.1 Segmentation into flight phases

Not yet implemented; see Sec. 4.1.

### 4.2 Phase-resolved observables

Not yet implemented; see Sec. 4.2.

### 4.3 Reproducing the reference results

Not yet implemented; see Sec. 4.3.

### 4.4 Modelling and simulation

Not yet implemented; see Sec. 4.4.

### 4.5 Tooling

Largely underway already: see Sec. 4.5 and the reproducibility map, Table 1.1.